Lesson 1.01	**Duration: 40 minutes**

Functions Introduction

Learning Objectives

- Define what a function is and explain its purpose in programming
- Identify the components of a function: name, parameters, and return value
- Write simple functions that perform specific tasks

Timing Breakdown

0-5 min	**Engage**	Show a video call in progress or recording. Ask: 'What technology makes this p...
5-15 min	**Explore**	Show different online meeting tools: Zoom, Google Meet, Jitsi. Demonstrate joini...
15-25 min	**Explain**	Define online meeting components: audio/video, internet connection, meeting link...
25-35 min	**Elaborate**	Task: 'Create a Google Form to survey your class about their favorite subject. I...
35-40 min	**Evaluate**	Exit ticket: '1. Name two online meeting tools. 2. What is the difference betwee...

Engage (5-7 minutes)

Teacher Activity

Show a video call in progress or recording. Ask: 'What technology makes this possible? What are the advantages over a phone call?' Discuss online meetings as a communication tool.

Student Activity

Identify features: video, screen sharing, chat, multiple participants. Discuss when video calls are better than phone calls.

Explore (10-12 minutes)

Teacher Activity

Show different online meeting tools: Zoom, Google Meet, Jitsi. Demonstrate joining a meeting, using mute/camera, screen share, chat. Compare features.

Student Activity

Observe the demonstration. Take notes on key features. Try Jitsi Meet no account needed if computers available.

Explain (10-12 minutes)

Teacher Activity

Define online meeting components: audio/video, internet connection, meeting link, host controls. Explain online workspaces: Google Workspace, Microsoft 365. Show how forms, sheets, and docs work collaboratively.

Student Activity

Take notes. Understand the difference between meetings and workspaces. See how collaboration works in real-time.

Elaborate (8-10 minutes)

Teacher Activity

Task: 'Create a Google Form to survey your class about their favorite subject. Include 5 questions with different input types multiple choice, text, rating. Share the form with a partner.'

Student Activity

Work independently. Create the form with varied question types. Test it by filling it out themselves. Share with partner.

Evaluate (5-8 minutes)

Teacher Activity

Exit ticket: '1. Name two online meeting tools. 2. What is the difference between a meeting and a workspace? 3. How do online workspaces support collaboration?'

Student Activity

Answer the questions. Discuss the benefits of collaborative tools.

Cross-Curricular Connections

Mathematics: Online tools use real-time data synchronization across networks — timing and protocols
Science: Video conferencing uses compression algorithms to reduce data transmission
Language: Professional communication in online meetings requires clear speaking and writing

Digital Citizen Reminder: Respect meeting etiquette — mute when not speaking, be punctual, and dress appropriately on camera.

Dormitory Task (Paper-and-Pencil)

Design a Google Form on paper for a school library survey. Write 5 questions with different input types text, multiple choice, yes/no, rating 1-5, dropdown. Label each input type.

Resources Needed:

- Computer with internet
- Google account
- survey examples
- form planning worksheet

Differentiation

Approaching: Use a pre-made form template and add questions
On Level: Create a survey form with 5 different input types independently
Advanced: Analyze survey results using spreadsheets and create a presentation of findings
SEND: Guided form creation with sentence starters, simplified question types

Formative Assessment

Method: Exit ticket and self-assessment
CT Skill Assessed: Decomposition — breaking a survey into question types, input methods, and analysis plan

Teacher Reflection (Post-Lesson)

What worked well:

What to improve:

Connection to next lesson:

Lesson 1.02	Duration: 40 minutes

Function Calls

Learning Objectives

- Distinguish between function definition and function invocation
- Pass arguments to functions and capture return values
- Trace the execution flow of a program with multiple function calls

Timing Breakdown

0-5 min	**Engage**	Show a video call in progress or recording. Ask: 'What technology makes this p...
5-15 min	**Explore**	Show different online meeting tools: Zoom, Google Meet, Jitsi. Demonstrate joini...
15-25 min	**Explain**	Define online meeting components: audio/video, internet connection, meeting link...
25-35 min	**Elaborate**	Task: 'Create a Google Form to survey your class about their favorite subject. I...
35-40 min	**Evaluate**	Exit ticket: '1. Name two online meeting tools. 2. What is the difference betwee...

Engage (5-7 minutes)

Teacher Activity

Show a video call in progress or recording. Ask: 'What technology makes this possible? What are the advantages over a phone call?' Discuss online meetings as a communication tool.

Student Activity

Identify features: video, screen sharing, chat, multiple participants. Discuss when video calls are better than phone calls.

Explore (10-12 minutes)

Teacher Activity

Show different online meeting tools: Zoom, Google Meet, Jitsi. Demonstrate joining a meeting, using mute/camera, screen share, chat. Compare features.

Student Activity

Observe the demonstration. Take notes on key features. Try Jitsi Meet no account needed if computers available.

Explain (10-12 minutes)

Teacher Activity

Define online meeting components: audio/video, internet connection, meeting link, host controls. Explain online workspaces: Google Workspace, Microsoft 365. Show how forms, sheets, and docs work collaboratively.

Student Activity

Take notes. Understand the difference between meetings and workspaces. See how collaboration works in real-time.

Elaborate (8-10 minutes)

Teacher Activity

Task: 'Create a Google Form to survey your class about their favorite subject. Include 5 questions with different input types multiple choice, text, rating. Share the form with a partner.'

Student Activity

Work independently. Create the form with varied question types. Test it by filling it out themselves. Share with partner.

Evaluate (5-8 minutes)

Teacher Activity

Exit ticket: '1. Name two online meeting tools. 2. What is the difference between a meeting and a workspace? 3. How do online workspaces support collaboration?'

Student Activity

Answer the questions. Discuss the benefits of collaborative tools.

Cross-Curricular Connections

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Science: Video conferencing uses compression algorithms to reduce data transmission
Language: Professional communication in online meetings requires clear speaking and writing

Digital Citizen Reminder: Respect meeting etiquette — mute when not speaking, be punctual, and dress appropriately on camera.

Dormitory Task (Paper-and-Pencil)

Design a Google Form on paper for a school library survey. Write 5 questions with different input types text, multiple choice, yes/no, rating 1-5, dropdown. Label each input type.

Resources Needed:

- Computer with internet
- Google account
- survey examples
- form planning worksheet

Differentiation

Approaching: Use a pre-made form template and add questions
On Level: Create a survey form with 5 different input types independently
Advanced: Analyze survey results using spreadsheets and create a presentation of findings
SEND: Guided form creation with sentence starters, simplified question types

Formative Assessment

Method: Exit ticket and self-assessment
CT Skill Assessed: Decomposition — breaking a survey into question types, input methods, and analysis plan

Teacher Reflection (Post-Lesson)

What worked well:

What to improve:

Connection to next lesson:

Lesson 1.03	Duration: 40 minutes

Arrays Concept

Learning Objectives

- Explain what an array is and why it is useful for storing multiple values
- Access and modify elements in an array using index positions
- Describe zero-based indexing and how it applies to arrays

Timing Breakdown

0-5 min	**Engage**	Show a video call in progress or recording. Ask: 'What technology makes this p...
5-15 min	**Explore**	Show different online meeting tools: Zoom, Google Meet, Jitsi. Demonstrate joini...
15-25 min	**Explain**	Define online meeting components: audio/video, internet connection, meeting link...
25-35 min	**Elaborate**	Task: 'Create a Google Form to survey your class about their favorite subject. I...
35-40 min	**Evaluate**	Exit ticket: '1. Name two online meeting tools. 2. What is the difference betwee...

Engage (5-7 minutes)

Teacher Activity

Show a video call in progress or recording. Ask: 'What technology makes this possible? What are the advantages over a phone call?' Discuss online meetings as a communication tool.

Student Activity

Identify features: video, screen sharing, chat, multiple participants. Discuss when video calls are better than phone calls.

Explore (10-12 minutes)

Teacher Activity

Show different online meeting tools: Zoom, Google Meet, Jitsi. Demonstrate joining a meeting, using mute/camera, screen share, chat. Compare features.

Student Activity

Observe the demonstration. Take notes on key features. Try Jitsi Meet no account needed if computers available.

Explain (10-12 minutes)

Teacher Activity

Define online meeting components: audio/video, internet connection, meeting link, host controls. Explain online workspaces: Google Workspace, Microsoft 365. Show how forms, sheets, and docs work collaboratively.

Student Activity

Take notes. Understand the difference between meetings and workspaces. See how collaboration works in real-time.

Elaborate (8-10 minutes)

Teacher Activity

Task: 'Create a Google Form to survey your class about their favorite subject. Include 5 questions with different input types multiple choice, text, rating. Share the form with a partner.'

Student Activity

Work independently. Create the form with varied question types. Test it by filling it out themselves. Share with partner.

Evaluate (5-8 minutes)

Teacher Activity

Exit ticket: '1. Name two online meeting tools. 2. What is the difference between a meeting and a workspace? 3. How do online workspaces support collaboration?'

Student Activity

Answer the questions. Discuss the benefits of collaborative tools.

Cross-Curricular Connections

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Digital Citizen Reminder: Respect meeting etiquette — mute when not speaking, be punctual, and dress appropriately on camera.

Dormitory Task (Paper-and-Pencil)

Design a Google Form on paper for a school library survey. Write 5 questions with different input types text, multiple choice, yes/no, rating 1-5, dropdown. Label each input type.

Resources Needed:
- Computer with internet
- Google account
- survey examples
- form planning worksheet

Differentiation

Approaching: Use a pre-made form template and add questions
On Level: Create a survey form with 5 different input types independently
Advanced: Analyze survey results using spreadsheets and create a presentation of findings
SEND: Guided form creation with sentence starters, simplified question types

Formative Assessment

Method: Exit ticket and self-assessment
CT Skill Assessed: Decomposition — breaking a survey into question types, input methods, and analysis plan

Teacher Reflection (Post-Lesson)

What worked well:

What to improve:

Connection to next lesson:

Lesson 1.04	Duration: 40 minutes

Array Operations

Learning Objectives

- Perform common array operations: insert, delete, search, and sort
- Analyze the efficiency of different array operations
- Implement algorithms that manipulate arrays to solve problems

Timing Breakdown

0-5 min	**Engage**	Show a video call in progress or recording. Ask: 'What technology makes this p...
5-15 min	**Explore**	Show different online meeting tools: Zoom, Google Meet, Jitsi. Demonstrate joini...
15-25 min	**Explain**	Define online meeting components: audio/video, internet connection, meeting link...
25-35 min	**Elaborate**	Task: 'Create a Google Form to survey your class about their favorite subject. I...
35-40 min	**Evaluate**	Exit ticket: '1. Name two online meeting tools. 2. What is the difference betwee...

Engage (5-7 minutes)

Teacher Activity

Show a video call in progress or recording. Ask: 'What technology makes this possible? What are the advantages over a phone call?' Discuss online meetings as a communication tool.

Student Activity

Identify features: video, screen sharing, chat, multiple participants. Discuss when video calls are better than phone calls.

Explore (10-12 minutes)

Teacher Activity

Show different online meeting tools: Zoom, Google Meet, Jitsi. Demonstrate joining a meeting, using mute/camera, screen share, chat. Compare features.

Student Activity

Observe the demonstration. Take notes on key features. Try Jitsi Meet no account needed if computers available.

Explain (10-12 minutes)

Teacher Activity

Define online meeting components: audio/video, internet connection, meeting link, host controls. Explain online workspaces: Google Workspace, Microsoft 365. Show how forms, sheets, and docs work collaboratively.

Student Activity

Take notes. Understand the difference between meetings and workspaces. See how collaboration works in real-time.

Elaborate (8-10 minutes)

Teacher Activity

Task: 'Create a Google Form to survey your class about their favorite subject. Include 5 questions with different input types multiple choice, text, rating. Share the form with a partner.'

Student Activity

Work independently. Create the form with varied question types. Test it by filling it out themselves. Share with partner.

Evaluate (5-8 minutes)

Teacher Activity

Exit ticket: '1. Name two online meeting tools. 2. What is the difference between a meeting and a workspace? 3. How do online workspaces support collaboration?'

Student Activity

Answer the questions. Discuss the benefits of collaborative tools.

Cross-Curricular Connections

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Science: Video conferencing uses compression algorithms to reduce data transmission
Language: Professional communication in online meetings requires clear speaking and writing

Digital Citizen Reminder: Respect meeting etiquette — mute when not speaking, be punctual, and dress appropriately on camera.

Dormitory Task (Paper-and-Pencil)

Design a Google Form on paper for a school library survey. Write 5 questions with different input types text, multiple choice, yes/no, rating 1-5, dropdown. Label each input type.

Resources Needed:
- Computer with internet
- Google account
- survey examples
- form planning worksheet

Differentiation

Approaching: Use a pre-made form template and add questions
On Level: Create a survey form with 5 different input types independently
Advanced: Analyze survey results using spreadsheets and create a presentation of findings
SEND: Guided form creation with sentence starters, simplified question types

Formative Assessment

Method: Exit ticket and self-assessment
CT Skill Assessed: Decomposition — breaking a survey into question types, input methods, and analysis plan

Teacher Reflection (Post-Lesson)

What worked well:

What to improve:

Connection to next lesson:

Lesson 1.05	Duration: 40 minutes

String Operations

Learning Objectives

- Manipulate strings using concatenation, slicing, and common methods
- Apply string functions to solve text processing problems
- Understand how strings are stored as character arrays internally

Timing Breakdown

0-5 min	**Engage**	Show a video call in progress or recording. Ask: 'What technology makes this p...
5-15 min	**Explore**	Show different online meeting tools: Zoom, Google Meet, Jitsi. Demonstrate joini...
15-25 min	**Explain**	Define online meeting components: audio/video, internet connection, meeting link...
25-35 min	**Elaborate**	Task: 'Create a Google Form to survey your class about their favorite subject. I...
35-40 min	**Evaluate**	Exit ticket: '1. Name two online meeting tools. 2. What is the difference betwee...

Engage (5-7 minutes)

Teacher Activity

Show a video call in progress or recording. Ask: 'What technology makes this possible? What are the advantages over a phone call?' Discuss online meetings as a communication tool.

Student Activity

Identify features: video, screen sharing, chat, multiple participants. Discuss when video calls are better than phone calls.

Explore (10-12 minutes)

Teacher Activity

Show different online meeting tools: Zoom, Google Meet, Jitsi. Demonstrate joining a meeting, using mute/camera, screen share, chat. Compare features.

Student Activity

Observe the demonstration. Take notes on key features. Try Jitsi Meet no account needed if computers available.

Explain (10-12 minutes)

Teacher Activity

Define online meeting components: audio/video, internet connection, meeting link, host controls. Explain online workspaces: Google Workspace, Microsoft 365. Show how forms, sheets, and docs work collaboratively.

Student Activity

Take notes. Understand the difference between meetings and workspaces. See how collaboration works in real-time.

Elaborate (8-10 minutes)

Teacher Activity

Task: 'Create a Google Form to survey your class about their favorite subject. Include 5 questions with different input types multiple choice, text, rating. Share the form with a partner.'

Student Activity

Work independently. Create the form with varied question types. Test it by filling it out themselves. Share with partner.

Evaluate (5-8 minutes)

Teacher Activity

Exit ticket: '1. Name two online meeting tools. 2. What is the difference between a meeting and a workspace? 3. How do online workspaces support collaboration?'

Student Activity

Answer the questions. Discuss the benefits of collaborative tools.

Cross-Curricular Connections

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Science: Video conferencing uses compression algorithms to reduce data transmission
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Digital Citizen Reminder: Respect meeting etiquette — mute when not speaking, be punctual, and dress appropriately on camera.

Dormitory Task (Paper-and-Pencil)

Design a Google Form on paper for a school library survey. Write 5 questions with different input types text, multiple choice, yes/no, rating 1-5, dropdown. Label each input type.

Resources Needed:
- Computer with internet
- Google account
- survey examples
- form planning worksheet

Differentiation

Approaching: Use a pre-made form template and add questions
On Level: Create a survey form with 5 different input types independently
Advanced: Analyze survey results using spreadsheets and create a presentation of findings
SEND: Guided form creation with sentence starters, simplified question types

Formative Assessment

Method: Exit ticket and self-assessment
CT Skill Assessed: Decomposition — breaking a survey into question types, input methods, and analysis plan

Teacher Reflection (Post-Lesson)

What worked well:

What to improve:

Connection to next lesson:

Lesson 1.06	**Duration: 40 minutes**

Nested Loops

Learning Objectives

- Construct nested loops and trace their execution step by step
- Apply nested loops to process multi-dimensional data patterns
- Analyze the time complexity of nested loop structures

Timing Breakdown

0-5 min	**Engage**	Show a video call in progress or recording. Ask: 'What technology makes this p...
5-15 min	**Explore**	Show different online meeting tools: Zoom, Google Meet, Jitsi. Demonstrate joini...
15-25 min	**Explain**	Define online meeting components: audio/video, internet connection, meeting link...
25-35 min	**Elaborate**	Task: 'Create a Google Form to survey your class about their favorite subject. I...
35-40 min	**Evaluate**	Exit ticket: '1. Name two online meeting tools. 2. What is the difference betwee...

Engage (5-7 minutes)

Teacher Activity

Show a video call in progress or recording. Ask: 'What technology makes this possible? What are the advantages over a phone call?' Discuss online meetings as a communication tool.

Student Activity

Identify features: video, screen sharing, chat, multiple participants. Discuss when video calls are better than phone calls.

Explore (10-12 minutes)

Teacher Activity

Show different online meeting tools: Zoom, Google Meet, Jitsi. Demonstrate joining a meeting, using mute/ camera, screen share, chat. Compare features.

Student Activity

Observe the demonstration. Take notes on key features. Try Jitsi Meet no account needed if computers available.

Explain (10-12 minutes)

Teacher Activity

Define online meeting components: audio/video, internet connection, meeting link, host controls. Explain online workspaces: Google Workspace, Microsoft 365. Show how forms, sheets, and docs work collaboratively.

Student Activity

Take notes. Understand the difference between meetings and workspaces. See how collaboration works in real-time.

Elaborate (8-10 minutes)

Teacher Activity

Task: ‘Create a Google Form to survey your class about their favorite subject. Include 5 questions with different input types multiple choice, text, rating. Share the form with a partner.’

Student Activity

Work independently. Create the form with varied question types. Test it by filling it out themselves. Share with partner.

Evaluate (5-8 minutes)

Teacher Activity

Exit ticket: ‘1. Name two online meeting tools. 2. What is the difference between a meeting and a workspace? 3. How do online workspaces support collaboration?’

Student Activity

Answer the questions. Discuss the benefits of collaborative tools.

Cross-Curricular Connections

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Science: Video conferencing uses compression algorithms to reduce data transmission
Language: Professional communication in online meetings requires clear speaking and writing

Digital Citizen Reminder: Respect meeting etiquette — mute when not speaking, be punctual, and dress appropriately on camera.

Dormitory Task (Paper-and-Pencil)

Design a Google Form on paper for a school library survey. Write 5 questions with different input types text, multiple choice, yes/no, rating 1-5, dropdown. Label each input type.

Resources Needed:
- Computer with internet
- Google account
- survey examples
- form planning worksheet

Differentiation

Approaching: Use a pre-made form template and add questions
On Level: Create a survey form with 5 different input types independently
Advanced: Analyze survey results using spreadsheets and create a presentation of findings
SEND: Guided form creation with sentence starters, simplified question types

Formative Assessment

Method: Exit ticket and self-assessment
CT Skill Assessed: Decomposition — breaking a survey into question types, input methods, and analysis plan

Teacher Reflection (Post-Lesson)

What worked well:

What to improve:

Connection to next lesson:

Lesson 1.07	Duration: 40 minutes

Recursion Basics

Learning Objectives

- Explain the concept of recursion and identify base cases
- Write recursive functions to solve mathematical and logical problems
- Compare recursive and iterative approaches for the same problem

Timing Breakdown

0-5 min	**Engage**	Show a video call in progress or recording. Ask: 'What technology makes this p...
5-15 min	**Explore**	Show different online meeting tools: Zoom, Google Meet, Jitsi. Demonstrate joini...
15-25 min	**Explain**	Define online meeting components: audio/video, internet connection, meeting link...
25-35 min	**Elaborate**	Task: 'Create a Google Form to survey your class about their favorite subject. I...
35-40 min	**Evaluate**	Exit ticket: '1. Name two online meeting tools. 2. What is the difference betwee...

Engage (5-7 minutes)

Teacher Activity

Show a video call in progress or recording. Ask: ‘What technology makes this possible? What are the advantages over a phone call?’ Discuss online meetings as a communication tool.

Student Activity

Identify features: video, screen sharing, chat, multiple participants. Discuss when video calls are better than phone calls.

Explore (10-12 minutes)

Teacher Activity

Show different online meeting tools: Zoom, Google Meet, Jitsi. Demonstrate joining a meeting, using mute/camera, screen share, chat. Compare features.

Student Activity

Observe the demonstration. Take notes on key features. Try Jitsi Meet no account needed if computers available.

Explain (10-12 minutes)

Teacher Activity

Define online meeting components: audio/video, internet connection, meeting link, host controls. Explain online workspaces: Google Workspace, Microsoft 365. Show how forms, sheets, and docs work collaboratively.

Student Activity

Take notes. Understand the difference between meetings and workspaces. See how collaboration works in real-time.

Elaborate (8-10 minutes)

Teacher Activity

Task: 'Create a Google Form to survey your class about their favorite subject. Include 5 questions with different input types multiple choice, text, rating. Share the form with a partner.'

Student Activity

Work independently. Create the form with varied question types. Test it by filling it out themselves. Share with partner.

Evaluate (5-8 minutes)

Teacher Activity

Exit ticket: '1. Name two online meeting tools. 2. What is the difference between a meeting and a workspace? 3. How do online workspaces support collaboration?'

Student Activity

Answer the questions. Discuss the benefits of collaborative tools.

Cross-Curricular Connections

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Dormitory Task (Paper-and-Pencil)

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Resources Needed:
- Computer with internet
- Google account
- survey examples
- form planning worksheet

Differentiation

Approaching: Use a pre-made form template and add questions
On Level: Create a survey form with 5 different input types independently
Advanced: Analyze survey results using spreadsheets and create a presentation of findings
SEND: Guided form creation with sentence starters, simplified question types

Formative Assessment

Method: Exit ticket and self-assessment
CT Skill Assessed: Decomposition — breaking a survey into question types, input methods, and analysis plan

Teacher Reflection (Post-Lesson)

What worked well:

What to improve:

Connection to next lesson:

Lesson 1.08	Duration: 40 minutes

Algorithm Design

Learning Objectives

- Break down complex problems into smaller, manageable steps
- Use pseudocode and flowcharts to design algorithms before coding
- Evaluate algorithm efficiency using time and space complexity concepts

Timing Breakdown

0-5 min	**Engage**	Show a video call in progress or recording. Ask: 'What technology makes this p...
5-15 min	**Explore**	Show different online meeting tools: Zoom, Google Meet, Jitsi. Demonstrate joini...
15-25 min	**Explain**	Define online meeting components: audio/video, internet connection, meeting link...
25-35 min	**Elaborate**	Task: 'Create a Google Form to survey your class about their favorite subject. I...
35-40 min	**Evaluate**	Exit ticket: '1. Name two online meeting tools. 2. What is the difference betwee...

Engage (5-7 minutes)

Teacher Activity

Show a video call in progress or recording. Ask: 'What technology makes this possible? What are the advantages over a phone call?' Discuss online meetings as a communication tool.

Student Activity

Identify features: video, screen sharing, chat, multiple participants. Discuss when video calls are better than phone calls.

Explore (10-12 minutes)

Teacher Activity

Show different online meeting tools: Zoom, Google Meet, Jitsi. Demonstrate joining a meeting, using mute/camera, screen share, chat. Compare features.

Student Activity

Observe the demonstration. Take notes on key features. Try Jitsi Meet no account needed if computers available.

Explain (10-12 minutes)

Teacher Activity

Define online meeting components: audio/video, internet connection, meeting link, host controls. Explain online workspaces: Google Workspace, Microsoft 365. Show how forms, sheets, and docs work collaboratively.

Student Activity

Take notes. Understand the difference between meetings and workspaces. See how collaboration works in real-time.

Elaborate (8-10 minutes)

Teacher Activity

Task: 'Create a Google Form to survey your class about their favorite subject. Include 5 questions with different input types multiple choice, text, rating. Share the form with a partner.'

Student Activity

Work independently. Create the form with varied question types. Test it by filling it out themselves. Share with partner.

Evaluate (5-8 minutes)

Teacher Activity

Exit ticket: '1. Name two online meeting tools. 2. What is the difference between a meeting and a workspace? 3. How do online workspaces support collaboration?'

Student Activity

Answer the questions. Discuss the benefits of collaborative tools.

Cross-Curricular Connections

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Digital Citizen Reminder: Respect meeting etiquette — mute when not speaking, be punctual, and dress appropriately on camera.

Dormitory Task (Paper-and-Pencil)

Design a Google Form on paper for a school library survey. Write 5 questions with different input types text, multiple choice, yes/no, rating 1-5, dropdown. Label each input type.

Resources Needed:

- Computer with internet
- Google account
- survey examples
- form planning worksheet

Differentiation

Approaching: Use a pre-made form template and add questions
On Level: Create a survey form with 5 different input types independently
Advanced: Analyze survey results using spreadsheets and create a presentation of findings
SEND: Guided form creation with sentence starters, simplified question types

Formative Assessment

Method: Exit ticket and self-assessment
CT Skill Assessed: Decomposition — breaking a survey into question types, input methods, and analysis plan

Teacher Reflection (Post-Lesson)

What worked well:

What to improve:

Connection to next lesson:

Lesson 1.09	Duration: 40 minutes

Problem Solving

Learning Objectives

- Apply systematic problem-solving strategies to programming challenges
- Decompose problems into subproblems and identify patterns
- Translate real-world scenarios into computational solutions

Timing Breakdown

0-5 min	Engage	Show a video call in progress or recording. Ask: 'What technology makes this p...
5-15 min	Explore	Show different online meeting tools: Zoom, Google Meet, Jitsi. Demonstrate joini...
15-25 min	Explain	Define online meeting components: audio/video, internet connection, meeting link...
25-35 min	Elaborate	Task: 'Create a Google Form to survey your class about their favorite subject. I...
35-40 min	Evaluate	Exit ticket: '1. Name two online meeting tools. 2. What is the difference betwee...

Engage (5-7 minutes)

Teacher Activity

Show a video call in progress or recording. Ask: 'What technology makes this possible? What are the advantages over a phone call?' Discuss online meetings as a communication tool.

Student Activity

Identify features: video, screen sharing, chat, multiple participants. Discuss when video calls are better than phone calls.

Explore (10-12 minutes)

Teacher Activity

Show different online meeting tools: Zoom, Google Meet, Jitsi. Demonstrate joining a meeting, using mute/camera, screen share, chat. Compare features.

Student Activity

Observe the demonstration. Take notes on key features. Try Jitsi Meet no account needed if computers available.

Explain (10-12 minutes)

Teacher Activity

Define online meeting components: audio/video, internet connection, meeting link, host controls. Explain online workspaces: Google Workspace, Microsoft 365. Show how forms, sheets, and docs work collaboratively.

Student Activity

Take notes. Understand the difference between meetings and workspaces. See how collaboration works in real-time.

Elaborate (8-10 minutes)

Teacher Activity

Task: 'Create a Google Form to survey your class about their favorite subject. Include 5 questions with different input types multiple choice, text, rating. Share the form with a partner.'

Student Activity

Work independently. Create the form with varied question types. Test it by filling it out themselves. Share with partner.

Evaluate (5-8 minutes)

Teacher Activity

Exit ticket: '1. Name two online meeting tools. 2. What is the difference between a meeting and a workspace? 3. How do online workspaces support collaboration?'

Student Activity

Answer the questions. Discuss the benefits of collaborative tools.

Cross-Curricular Connections

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Science: Video conferencing uses compression algorithms to reduce data transmission
Language: Professional communication in online meetings requires clear speaking and writing

Digital Citizen Reminder: Respect meeting etiquette — mute when not speaking, be punctual, and dress appropriately on camera.

Dormitory Task (Paper-and-Pencil)

Design a Google Form on paper for a school library survey. Write 5 questions with different input types text, multiple choice, yes/no, rating 1-5, dropdown. Label each input type.

Resources Needed:

- Computer with internet
- Google account
- survey examples
- form planning worksheet

Differentiation

Approaching: Use a pre-made form template and add questions
On Level: Create a survey form with 5 different input types independently
Advanced: Analyze survey results using spreadsheets and create a presentation of findings
SEND: Guided form creation with sentence starters, simplified question types

Formative Assessment

Method: Exit ticket and self-assessment
CT Skill Assessed: Decomposition — breaking a survey into question types, input methods, and analysis plan

Teacher Reflection (Post-Lesson)

What worked well:

What to improve:

Connection to next lesson:

Lesson 1.10	Duration: 40 minutes

Debugging

Learning Objectives

- Identify common types of programming errors: syntax, runtime, and logic errors
- Use systematic debugging techniques to locate and fix bugs
- Apply defensive programming practices to prevent errors proactively

Timing Breakdown

0-5 min	Engage	Show a video call in progress or recording. Ask: 'What technology makes this p...
5-15 min	**Explore**	Show different online meeting tools: Zoom, Google Meet, Jitsi. Demonstrate joini...
15-25 min	**Explain**	Define online meeting components: audio/video, internet connection, meeting link...
25-35 min	**Elaborate**	Task: 'Create a Google Form to survey your class about their favorite subject. I...
35-40 min	**Evaluate**	Exit ticket: '1. Name two online meeting tools. 2. What is the difference betwee...

Engage (5-7 minutes)

Teacher Activity

Show a video call in progress or recording. Ask: ‘What technology makes this possible? What are the advantages over a phone call?’ Discuss online meetings as a communication tool.

Student Activity

Identify features: video, screen sharing, chat, multiple participants. Discuss when video calls are better than phone calls.

Explore (10-12 minutes)

Teacher Activity

Show different online meeting tools: Zoom, Google Meet, Jitsi. Demonstrate joining a meeting, using mute/camera, screen share, chat. Compare features.

Student Activity

Observe the demonstration. Take notes on key features. Try Jitsi Meet no account needed if computers available.

Explain (10-12 minutes)

Teacher Activity

Define online meeting components: audio/video, internet connection, meeting link, host controls. Explain online workspaces: Google Workspace, Microsoft 365. Show how forms, sheets, and docs work collaboratively.

Student Activity

Take notes. Understand the difference between meetings and workspaces. See how collaboration works in real-time.

Elaborate (8-10 minutes)

Teacher Activity

Task: 'Create a Google Form to survey your class about their favorite subject. Include 5 questions with different input types multiple choice, text, rating. Share the form with a partner.'

Student Activity

Work independently. Create the form with varied question types. Test it by filling it out themselves. Share with partner.

Evaluate (5-8 minutes)

Teacher Activity

Exit ticket: '1. Name two online meeting tools. 2. What is the difference between a meeting and a workspace? 3. How do online workspaces support collaboration?'

Student Activity

Answer the questions. Discuss the benefits of collaborative tools.

Cross-Curricular Connections

Mathematics: Online tools use real-time data synchronization across networks — timing and protocols
Science: Video conferencing uses compression algorithms to reduce data transmission
Language: Professional communication in online meetings requires clear speaking and writing

Digital Citizen Reminder: Respect meeting etiquette — mute when not speaking, be punctual, and dress appropriately on camera.

Dormitory Task (Paper-and-Pencil)

Design a Google Form on paper for a school library survey. Write 5 questions with different input types text, multiple choice, yes/no, rating 1-5, dropdown. Label each input type.

Resources Needed:

- Computer with internet
- Google account
- survey examples
- form planning worksheet

Differentiation

Approaching: Use a pre-made form template and add questions
On Level: Create a survey form with 5 different input types independently
Advanced: Analyze survey results using spreadsheets and create a presentation of findings
SEND: Guided form creation with sentence starters, simplified question types

Formative Assessment

Method: Exit ticket and self-assessment
CT Skill Assessed: Decomposition — breaking a survey into question types, input methods, and analysis plan

Teacher Reflection (Post-Lesson)

What worked well:

What to improve:

Connection to next lesson:

Lesson 1.11	Duration: 40 minutes

Project - Calculator

Learning Objectives

- Design and implement a functional calculator using functions and control structures
- Apply modular programming principles by separating concerns into functions
- Handle user input, validate data, and implement error handling

Timing Breakdown

0-5 min	**Engage**	Show a video call in progress or recording. Ask: 'What technology makes this p...
5-15 min	**Explore**	Show different online meeting tools: Zoom, Google Meet, Jitsi. Demonstrate joini...
15-25 min	**Explain**	Define online meeting components: audio/video, internet connection, meeting link...
25-35 min	**Elaborate**	Task: 'Create a Google Form to survey your class about their favorite subject. I...
35-40 min	**Evaluate**	Exit ticket: '1. Name two online meeting tools. 2. What is the difference betwee...

Engage (5-7 minutes)

Teacher Activity

Show a video call in progress or recording. Ask: 'What technology makes this possible? What are the advantages over a phone call?' Discuss online meetings as a communication tool.

Student Activity

Identify features: video, screen sharing, chat, multiple participants. Discuss when video calls are better than phone calls.

Explore (10-12 minutes)

Teacher Activity

Show different online meeting tools: Zoom, Google Meet, Jitsi. Demonstrate joining a meeting, using mute/camera, screen share, chat. Compare features.

Student Activity

Observe the demonstration. Take notes on key features. Try Jitsi Meet no account needed if computers available.

Explain (10-12 minutes)

Teacher Activity

Define online meeting components: audio/video, internet connection, meeting link, host controls. Explain online workspaces: Google Workspace, Microsoft 365. Show how forms, sheets, and docs work collaboratively.

Student Activity

Take notes. Understand the difference between meetings and workspaces. See how collaboration works in real-time.

Elaborate (8-10 minutes)

Teacher Activity

Task: 'Create a Google Form to survey your class about their favorite subject. Include 5 questions with different input types multiple choice, text, rating. Share the form with a partner.'

Student Activity

Work independently. Create the form with varied question types. Test it by filling it out themselves. Share with partner.

Evaluate (5-8 minutes)

Teacher Activity

Exit ticket: '1. Name two online meeting tools. 2. What is the difference between a meeting and a workspace? 3. How do online workspaces support collaboration?'

Student Activity

Answer the questions. Discuss the benefits of collaborative tools.

Cross-Curricular Connections

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Resources Needed:
- Computer with internet
- Google account
- survey examples
- form planning worksheet

Differentiation

Approaching: Use a pre-made form template and add questions
On Level: Create a survey form with 5 different input types independently
Advanced: Analyze survey results using spreadsheets and create a presentation of findings
SEND: Guided form creation with sentence starters, simplified question types

Formative Assessment

Method: Exit ticket and self-assessment
CT Skill Assessed: Decomposition — breaking a survey into question types, input methods, and analysis plan

Teacher Reflection (Post-Lesson)

What worked well:

What to improve:

Connection to next lesson:

Lesson 1.12	Duration: 40 minutes

Chapter 1 Review

Learning Objectives

- Consolidate understanding of functions, arrays, loops, recursion, and algorithms
- Solve 综合 problems that combine multiple programming concepts
- Reflect on learning progress and identify areas for improvement

Timing Breakdown

0-5 min	Engage	Show a video call in progress or recording. Ask: 'What technology makes this p...
5-15 min	Explore	Show different online meeting tools: Zoom, Google Meet, Jitsi. Demonstrate joini...
15-25 min	Explain	Define online meeting components: audio/video, internet connection, meeting link...
25-35 min	Elaborate	Task: 'Create a Google Form to survey your class about their favorite subject. I...
35-40 min	Evaluate	Exit ticket: '1. Name two online meeting tools. 2. What is the difference betwee...

Engage (5-7 minutes)

Teacher Activity

Show a video call in progress or recording. Ask: 'What technology makes this possible? What are the advantages over a phone call?' Discuss online meetings as a communication tool.

Student Activity

Identify features: video, screen sharing, chat, multiple participants. Discuss when video calls are better than phone calls.

Explore (10-12 minutes)

Teacher Activity

Show different online meeting tools: Zoom, Google Meet, Jitsi. Demonstrate joining a meeting, using mute/camera, screen share, chat. Compare features.

Student Activity

Observe the demonstration. Take notes on key features. Try Jitsi Meet no account needed if computers available.

Explain (10-12 minutes)

Teacher Activity

Define online meeting components: audio/video, internet connection, meeting link, host controls. Explain online workspaces: Google Workspace, Microsoft 365. Show how forms, sheets, and docs work collaboratively.

Student Activity

Take notes. Understand the difference between meetings and workspaces. See how collaboration works in real-time.

Elaborate (8-10 minutes)

Teacher Activity

Task: 'Create a Google Form to survey your class about their favorite subject. Include 5 questions with different input types multiple choice, text, rating. Share the form with a partner.'

Student Activity

Work independently. Create the form with varied question types. Test it by filling it out themselves. Share with partner.

Evaluate (5-8 minutes)

Teacher Activity

Exit ticket: '1. Name two online meeting tools. 2. What is the difference between a meeting and a workspace? 3. How do online workspaces support collaboration?'

Student Activity

Answer the questions. Discuss the benefits of collaborative tools.

Cross-Curricular Connections

Mathematics: Online tools use real-time data synchronization across networks — timing and protocols
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Digital Citizen Reminder: Respect meeting etiquette — mute when not speaking, be punctual, and dress appropriately on camera.

Dormitory Task (Paper-and-Pencil)

Design a Google Form on paper for a school library survey. Write 5 questions with different input types text, multiple choice, yes/no, rating 1-5, dropdown. Label each input type.

Resources Needed:
- Computer with internet
- Google account
- survey examples
- form planning worksheet

Differentiation

Approaching: Use a pre-made form template and add questions
On Level: Create a survey form with 5 different input types independently
Advanced: Analyze survey results using spreadsheets and create a presentation of findings
SEND: Guided form creation with sentence starters, simplified question types

Formative Assessment

Method: Exit ticket and self-assessment
CT Skill Assessed: Decomposition — breaking a survey into question types, input methods, and analysis plan

Teacher Reflection (Post-Lesson)

What worked well:

What to improve:

Connection to next lesson:

Lesson 2.01	Duration: 40 minutes

Lists Introduction

Learning Objectives

- Define what a list is and explain how it differs from arrays
- Create lists, access elements, and modify list contents dynamically
- Identify situations where lists are preferred over arrays

Timing Breakdown

0-5 min	Engage	Show a 2D image and its 3D-effect version. Ask: 'What makes the second image loo...
5-15 min	Explore	Open GIMP. Demonstrate: add text layer, create layer group, apply layer mask, ad...
15-25 min	Explain	Define 3D concepts: three dimensions length, width, depth. 3D text technique: ...
25-35 min	Elaborate	Task: 'Create a 3D text effect for a school poster. Add shadow, highlight, and d...
35-40 min	Evaluate	Exit ticket: '1. What makes a 2D image look 3D? 2. What is a layer mask? 3. Name...

Engage (5-7 minutes)

Teacher Activity

Show a 2D image and its 3D-effect version. Ask: 'What makes the second image look three-dimensional?' Discuss shadows, highlights, depth.

Student Activity

Compare images. Identify 3D cues: shadows, highlights, perspective. Discuss how these create depth perception.

Explore (10-12 minutes)

Teacher Activity

Open GIMP. Demonstrate: add text layer, create layer group, apply layer mask, add shadow effect for 3D look. Students follow along.

Student Activity

Follow along. Create text with shadow effect. Experiment with layer masks. Apply 3D effects.

Explain (10-12 minutes)

Teacher Activity

Define 3D concepts: three dimensions length, width, depth. 3D text technique: duplicate layer, offset, darken. Layer masks: control visibility without deleting. Layer groups: organize complex projects.

Student Activity

Take notes. Understand how 2D images can simulate 3D. Practice layer mask concepts.

Elaborate (8-10 minutes)

Teacher Activity

Task: 'Create a 3D text effect for a school poster. Add shadow, highlight, and depth to make text pop.' Show examples of professional 3D text.

Student Activity

Work independently. Apply all techniques. Get peer feedback on the 3D effect quality.

Evaluate (5-8 minutes)

Teacher Activity

Exit ticket: '1. What makes a 2D image look 3D? 2. What is a layer mask? 3. Name two uses of layer groups.'

Student Activity

Answer the questions. Show their work to a partner.

Cross-Curricular Connections

Mathematics: 3D coordinates use x, y, z axes; perspective involves geometric projections
Science: Light and shadow follow physics principles — angles, intensity, diffusion
Language: Visual hierarchy in design guides the viewer's eye through content

Digital Citizen Reminder: Credit the original creator when using their techniques or assets in your work.

Dormitory Task (Paper-and-Pencil)

On paper, draw a 3D letter 'A'. Start with a flat 2D 'A', then add shadow lines, depth lines, and shading to make it look three-dimensional. Label each technique used.

Resources Needed:

- Computer with GIMP
- text effect samples
- layer mask tutorial
- grid paper

Differentiation

Approaching: Follow a step-by-step tutorial to create 3D text effect
On Level: Create 3D text effects independently using all techniques
Advanced: Design a complete 3D poster with multiple text elements and effects
SEND: Pre-made 3D templates to modify, guided steps with visual examples

Formative Assessment

Method: Exit ticket and self-assessment
CT Skill Assessed: Abstraction — understanding how 2D techniques simulate 3D perception

Teacher Reflection (Post-Lesson)

What worked well:

What to improve:

Connection to next lesson:

Lesson 2.02	Duration: 40 minutes

List Operations

Learning Objectives

- Perform advanced list operations including slicing, sorting, and searching
- Use list methods to transform and analyze collections of data
- Implement algorithms that operate on lists efficiently

Timing Breakdown

0-5 min	**Engage**	Show a 2D image and its 3D-effect version. Ask: 'What makes the second image loo...
5-15 min	**Explore**	Open GIMP. Demonstrate: add text layer, create layer group, apply layer mask, ad...
15-25 min	**Explain**	Define 3D concepts: three dimensions length, width, depth. 3D text technique: ...
25-35 min	**Elaborate**	Task: 'Create a 3D text effect for a school poster. Add shadow, highlight, and d...
35-40 min	**Evaluate**	Exit ticket: '1. What makes a 2D image look 3D? 2. What is a layer mask? 3. Name...

Engage (5-7 minutes)

Teacher Activity

Show a 2D image and its 3D-effect version. Ask: 'What makes the second image look three-dimensional?' Discuss shadows, highlights, depth.

Student Activity

Compare images. Identify 3D cues: shadows, highlights, perspective. Discuss how these create depth perception.

Explore (10-12 minutes)

Teacher Activity

Open GIMP. Demonstrate: add text layer, create layer group, apply layer mask, add shadow effect for 3D look. Students follow along.

Student Activity

Follow along. Create text with shadow effect. Experiment with layer masks. Apply 3D effects.

Explain (10-12 minutes)

Teacher Activity

Define 3D concepts: three dimensions length, width, depth. 3D text technique: duplicate layer, offset, darken. Layer masks: control visibility without deleting. Layer groups: organize complex projects.

Student Activity

Take notes. Understand how 2D images can simulate 3D. Practice layer mask concepts.

Elaborate (8-10 minutes)

Teacher Activity

Task: 'Create a 3D text effect for a school poster. Add shadow, highlight, and depth to make text pop.' Show examples of professional 3D text.

Student Activity

Work independently. Apply all techniques. Get peer feedback on the 3D effect quality.

Evaluate (5-8 minutes)

Teacher Activity

Exit ticket: '1. What makes a 2D image look 3D? 2. What is a layer mask? 3. Name two uses of layer groups.'

Student Activity

Answer the questions. Show their work to a partner.

Cross-Curricular Connections

Mathematics: 3D coordinates use x, y, z axes; perspective involves geometric projections
Science: Light and shadow follow physics principles — angles, intensity, diffusion
Language: Visual hierarchy in design guides the viewer's eye through content

Digital Citizen Reminder: Credit the original creator when using their techniques or assets in your work.

Dormitory Task (Paper-and-Pencil)

On paper, draw a 3D letter 'A'. Start with a flat 2D 'A', then add shadow lines, depth lines, and shading to make it look three-dimensional. Label each technique used.

Resources Needed:
- Computer with GIMP
- text effect samples
- layer mask tutorial
- grid paper

Differentiation

Approaching: Follow a step-by-step tutorial to create 3D text effect
On Level: Create 3D text effects independently using all techniques
Advanced: Design a complete 3D poster with multiple text elements and effects
SEND: Pre-made 3D templates to modify, guided steps with visual examples

Formative Assessment

Method: Exit ticket and self-assessment
CT Skill Assessed: Abstraction — understanding how 2D techniques simulate 3D perception

Teacher Reflection (Post-Lesson)

What worked well:

What to improve:

Connection to next lesson:

Lesson 2.03	Duration: 40 minutes

List Comprehension

Learning Objectives

- Write list comprehensions to create lists concisely and efficiently
- Apply conditions and nested logic within list comprehensions
- Compare list comprehensions with traditional loops for readability and performance

Timing Breakdown

0-5 min	Engage	Show a 2D image and its 3D-effect version. Ask: 'What makes the second image loo...
5-15 min	**Explore**	Open GIMP. Demonstrate: add text layer, create layer group, apply layer mask, ad...
15-25 min	**Explain**	Define 3D concepts: three dimensions length, width, depth. 3D text technique: ...
25-35 min	**Elaborate**	Task: 'Create a 3D text effect for a school poster. Add shadow, highlight, and d...
35-40 min	**Evaluate**	Exit ticket: '1. What makes a 2D image look 3D? 2. What is a layer mask? 3. Name...

Engage (5-7 minutes)

Teacher Activity

Show a 2D image and its 3D-effect version. Ask: 'What makes the second image look three-dimensional?' Discuss shadows, highlights, depth.

Student Activity

Compare images. Identify 3D cues: shadows, highlights, perspective. Discuss how these create depth perception.

Explore (10-12 minutes)

Teacher Activity

Open GIMP. Demonstrate: add text layer, create layer group, apply layer mask, add shadow effect for 3D look. Students follow along.

Student Activity

Follow along. Create text with shadow effect. Experiment with layer masks. Apply 3D effects.

Explain (10-12 minutes)

Teacher Activity

Define 3D concepts: three dimensions length, width, depth. 3D text technique: duplicate layer, offset, darken. Layer masks: control visibility without deleting. Layer groups: organize complex projects.

Student Activity

Take notes. Understand how 2D images can simulate 3D. Practice layer mask concepts.

Elaborate (8-10 minutes)

Teacher Activity

Task: 'Create a 3D text effect for a school poster. Add shadow, highlight, and depth to make text pop.' Show examples of professional 3D text.

Student Activity

Work independently. Apply all techniques. Get peer feedback on the 3D effect quality.

Evaluate (5-8 minutes)

Teacher Activity

Exit ticket: '1. What makes a 2D image look 3D? 2. What is a layer mask? 3. Name two uses of layer groups.'

Student Activity

Answer the questions. Show their work to a partner.

Cross-Curricular Connections

Mathematics: 3D coordinates use x, y, z axes; perspective involves geometric projections
Science: Light and shadow follow physics principles — angles, intensity, diffusion
Language: Visual hierarchy in design guides the viewer's eye through content

Digital Citizen Reminder: Credit the original creator when using their techniques or assets in your work.

Dormitory Task (Paper-and-Pencil)

On paper, draw a 3D letter 'A'. Start with a flat 2D 'A', then add shadow lines, depth lines, and shading to make it look three-dimensional. Label each technique used.

Resources Needed:
- Computer with GIMP
- text effect samples
- layer mask tutorial
- grid paper

Differentiation

Approaching: Follow a step-by-step tutorial to create 3D text effect
On Level: Create 3D text effects independently using all techniques
Advanced: Design a complete 3D poster with multiple text elements and effects
SEND: Pre-made 3D templates to modify, guided steps with visual examples

Formative Assessment

Method: Exit ticket and self-assessment
CT Skill Assessed: Abstraction — understanding how 2D techniques simulate 3D perception

Teacher Reflection (Post-Lesson)

What worked well:

What to improve:

Connection to next lesson:

Lesson 2.04	Duration: 40 minutes

Dictionaries

Learning Objectives

- Define dictionaries as key-value pairs and explain their advantages over lists
- Create dictionaries, access values by key, and add or remove entries
- Choose appropriate data structures for different problem scenarios

Timing Breakdown

0-5 min	Engage	Show a 2D image and its 3D-effect version. Ask: 'What makes the second image loo...
5-15 min	**Explore**	Open GIMP. Demonstrate: add text layer, create layer group, apply layer mask, ad...
15-25 min	**Explain**	Define 3D concepts: three dimensions length, width, depth. 3D text technique: ...
25-35 min	**Elaborate**	Task: 'Create a 3D text effect for a school poster. Add shadow, highlight, and d...
35-40 min	**Evaluate**	Exit ticket: '1. What makes a 2D image look 3D? 2. What is a layer mask? 3. Name...

Engage (5-7 minutes)

Teacher Activity

Show a 2D image and its 3D-effect version. Ask: 'What makes the second image look three-dimensional?' Discuss shadows, highlights, depth.

Student Activity

Compare images. Identify 3D cues: shadows, highlights, perspective. Discuss how these create depth perception.

Explore (10-12 minutes)

Teacher Activity

Open GIMP. Demonstrate: add text layer, create layer group, apply layer mask, add shadow effect for 3D look. Students follow along.

Student Activity

Follow along. Create text with shadow effect. Experiment with layer masks. Apply 3D effects.

Explain (10-12 minutes)

Teacher Activity

Define 3D concepts: three dimensions length, width, depth. 3D text technique: duplicate layer, offset, darken. Layer masks: control visibility without deleting. Layer groups: organize complex projects.

Student Activity

Take notes. Understand how 2D images can simulate 3D. Practice layer mask concepts.

Elaborate (8-10 minutes)

Teacher Activity

Task: 'Create a 3D text effect for a school poster. Add shadow, highlight, and depth to make text pop.' Show examples of professional 3D text.

Student Activity

Work independently. Apply all techniques. Get peer feedback on the 3D effect quality.

Evaluate (5-8 minutes)

Teacher Activity

Exit ticket: '1. What makes a 2D image look 3D? 2. What is a layer mask? 3. Name two uses of layer groups.'

Student Activity

Answer the questions. Show their work to a partner.

Cross-Curricular Connections

Mathematics: 3D coordinates use x, y, z axes; perspective involves geometric projections
Science: Light and shadow follow physics principles — angles, intensity, diffusion
Language: Visual hierarchy in design guides the viewer's eye through content

Digital Citizen Reminder: Credit the original creator when using their techniques or assets in your work.

Dormitory Task (Paper-and-Pencil)

On paper, draw a 3D letter 'A'. Start with a flat 2D 'A', then add shadow lines, depth lines, and shading to make it look three-dimensional. Label each technique used.

Resources Needed:
- Computer with GIMP
- text effect samples
- layer mask tutorial
- grid paper

Differentiation

Approaching: Follow a step-by-step tutorial to create 3D text effect
On Level: Create 3D text effects independently using all techniques
Advanced: Design a complete 3D poster with multiple text elements and effects
SEND: Pre-made 3D templates to modify, guided steps with visual examples

Formative Assessment

Method: Exit ticket and self-assessment
CT Skill Assessed: Abstraction — understanding how 2D techniques simulate 3D perception

Teacher Reflection (Post-Lesson)

What worked well:

What to improve:

Connection to next lesson:

Lesson 2.05	Duration: 40 minutes

Dictionary Operations

Learning Objectives

- Perform advanced dictionary operations: iteration, merging, and comprehension
- Use dictionary methods to extract and transform data efficiently
- Solve real-world data processing problems using dictionaries

Timing Breakdown

0-5 min	**Engage**	Show a 2D image and its 3D-effect version. Ask: 'What makes the second image loo...
5-15 min	**Explore**	Open GIMP. Demonstrate: add text layer, create layer group, apply layer mask, ad...
15-25 min	**Explain**	Define 3D concepts: three dimensions length, width, depth. 3D text technique: ...
25-35 min	**Elaborate**	Task: 'Create a 3D text effect for a school poster. Add shadow, highlight, and d...
35-40 min	**Evaluate**	Exit ticket: '1. What makes a 2D image look 3D? 2. What is a layer mask? 3. Name...

Engage (5-7 minutes)

Teacher Activity

Show a 2D image and its 3D-effect version. Ask: 'What makes the second image look three-dimensional?' Discuss shadows, highlights, depth.

Student Activity

Compare images. Identify 3D cues: shadows, highlights, perspective. Discuss how these create depth perception.

Explore (10-12 minutes)

Teacher Activity

Open GIMP. Demonstrate: add text layer, create layer group, apply layer mask, add shadow effect for 3D look. Students follow along.

Student Activity

Follow along. Create text with shadow effect. Experiment with layer masks. Apply 3D effects.

Explain (10-12 minutes)

Teacher Activity

Define 3D concepts: three dimensions length, width, depth. 3D text technique: duplicate layer, offset, darken. Layer masks: control visibility without deleting. Layer groups: organize complex projects.

Student Activity

Take notes. Understand how 2D images can simulate 3D. Practice layer mask concepts.

Elaborate (8-10 minutes)

Teacher Activity

Task: 'Create a 3D text effect for a school poster. Add shadow, highlight, and depth to make text pop.' Show examples of professional 3D text.

Student Activity

Work independently. Apply all techniques. Get peer feedback on the 3D effect quality.

Evaluate (5-8 minutes)

Teacher Activity

Exit ticket: '1. What makes a 2D image look 3D? 2. What is a layer mask? 3. Name two uses of layer groups.'

Student Activity

Answer the questions. Show their work to a partner.

Cross-Curricular Connections

Mathematics: 3D coordinates use x, y, z axes; perspective involves geometric projections
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Digital Citizen Reminder: Credit the original creator when using their techniques or assets in your work.

Dormitory Task (Paper-and-Pencil)

On paper, draw a 3D letter 'A'. Start with a flat 2D 'A', then add shadow lines, depth lines, and shading to make it look three-dimensional. Label each technique used.

Resources Needed:

- Computer with GIMP
- text effect samples
- layer mask tutorial
- grid paper

Differentiation

Approaching: Follow a step-by-step tutorial to create 3D text effect
On Level: Create 3D text effects independently using all techniques
Advanced: Design a complete 3D poster with multiple text elements and effects
SEND: Pre-made 3D templates to modify, guided steps with visual examples

Formative Assessment

Method: Exit ticket and self-assessment
CT Skill Assessed: Abstraction — understanding how 2D techniques simulate 3D perception

Teacher Reflection (Post-Lesson)

What worked well:

What to improve:

Connection to next lesson:

Lesson 2.06	Duration: 40 minutes

Tuples

Learning Objectives

- Define tuples and explain how they differ from lists in mutability and use cases
- Use tuples for function returns, dictionary keys, and data protection
- Apply tuple unpacking and understand named tuples for cleaner code

Timing Breakdown

0-5 min	Engage	Show a 2D image and its 3D-effect version. Ask: 'What makes the second image loo...
5-15 min	Explore	Open GIMP. Demonstrate: add text layer, create layer group, apply layer mask, ad...
15-25 min	Explain	Define 3D concepts: three dimensions length, width, depth. 3D text technique: ...
25-35 min	Elaborate	Task: 'Create a 3D text effect for a school poster. Add shadow, highlight, and d...
35-40 min	Evaluate	Exit ticket: '1. What makes a 2D image look 3D? 2. What is a layer mask? 3. Name...

Engage (5-7 minutes)

Teacher Activity

Show a 2D image and its 3D-effect version. Ask: 'What makes the second image look three-dimensional?' Discuss shadows, highlights, depth.

Student Activity

Compare images. Identify 3D cues: shadows, highlights, perspective. Discuss how these create depth perception.

Explore (10-12 minutes)

Teacher Activity

Open GIMP. Demonstrate: add text layer, create layer group, apply layer mask, add shadow effect for 3D look. Students follow along.

Student Activity

Follow along. Create text with shadow effect. Experiment with layer masks. Apply 3D effects.

Explain (10-12 minutes)

Teacher Activity

Define 3D concepts: three dimensions length, width, depth. 3D text technique: duplicate layer, offset, darken. Layer masks: control visibility without deleting. Layer groups: organize complex projects.

Student Activity

Take notes. Understand how 2D images can simulate 3D. Practice layer mask concepts.

Elaborate (8-10 minutes)

Teacher Activity

Task: 'Create a 3D text effect for a school poster. Add shadow, highlight, and depth to make text pop.' Show examples of professional 3D text.

Student Activity

Work independently. Apply all techniques. Get peer feedback on the 3D effect quality.

Evaluate (5-8 minutes)

Teacher Activity

Exit ticket: '1. What makes a 2D image look 3D? 2. What is a layer mask? 3. Name two uses of layer groups.'

Student Activity

Answer the questions. Show their work to a partner.

Cross-Curricular Connections

Mathematics: 3D coordinates use x, y, z axes; perspective involves geometric projections
Science: Light and shadow follow physics principles — angles, intensity, diffusion
Language: Visual hierarchy in design guides the viewer's eye through content

Digital Citizen Reminder: Credit the original creator when using their techniques or assets in your work.

Dormitory Task (Paper-and-Pencil)

On paper, draw a 3D letter 'A'. Start with a flat 2D 'A', then add shadow lines, depth lines, and shading to make it look three-dimensional. Label each technique used.

Resources Needed:

- Computer with GIMP
- text effect samples
- layer mask tutorial
- grid paper

Differentiation

Approaching: Follow a step-by-step tutorial to create 3D text effect
On Level: Create 3D text effects independently using all techniques
Advanced: Design a complete 3D poster with multiple text elements and effects
SEND: Pre-made 3D templates to modify, guided steps with visual examples

Formative Assessment

Method: Exit ticket and self-assessment
CT Skill Assessed: Abstraction — understanding how 2D techniques simulate 3D perception

Teacher Reflection (Post-Lesson)

What worked well:

What to improve:

Connection to next lesson:

Lesson 2.07	**Duration: 40 minutes**

Sets

Learning Objectives

- Define sets and explain their properties of uniqueness and unordered storage
- Perform set operations: union, intersection, difference, and symmetric difference
- Apply sets to solve problems involving membership testing and deduplication

Timing Breakdown

0-5 min	**Engage**	Show a 2D image and its 3D-effect version. Ask: 'What makes the second image loo...
5-15 min	**Explore**	Open GIMP. Demonstrate: add text layer, create layer group, apply layer mask, ad...
15-25 min	**Explain**	Define 3D concepts: three dimensions length, width, depth. 3D text technique: ...
25-35 min	**Elaborate**	Task: 'Create a 3D text effect for a school poster. Add shadow, highlight, and d...
35-40 min	**Evaluate**	Exit ticket: '1. What makes a 2D image look 3D? 2. What is a layer mask? 3. Name...

Engage (5-7 minutes)

Teacher Activity

Show a 2D image and its 3D-effect version. Ask: ‘What makes the second image look three-dimensional?’ Discuss shadows, highlights, depth.

Student Activity

Compare images. Identify 3D cues: shadows, highlights, perspective. Discuss how these create depth perception.

Explore (10-12 minutes)

Teacher Activity

Open GIMP. Demonstrate: add text layer, create layer group, apply layer mask, add shadow effect for 3D look. Students follow along.

Student Activity

Follow along. Create text with shadow effect. Experiment with layer masks. Apply 3D effects.

Explain (10-12 minutes)

Teacher Activity

Define 3D concepts: three dimensions length, width, depth. 3D text technique: duplicate layer, offset, darken. Layer masks: control visibility without deleting. Layer groups: organize complex projects.

Student Activity

Take notes. Understand how 2D images can simulate 3D. Practice layer mask concepts.

Elaborate (8-10 minutes)

Teacher Activity

Task: 'Create a 3D text effect for a school poster. Add shadow, highlight, and depth to make text pop.' Show examples of professional 3D text.

Student Activity

Work independently. Apply all techniques. Get peer feedback on the 3D effect quality.

Evaluate (5-8 minutes)

Teacher Activity

Exit ticket: '1. What makes a 2D image look 3D? 2. What is a layer mask? 3. Name two uses of layer groups.'

Student Activity

Answer the questions. Show their work to a partner.

Cross-Curricular Connections

Mathematics: 3D coordinates use x, y, z axes; perspective involves geometric projections
Science: Light and shadow follow physics principles — angles, intensity, diffusion
Language: Visual hierarchy in design guides the viewer's eye through content

Digital Citizen Reminder: Credit the original creator when using their techniques or assets in your work.

Dormitory Task (Paper-and-Pencil)

On paper, draw a 3D letter 'A'. Start with a flat 2D 'A', then add shadow lines, depth lines, and shading to make it look three-dimensional. Label each technique used.

Resources Needed:

- Computer with GIMP
- text effect samples
- layer mask tutorial
- grid paper

Differentiation

Approaching: Follow a step-by-step tutorial to create 3D text effect
On Level: Create 3D text effects independently using all techniques
Advanced: Design a complete 3D poster with multiple text elements and effects
SEND: Pre-made 3D templates to modify, guided steps with visual examples

Formative Assessment

Method: Exit ticket and self-assessment
CT Skill Assessed: Abstraction — understanding how 2D techniques simulate 3D perception

Teacher Reflection (Post-Lesson)

What worked well:

What to improve:

Connection to next lesson:

Lesson 2.08	Duration: 40 minutes

Data Organization

Learning Objectives

- Choose appropriate data structures for different types of problems
- Organize data efficiently using combinations of lists, dictionaries, and tuples
- Design data models that represent real-world entities and relationships

Timing Breakdown

0-5 min	Engage	Show a 2D image and its 3D-effect version. Ask: 'What makes the second image loo...
5-15 min	**Explore**	Open GIMP. Demonstrate: add text layer, create layer group, apply layer mask, ad...
15-25 min	**Explain**	Define 3D concepts: three dimensions length, width, depth. 3D text technique: ...
25-35 min	**Elaborate**	Task: 'Create a 3D text effect for a school poster. Add shadow, highlight, and d...
35-40 min	**Evaluate**	Exit ticket: '1. What makes a 2D image look 3D? 2. What is a layer mask? 3. Name...

Engage (5-7 minutes)

Teacher Activity

Show a 2D image and its 3D-effect version. Ask: ‘What makes the second image look three-dimensional?’ Discuss shadows, highlights, depth.

Student Activity

Compare images. Identify 3D cues: shadows, highlights, perspective. Discuss how these create depth perception.

Explore (10-12 minutes)

Teacher Activity

Open GIMP. Demonstrate: add text layer, create layer group, apply layer mask, add shadow effect for 3D look. Students follow along.

Student Activity

Follow along. Create text with shadow effect. Experiment with layer masks. Apply 3D effects.

Explain (10-12 minutes)

Teacher Activity

Define 3D concepts: three dimensions length, width, depth. 3D text technique: duplicate layer, offset, darken. Layer masks: control visibility without deleting. Layer groups: organize complex projects.

Student Activity

Take notes. Understand how 2D images can simulate 3D. Practice layer mask concepts.

Elaborate (8-10 minutes)

Teacher Activity

Task: 'Create a 3D text effect for a school poster. Add shadow, highlight, and depth to make text pop.' Show examples of professional 3D text.

Student Activity

Work independently. Apply all techniques. Get peer feedback on the 3D effect quality.

Evaluate (5-8 minutes)

Teacher Activity

Exit ticket: '1. What makes a 2D image look 3D? 2. What is a layer mask? 3. Name two uses of layer groups.'

Student Activity

Answer the questions. Show their work to a partner.

Cross-Curricular Connections

Mathematics: 3D coordinates use x, y, z axes; perspective involves geometric projections
Science: Light and shadow follow physics principles — angles, intensity, diffusion
Language: Visual hierarchy in design guides the viewer's eye through content

Digital Citizen Reminder: Credit the original creator when using their techniques or assets in your work.

Dormitory Task (Paper-and-Pencil)

On paper, draw a 3D letter 'A'. Start with a flat 2D 'A', then add shadow lines, depth lines, and shading to make it look three-dimensional. Label each technique used.

Resources Needed:
- Computer with GIMP
- text effect samples
- layer mask tutorial
- grid paper

Differentiation

Approaching: Follow a step-by-step tutorial to create 3D text effect
On Level: Create 3D text effects independently using all techniques
Advanced: Design a complete 3D poster with multiple text elements and effects
SEND: Pre-made 3D templates to modify, guided steps with visual examples

Formative Assessment

Method: Exit ticket and self-assessment
CT Skill Assessed: Abstraction — understanding how 2D techniques simulate 3D perception

Teacher Reflection (Post-Lesson)

What worked well:

What to improve:

Connection to next lesson:

Lesson 2.09 **Duration: 40 minutes**

Nested Structures

Learning Objectives

- Work with nested data structures: lists of dictionaries, dictionaries of lists, and deeper nesting
- Navigate and manipulate complex nested data efficiently
- Design hierarchical data models for real-world applications

Timing Breakdown

0-5 min	**Engage**	Show a 2D image and its 3D-effect version. Ask: 'What makes the second image loo...
5-15 min	**Explore**	Open GIMP. Demonstrate: add text layer, create layer group, apply layer mask, ad...
15-25 min	**Explain**	Define 3D concepts: three dimensions length, width, depth. 3D text technique: ...
25-35 min	**Elaborate**	Task: 'Create a 3D text effect for a school poster. Add shadow, highlight, and d...
35-40 min	**Evaluate**	Exit ticket: '1. What makes a 2D image look 3D? 2. What is a layer mask? 3. Name...

Engage (5-7 minutes)

Teacher Activity

Show a 2D image and its 3D-effect version. Ask: 'What makes the second image look three-dimensional?' Discuss shadows, highlights, depth.

Student Activity

Compare images. Identify 3D cues: shadows, highlights, perspective. Discuss how these create depth perception.

Explore (10-12 minutes)

Teacher Activity

Open GIMP. Demonstrate: add text layer, create layer group, apply layer mask, add shadow effect for 3D look. Students follow along.

Student Activity

Follow along. Create text with shadow effect. Experiment with layer masks. Apply 3D effects.

Explain (10-12 minutes)

Teacher Activity

Define 3D concepts: three dimensions length, width, depth. 3D text technique: duplicate layer, offset, darken. Layer masks: control visibility without deleting. Layer groups: organize complex projects.

Student Activity

Take notes. Understand how 2D images can simulate 3D. Practice layer mask concepts.

Elaborate (8-10 minutes)

Teacher Activity

Task: 'Create a 3D text effect for a school poster. Add shadow, highlight, and depth to make text pop.' Show examples of professional 3D text.

Student Activity

Work independently. Apply all techniques. Get peer feedback on the 3D effect quality.

Evaluate (5-8 minutes)

Teacher Activity

Exit ticket: '1. What makes a 2D image look 3D? 2. What is a layer mask? 3. Name two uses of layer groups.'

Student Activity

Answer the questions. Show their work to a partner.

Cross-Curricular Connections

Mathematics: 3D coordinates use x, y, z axes; perspective involves geometric projections
Science: Light and shadow follow physics principles — angles, intensity, diffusion
Language: Visual hierarchy in design guides the viewer's eye through content

Digital Citizen Reminder: Credit the original creator when using their techniques or assets in your work.

Dormitory Task (Paper-and-Pencil)

On paper, draw a 3D letter 'A'. Start with a flat 2D 'A', then add shadow lines, depth lines, and shading to make it look three-dimensional. Label each technique used.

Resources Needed:

- Computer with GIMP
- text effect samples
- layer mask tutorial
- grid paper

Differentiation

Approaching: Follow a step-by-step tutorial to create 3D text effect
On Level: Create 3D text effects independently using all techniques
Advanced: Design a complete 3D poster with multiple text elements and effects
SEND: Pre-made 3D templates to modify, guided steps with visual examples

Formative Assessment

Method: Exit ticket and self-assessment
CT Skill Assessed: Abstraction — understanding how 2D techniques simulate 3D perception

Teacher Reflection (Post-Lesson)

What worked well:

What to improve:

Connection to next lesson:

Lesson 2.10	Duration: 40 minutes

Data Processing

Learning Objectives

- Process collections of data using iteration, filtering, and aggregation
- Apply statistical operations to datasets using appropriate data structures
- Transform raw data into meaningful information through structured processing

Timing Breakdown

0-5 min	Engage	Show a 2D image and its 3D-effect version. Ask: 'What makes the second image loo...
5-15 min	Explore	Open GIMP. Demonstrate: add text layer, create layer group, apply layer mask, ad...
15-25 min	Explain	Define 3D concepts: three dimensions length, width, depth. 3D text technique: ...
25-35 min	Elaborate	Task: 'Create a 3D text effect for a school poster. Add shadow, highlight, and d...
35-40 min	Evaluate	Exit ticket: '1. What makes a 2D image look 3D? 2. What is a layer mask? 3. Name...

Engage (5-7 minutes)

Teacher Activity

Show a 2D image and its 3D-effect version. Ask: 'What makes the second image look three-dimensional?' Discuss shadows, highlights, depth.

Student Activity

Compare images. Identify 3D cues: shadows, highlights, perspective. Discuss how these create depth perception.

Explore (10-12 minutes)

Teacher Activity

Open GIMP. Demonstrate: add text layer, create layer group, apply layer mask, add shadow effect for 3D look. Students follow along.

Student Activity

Follow along. Create text with shadow effect. Experiment with layer masks. Apply 3D effects.

Explain (10-12 minutes)

Teacher Activity

Define 3D concepts: three dimensions length, width, depth. 3D text technique: duplicate layer, offset, darken. Layer masks: control visibility without deleting. Layer groups: organize complex projects.

Student Activity

Take notes. Understand how 2D images can simulate 3D. Practice layer mask concepts.

Elaborate (8-10 minutes)

Teacher Activity

Task: 'Create a 3D text effect for a school poster. Add shadow, highlight, and depth to make text pop.' Show examples of professional 3D text.

Student Activity

Work independently. Apply all techniques. Get peer feedback on the 3D effect quality.

Evaluate (5-8 minutes)

Teacher Activity

Exit ticket: '1. What makes a 2D image look 3D? 2. What is a layer mask? 3. Name two uses of layer groups.'

Student Activity

Answer the questions. Show their work to a partner.

Cross-Curricular Connections

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Digital Citizen Reminder: Credit the original creator when using their techniques or assets in your work.

Dormitory Task (Paper-and-Pencil)

On paper, draw a 3D letter 'A'. Start with a flat 2D 'A', then add shadow lines, depth lines, and shading to make it look three-dimensional. Label each technique used.

Resources Needed:
- Computer with GIMP
- text effect samples
- layer mask tutorial
- grid paper

Differentiation

Approaching: Follow a step-by-step tutorial to create 3D text effect
On Level: Create 3D text effects independently using all techniques
Advanced: Design a complete 3D poster with multiple text elements and effects
SEND: Pre-made 3D templates to modify, guided steps with visual examples

Formative Assessment

Method: Exit ticket and self-assessment
CT Skill Assessed: Abstraction — understanding how 2D techniques simulate 3D perception

Teacher Reflection (Post-Lesson)

What worked well:

What to improve:

Connection to next lesson:

Lesson 2.11	Duration: 40 minutes

Project - Student Database

Learning Objectives

- Design and implement a complete student database system using multiple data structures
- Apply CRUD operations: create, read, update, and delete student records
- Build search, reporting, and data export features with proper error handling

Timing Breakdown

0-5 min	Engage	Show a 2D image and its 3D-effect version. Ask: 'What makes the second image loo...
5-15 min	**Explore**	Open GIMP. Demonstrate: add text layer, create layer group, apply layer mask, ad...
15-25 min	**Explain**	Define 3D concepts: three dimensions length, width, depth. 3D text technique: ...
25-35 min	**Elaborate**	Task: 'Create a 3D text effect for a school poster. Add shadow, highlight, and d...
35-40 min	**Evaluate**	Exit ticket: '1. What makes a 2D image look 3D? 2. What is a layer mask? 3. Name...

Engage (5-7 minutes)

Teacher Activity

Show a 2D image and its 3D-effect version. Ask: 'What makes the second image look three-dimensional?' Discuss shadows, highlights, depth.

Student Activity

Compare images. Identify 3D cues: shadows, highlights, perspective. Discuss how these create depth perception.

Explore (10-12 minutes)

Teacher Activity

Open GIMP. Demonstrate: add text layer, create layer group, apply layer mask, add shadow effect for 3D look. Students follow along.

Student Activity

Follow along. Create text with shadow effect. Experiment with layer masks. Apply 3D effects.

Explain (10-12 minutes)

Teacher Activity

Define 3D concepts: three dimensions length, width, depth. 3D text technique: duplicate layer, offset, darken. Layer masks: control visibility without deleting. Layer groups: organize complex projects.

Student Activity

Take notes. Understand how 2D images can simulate 3D. Practice layer mask concepts.

Elaborate (8-10 minutes)

Teacher Activity

Task: 'Create a 3D text effect for a school poster. Add shadow, highlight, and depth to make text pop.' Show examples of professional 3D text.

Student Activity

Work independently. Apply all techniques. Get peer feedback on the 3D effect quality.

Evaluate (5-8 minutes)

Teacher Activity

Exit ticket: '1. What makes a 2D image look 3D? 2. What is a layer mask? 3. Name two uses of layer groups.'

Student Activity

Answer the questions. Show their work to a partner.

Cross-Curricular Connections

Mathematics: 3D coordinates use x, y, z axes; perspective involves geometric projections
Science: Light and shadow follow physics principles — angles, intensity, diffusion
Language: Visual hierarchy in design guides the viewer's eye through content

Digital Citizen Reminder: Credit the original creator when using their techniques or assets in your work.

Dormitory Task (Paper-and-Pencil)

On paper, draw a 3D letter 'A'. Start with a flat 2D 'A', then add shadow lines, depth lines, and shading to make it look three-dimensional. Label each technique used.

Resources Needed:

- Computer with GIMP
- text effect samples
- layer mask tutorial
- grid paper

Differentiation

Approaching: Follow a step-by-step tutorial to create 3D text effect
On Level: Create 3D text effects independently using all techniques
Advanced: Design a complete 3D poster with multiple text elements and effects
SEND: Pre-made 3D templates to modify, guided steps with visual examples

Formative Assessment

Method: Exit ticket and self-assessment
CT Skill Assessed: Abstraction — understanding how 2D techniques simulate 3D perception

Teacher Reflection (Post-Lesson)

What worked well:

What to improve:

Connection to next lesson:

Lesson 2.12 **Duration: 40 minutes**

Chapter 2 Review

Learning Objectives

- Consolidate understanding of all data structures: lists, dictionaries, tuples, and sets
- Solve 综合 problems that require selecting and combining multiple data structures
- Evaluate design decisions and justify data structure choices for given problems

Timing Breakdown

0-5 min	**Engage**	Show a 2D image and its 3D-effect version. Ask: 'What makes the second image loo...
5-15 min	**Explore**	Open GIMP. Demonstrate: add text layer, create layer group, apply layer mask, ad...
15-25 min	**Explain**	Define 3D concepts: three dimensions length, width, depth. 3D text technique: ...
25-35 min	**Elaborate**	Task: 'Create a 3D text effect for a school poster. Add shadow, highlight, and d...
35-40 min	**Evaluate**	Exit ticket: '1. What makes a 2D image look 3D? 2. What is a layer mask? 3. Name...

Engage (5-7 minutes)

Teacher Activity

Show a 2D image and its 3D-effect version. Ask: 'What makes the second image look three-dimensional?' Discuss shadows, highlights, depth.

Student Activity

Compare images. Identify 3D cues: shadows, highlights, perspective. Discuss how these create depth perception.

Explore (10-12 minutes)

Teacher Activity

Open GIMP. Demonstrate: add text layer, create layer group, apply layer mask, add shadow effect for 3D look. Students follow along.

Student Activity

Follow along. Create text with shadow effect. Experiment with layer masks. Apply 3D effects.

Explain (10-12 minutes)

Teacher Activity

Define 3D concepts: three dimensions length, width, depth. 3D text technique: duplicate layer, offset, darken. Layer masks: control visibility without deleting. Layer groups: organize complex projects.

Student Activity

Take notes. Understand how 2D images can simulate 3D. Practice layer mask concepts.

Elaborate (8-10 minutes)

Teacher Activity

Task: ‘Create a 3D text effect for a school poster. Add shadow, highlight, and depth to make text pop.’ Show examples of professional 3D text.

Student Activity

Work independently. Apply all techniques. Get peer feedback on the 3D effect quality.

Evaluate (5-8 minutes)

Teacher Activity

Exit ticket: ‘1. What makes a 2D image look 3D? 2. What is a layer mask? 3. Name two uses of layer groups.’

Student Activity

Answer the questions. Show their work to a partner.

Cross-Curricular Connections

Mathematics: 3D coordinates use x, y, z axes; perspective involves geometric projections
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Digital Citizen Reminder: Credit the original creator when using their techniques or assets in your work.

Dormitory Task (Paper-and-Pencil)

On paper, draw a 3D letter ‘A’. Start with a flat 2D ‘A’, then add shadow lines, depth lines, and shading to make it look three-dimensional. Label each technique used.

Resources Needed:

- Computer with GIMP
- text effect samples
- layer mask tutorial
- grid paper

Differentiation

Approaching: Follow a step-by-step tutorial to create 3D text effect
On Level: Create 3D text effects independently using all techniques
Advanced: Design a complete 3D poster with multiple text elements and effects
SEND: Pre-made 3D templates to modify, guided steps with visual examples

Formative Assessment

Method: Exit ticket and self-assessment
CT Skill Assessed: Abstraction — understanding how 2D techniques simulate 3D perception

Teacher Reflection (Post-Lesson)

What worked well:

What to improve:

Connection to next lesson:

Lesson 3.01	Duration: 40 minutes

What is a Database?

Learning Objectives

- Define a database and explain its purpose in real-world applications
- Differentiate between flat-file and relational databases

Timing Breakdown

0-5 min	Engage	Show a sales report with 100 rows. Ask: 'How would you find the top 10 sales? Th...
5-15 min	Explore	Demonstrate AutoFill: enter 1, 2, select, drag to 20. Show number series, odd nu...
15-25 min	Explain	Define referencing types: relative changes when copied, absolute \$ sign, stay...
25-35 min	Elaborate	Task: 'Create a monthly expense tracker. Use AutoFill for dates, absolute refere...
35-40 min	Evaluate	Exit ticket: '1. What is the difference between relative and absolute referencin...

Engage (5-7 minutes)

Teacher Activity

Show a sales report with 100 rows. Ask: 'How would you find the top 10 sales? The average? The trend over time?' Demonstrate AutoFill and charts.

Student Activity

Consider the challenge. Recognize manual methods would be too slow. Express interest in automated solutions.

Explore (10-12 minutes)

Teacher Activity

Demonstrate AutoFill: enter 1, 2, select, drag to 20. Show number series, odd numbers, days, months. Then show relative vs absolute referencing with a multiplication table.

Student Activity

Follow along. Try AutoFill with different patterns. Create a multiplication table using absolute references.

Explain (10-12 minutes)

Teacher Activity

Define referencing types: relative changes when copied, absolute $ sign, stays fixed, mixed $A1 or A$1. Explain linking: cells within same sheet, between sheets, between workbooks. Show chart types: column, bar, line, pie.

Student Activity

Take notes. Understand when to use each referencing type. See how charts visualize data.

Elaborate (8-10 minutes)

Teacher Activity

Task: 'Create a monthly expense tracker. Use AutoFill for dates, absolute reference for tax rate, link to a summary sheet, and create a pie chart of expense categories.'

Student Activity

Work independently. Apply all techniques. Create the complete tracker with charts.

Evaluate (5-8 minutes)

Teacher Activity

Exit ticket: '1. What is the difference between relative and absolute referencing? 2. When would you use a pie chart vs a bar chart? 3. How do you link cells between sheets?'

Student Activity

Answer the questions. Review their spreadsheet for completeness.

Cross-Curricular Connections

Mathematics: Absolute referencing uses the \$ symbol like algebraic constants; charts use coordinate systems
Science: Scientists use spreadsheets for data analysis — graphs show experimental trends
Language: Presenting data in charts supports evidence-based arguments in reports

Digital Citizen Reminder: Back up important spreadsheets regularly — data loss can be devastating.

Dormitory Task (Paper-and-Pencil)

On paper, create a multiplication table from 1x1 to 5x5. Mark which cells would need absolute references if you built this in a spreadsheet. Write the formula for cell B2 2x2 using absolute references.

Resources Needed:
- Computer with spreadsheet
- expense tracker template
- chart examples
- formula reference

Differentiation

Approaching: Use a pre-made template with AutoFill already demonstrated
On Level: Create the expense tracker with all features independently
Advanced: Add pivot table analysis and multiple chart types with formatting
SEND: Paper-based spreadsheet simulation, guided formula writing

Formative Assessment

Method: Exit ticket and self-assessment
CT Skill Assessed: Algorithm Design — planning the spreadsheet structure before building it

Teacher Reflection (Post-Lesson)

What worked well:

What to improve:

Connection to next lesson:

Lesson 3.02	Duration: 40 minutes

Database Concepts

Learning Objectives

- Explain the concepts of tables, records, fields, and primary keys
- Describe how relationships connect data across tables
- Distinguish between data types used in database fields

Timing Breakdown

0-5 min	**Engage**	Show a sales report with 100 rows. Ask: 'How would you find the top 10 sales? Th...
5-15 min	**Explore**	Demonstrate AutoFill: enter 1, 2, select, drag to 20. Show number series, odd nu...
15-25 min	**Explain**	Define referencing types: relative changes when copied, absolute \$ sign, stay...
25-35 min	**Elaborate**	Task: 'Create a monthly expense tracker. Use AutoFill for dates, absolute refere...
35-40 min	**Evaluate**	Exit ticket: '1. What is the difference between relative and absolute referencin...

Engage (5-7 minutes)

Teacher Activity

Show a sales report with 100 rows. Ask: 'How would you find the top 10 sales? The average? The trend over time?' Demonstrate AutoFill and charts.

Student Activity

Consider the challenge. Recognize manual methods would be too slow. Express interest in automated solutions.

Explore (10-12 minutes)

Teacher Activity

Demonstrate AutoFill: enter 1, 2, select, drag to 20. Show number series, odd numbers, days, months. Then show relative vs absolute referencing with a multiplication table.

Student Activity

Follow along. Try AutoFill with different patterns. Create a multiplication table using absolute references.

Explain (10-12 minutes)

Teacher Activity

Define referencing types: relative changes when copied, absolute $ sign, stays fixed, mixed $A1 or A$1. Explain linking: cells within same sheet, between sheets, between workbooks. Show chart types: column, bar, line, pie.

Student Activity

Take notes. Understand when to use each referencing type. See how charts visualize data.

Elaborate (8-10 minutes)

Teacher Activity

Task: ‘Create a monthly expense tracker. Use AutoFill for dates, absolute reference for tax rate, link to a summary sheet, and create a pie chart of expense categories.’

Student Activity

Work independently. Apply all techniques. Create the complete tracker with charts.

Evaluate (5-8 minutes)

Teacher Activity

Exit ticket: ‘1. What is the difference between relative and absolute referencing? 2. When would you use a pie chart vs a bar chart? 3. How do you link cells between sheets?’

Student Activity

Answer the questions. Review their spreadsheet for completeness.

Cross-Curricular Connections

Mathematics: Absolute referencing uses the \$ symbol like algebraic constants; charts use coordinate systems
Science: Scientists use spreadsheets for data analysis — graphs show experimental trends
Language: Presenting data in charts supports evidence-based arguments in reports

Digital Citizen Reminder: Back up important spreadsheets regularly — data loss can be devastating.

Dormitory Task (Paper-and-Pencil)

On paper, create a multiplication table from 1x1 to 5x5. Mark which cells would need absolute references if you built this in a spreadsheet. Write the formula for cell B2 2x2 using absolute references.

Resources Needed:
- Computer with spreadsheet
- expense tracker template
- chart examples
- formula reference

Differentiation

Approaching: Use a pre-made template with AutoFill already demonstrated
On Level: Create the expense tracker with all features independently
Advanced: Add pivot table analysis and multiple chart types with formatting
SEND: Paper-based spreadsheet simulation, guided formula writing

Formative Assessment

Method: Exit ticket and self-assessment
CT Skill Assessed: Algorithm Design — planning the spreadsheet structure before building it

Teacher Reflection (Post-Lesson)

What worked well:

What to improve:

Connection to next lesson:

Lesson 3.03	Duration: 40 minutes

SQL Introduction

Learning Objectives

- Define SQL and its role as a standard database language
- Identify the main categories of SQL commands
- Write and execute a basic SQL statement to create a table

Timing Breakdown

0-5 min	Engage	Show a sales report with 100 rows. Ask: 'How would you find the top 10 sales? Th...
5-15 min	**Explore**	Demonstrate AutoFill: enter 1, 2, select, drag to 20. Show number series, odd nu...
15-25 min	**Explain**	Define referencing types: relative changes when copied, absolute \$ sign, stay...
25-35 min	**Elaborate**	Task: 'Create a monthly expense tracker. Use AutoFill for dates, absolute refere...
35-40 min	**Evaluate**	Exit ticket: '1. What is the difference between relative and absolute referencin...

Engage (5-7 minutes)

Teacher Activity

Show a sales report with 100 rows. Ask: 'How would you find the top 10 sales? The average? The trend over time?' Demonstrate AutoFill and charts.

Student Activity

Consider the challenge. Recognize manual methods would be too slow. Express interest in automated solutions.

Explore (10-12 minutes)

Teacher Activity

Demonstrate AutoFill: enter 1, 2, select, drag to 20. Show number series, odd numbers, days, months. Then show relative vs absolute referencing with a multiplication table.

Student Activity

Follow along. Try AutoFill with different patterns. Create a multiplication table using absolute references.

Explain (10-12 minutes)

Teacher Activity

Define referencing types: relative changes when copied, absolute $ sign, stays fixed, mixed $A1 or A$1. Explain linking: cells within same sheet, between sheets, between workbooks. Show chart types: column, bar, line, pie.

Student Activity

Take notes. Understand when to use each referencing type. See how charts visualize data.

Elaborate (8-10 minutes)

Teacher Activity

Task: ‘Create a monthly expense tracker. Use AutoFill for dates, absolute reference for tax rate, link to a summary sheet, and create a pie chart of expense categories.’

Student Activity

Work independently. Apply all techniques. Create the complete tracker with charts.

Evaluate (5-8 minutes)

Teacher Activity

Exit ticket: ‘1. What is the difference between relative and absolute referencing? 2. When would you use a pie chart vs a bar chart? 3. How do you link cells between sheets?’

Student Activity

Answer the questions. Review their spreadsheet for completeness.

Cross-Curricular Connections

Mathematics: Absolute referencing uses the \$ symbol like algebraic constants; charts use coordinate systems
Science: Scientists use spreadsheets for data analysis — graphs show experimental trends
Language: Presenting data in charts supports evidence-based arguments in reports

Digital Citizen Reminder: Back up important spreadsheets regularly — data loss can be devastating.

Dormitory Task (Paper-and-Pencil)

On paper, create a multiplication table from 1x1 to 5x5. Mark which cells would need absolute references if you built this in a spreadsheet. Write the formula for cell B2 2x2 using absolute references.

Resources Needed:

- Computer with spreadsheet
- expense tracker template
- chart examples
- formula reference

Differentiation

Approaching: Use a pre-made template with AutoFill already demonstrated
On Level: Create the expense tracker with all features independently
Advanced: Add pivot table analysis and multiple chart types with formatting
SEND: Paper-based spreadsheet simulation, guided formula writing

Formative Assessment

Method: Exit ticket and self-assessment
CT Skill Assessed: Algorithm Design — planning the spreadsheet structure before building it

Teacher Reflection (Post-Lesson)

What worked well:

What to improve:

Connection to next lesson:

Lesson 3.04	Duration: 40 minutes

SELECT Queries

Learning Objectives

- Write SELECT statements to retrieve data from a single table
- Use SELECT DISTINCT to eliminate duplicate records
- Apply column aliases and ORDER BY to format query results

Timing Breakdown

0-5 min	Engage	Show a sales report with 100 rows. Ask: 'How would you find the top 10 sales? Th...
5-15 min	**Explore**	Demonstrate AutoFill: enter 1, 2, select, drag to 20. Show number series, odd nu...
15-25 min	**Explain**	Define referencing types: relative changes when copied, absolute \$ sign, stay...
25-35 min	**Elaborate**	Task: 'Create a monthly expense tracker. Use AutoFill for dates, absolute refere...
35-40 min	**Evaluate**	Exit ticket: '1. What is the difference between relative and absolute referencin...

Engage (5-7 minutes)

Teacher Activity

Show a sales report with 100 rows. Ask: 'How would you find the top 10 sales? The average? The trend over time?' Demonstrate AutoFill and charts.

Student Activity

Consider the challenge. Recognize manual methods would be too slow. Express interest in automated solutions.

Explore (10-12 minutes)

Teacher Activity

Demonstrate AutoFill: enter 1, 2, select, drag to 20. Show number series, odd numbers, days, months. Then show relative vs absolute referencing with a multiplication table.

Student Activity

Follow along. Try AutoFill with different patterns. Create a multiplication table using absolute references.

Explain (10-12 minutes)

Teacher Activity

Define referencing types: relative changes when copied, absolute $ sign, stays fixed, mixed $A1 or A$1. Explain linking: cells within same sheet, between sheets, between workbooks. Show chart types: column, bar, line, pie.

Student Activity

Take notes. Understand when to use each referencing type. See how charts visualize data.

Elaborate (8-10 minutes)

Teacher Activity

Task: 'Create a monthly expense tracker. Use AutoFill for dates, absolute reference for tax rate, link to a summary sheet, and create a pie chart of expense categories.'

Student Activity

Work independently. Apply all techniques. Create the complete tracker with charts.

Evaluate (5-8 minutes)

Teacher Activity

Exit ticket: '1. What is the difference between relative and absolute referencing? 2. When would you use a pie chart vs a bar chart? 3. How do you link cells between sheets?'

Student Activity

Answer the questions. Review their spreadsheet for completeness.

Cross-Curricular Connections

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Digital Citizen Reminder: Back up important spreadsheets regularly — data loss can be devastating.

Dormitory Task (Paper-and-Pencil)

On paper, create a multiplication table from 1x1 to 5x5. Mark which cells would need absolute references if you built this in a spreadsheet. Write the formula for cell B2 2x2 using absolute references.

Resources Needed:
- Computer with spreadsheet
- expense tracker template
- chart examples
- formula reference

Differentiation

Approaching: Use a pre-made template with AutoFill already demonstrated
On Level: Create the expense tracker with all features independently
Advanced: Add pivot table analysis and multiple chart types with formatting
SEND: Paper-based spreadsheet simulation, guided formula writing

Formative Assessment

Method: Exit ticket and self-assessment
CT Skill Assessed: Algorithm Design — planning the spreadsheet structure before building it

Teacher Reflection (Post-Lesson)

What worked well:

What to improve:

Connection to next lesson:

Lesson 3.05	Duration: 40 minutes

INSERT Data

Learning Objectives

- Write INSERT INTO statements to add single records to a table
- Insert multiple rows using a single INSERT command
- Handle common errors when inserting data into tables

Timing Breakdown

0-5 min	Engage	Show a sales report with 100 rows. Ask: 'How would you find the top 10 sales? Th...
5-15 min	Explore	Demonstrate AutoFill: enter 1, 2, select, drag to 20. Show number series, odd nu...
15-25 min	Explain	Define referencing types: relative changes when copied, absolute \$ sign, stay...
25-35 min	Elaborate	Task: 'Create a monthly expense tracker. Use AutoFill for dates, absolute refere...
35-40 min	Evaluate	Exit ticket: '1. What is the difference between relative and absolute referencin...

Engage (5-7 minutes)

Teacher Activity

Show a sales report with 100 rows. Ask: 'How would you find the top 10 sales? The average? The trend over time?' Demonstrate AutoFill and charts.

Student Activity

Consider the challenge. Recognize manual methods would be too slow. Express interest in automated solutions.

Explore (10-12 minutes)

Teacher Activity

Demonstrate AutoFill: enter 1, 2, select, drag to 20. Show number series, odd numbers, days, months. Then show relative vs absolute referencing with a multiplication table.

Student Activity

Follow along. Try AutoFill with different patterns. Create a multiplication table using absolute references.

Explain (10-12 minutes)

Teacher Activity

Define referencing types: relative changes when copied, absolute $ sign, stays fixed, mixed $A1 or A$1. Explain linking: cells within same sheet, between sheets, between workbooks. Show chart types: column, bar, line, pie.

Student Activity

Take notes. Understand when to use each referencing type. See how charts visualize data.

Elaborate (8-10 minutes)

Teacher Activity

Task: 'Create a monthly expense tracker. Use AutoFill for dates, absolute reference for tax rate, link to a summary sheet, and create a pie chart of expense categories.'

Student Activity

Work independently. Apply all techniques. Create the complete tracker with charts.

Evaluate (5-8 minutes)

Teacher Activity

Exit ticket: '1. What is the difference between relative and absolute referencing? 2. When would you use a pie chart vs a bar chart? 3. How do you link cells between sheets?'

Student Activity

Answer the questions. Review their spreadsheet for completeness.

Cross-Curricular Connections

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Language: Presenting data in charts supports evidence-based arguments in reports

Digital Citizen Reminder: Back up important spreadsheets regularly — data loss can be devastating.

Dormitory Task (Paper-and-Pencil)

On paper, create a multiplication table from 1x1 to 5x5. Mark which cells would need absolute references if you built this in a spreadsheet. Write the formula for cell B2 2x2 using absolute references.

Resources Needed:

- Computer with spreadsheet
- expense tracker template
- chart examples
- formula reference

Differentiation

Approaching: Use a pre-made template with AutoFill already demonstrated
On Level: Create the expense tracker with all features independently
Advanced: Add pivot table analysis and multiple chart types with formatting
SEND: Paper-based spreadsheet simulation, guided formula writing

Formative Assessment

Method: Exit ticket and self-assessment
CT Skill Assessed: Algorithm Design — planning the spreadsheet structure before building it

Teacher Reflection (Post-Lesson)

What worked well:

What to improve:

Connection to next lesson:

Lesson 3.06	Duration: 40 minutes

UPDATE Data

Learning Objectives

- Write UPDATE statements to modify existing records
- Use WHERE clauses to target specific rows for updates
- Understand the risks of updating without a WHERE condition

Timing Breakdown

0-5 min	Engage	Show a sales report with 100 rows. Ask: 'How would you find the top 10 sales? Th...
5-15 min	Explore	Demonstrate AutoFill: enter 1, 2, select, drag to 20. Show number series, odd nu...
15-25 min	Explain	Define referencing types: relative changes when copied, absolute \$ sign, stay...
25-35 min	Elaborate	Task: 'Create a monthly expense tracker. Use AutoFill for dates, absolute refere...
35-40 min	Evaluate	Exit ticket: '1. What is the difference between relative and absolute referencin...

Engage (5-7 minutes)

Teacher Activity

Show a sales report with 100 rows. Ask: 'How would you find the top 10 sales? The average? The trend over time?' Demonstrate AutoFill and charts.

Student Activity

Consider the challenge. Recognize manual methods would be too slow. Express interest in automated solutions.

Explore (10-12 minutes)

Teacher Activity

Demonstrate AutoFill: enter 1, 2, select, drag to 20. Show number series, odd numbers, days, months. Then show relative vs absolute referencing with a multiplication table.

Student Activity

Follow along. Try AutoFill with different patterns. Create a multiplication table using absolute references.

Explain (10-12 minutes)

Teacher Activity

Define referencing types: relative changes when copied, absolute $ sign, stays fixed, mixed $A1 or A$1. Explain linking: cells within same sheet, between sheets, between workbooks. Show chart types: column, bar, line, pie.

Student Activity

Take notes. Understand when to use each referencing type. See how charts visualize data.

Elaborate (8-10 minutes)

Teacher Activity

Task: ‘Create a monthly expense tracker. Use AutoFill for dates, absolute reference for tax rate, link to a summary sheet, and create a pie chart of expense categories.’

Student Activity

Work independently. Apply all techniques. Create the complete tracker with charts.

Evaluate (5-8 minutes)

Teacher Activity

Exit ticket: ‘1. What is the difference between relative and absolute referencing? 2. When would you use a pie chart vs a bar chart? 3. How do you link cells between sheets?’

Student Activity

Answer the questions. Review their spreadsheet for completeness.

Cross-Curricular Connections

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Digital Citizen Reminder: Back up important spreadsheets regularly — data loss can be devastating.

Dormitory Task (Paper-and-Pencil)

On paper, create a multiplication table from 1x1 to 5x5. Mark which cells would need absolute references if you built this in a spreadsheet. Write the formula for cell B2 2x2 using absolute references.

Resources Needed:
- Computer with spreadsheet
- expense tracker template
- chart examples
- formula reference

Differentiation

Approaching: Use a pre-made template with AutoFill already demonstrated
On Level: Create the expense tracker with all features independently
Advanced: Add pivot table analysis and multiple chart types with formatting
SEND: Paper-based spreadsheet simulation, guided formula writing

Formative Assessment

Method: Exit ticket and self-assessment
CT Skill Assessed: Algorithm Design — planning the spreadsheet structure before building it

Teacher Reflection (Post-Lesson)

What worked well:

What to improve:

Connection to next lesson:

Lesson 3.07	**Duration: 40 minutes**

DELETE Data

Learning Objectives

- Write DELETE statements to remove specific records from a table
- Use WHERE clauses to prevent accidental deletion of all data
- Explain the difference between DELETE and DROP commands

Timing Breakdown

0-5 min	**Engage**	Show a sales report with 100 rows. Ask: 'How would you find the top 10 sales? Th...
5-15 min	**Explore**	Demonstrate AutoFill: enter 1, 2, select, drag to 20. Show number series, odd nu...
15-25 min	**Explain**	Define referencing types: relative changes when copied, absolute \$ sign, stay...
25-35 min	**Elaborate**	Task: 'Create a monthly expense tracker. Use AutoFill for dates, absolute refere...
35-40 min	**Evaluate**	Exit ticket: '1. What is the difference between relative and absolute referencin...

Engage (5-7 minutes)

Teacher Activity

Show a sales report with 100 rows. Ask: 'How would you find the top 10 sales? The average? The trend over time?' Demonstrate AutoFill and charts.

Student Activity

Consider the challenge. Recognize manual methods would be too slow. Express interest in automated solutions.

Explore (10-12 minutes)

Teacher Activity

Demonstrate AutoFill: enter 1, 2, select, drag to 20. Show number series, odd numbers, days, months. Then show relative vs absolute referencing with a multiplication table.

Student Activity

Follow along. Try AutoFill with different patterns. Create a multiplication table using absolute references.

Explain (10-12 minutes)

Teacher Activity

Define referencing types: relative changes when copied, absolute $ sign, stays fixed, mixed $A1 or A$1. Explain linking: cells within same sheet, between sheets, between workbooks. Show chart types: column, bar, line, pie.

Student Activity

Take notes. Understand when to use each referencing type. See how charts visualize data.

Elaborate (8-10 minutes)

Teacher Activity

Task: 'Create a monthly expense tracker. Use AutoFill for dates, absolute reference for tax rate, link to a summary sheet, and create a pie chart of expense categories.'

Student Activity

Work independently. Apply all techniques. Create the complete tracker with charts.

Evaluate (5-8 minutes)

Teacher Activity

Exit ticket: '1. What is the difference between relative and absolute referencing? 2. When would you use a pie chart vs a bar chart? 3. How do you link cells between sheets?'

Student Activity

Answer the questions. Review their spreadsheet for completeness.

Cross-Curricular Connections

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Digital Citizen Reminder: Back up important spreadsheets regularly — data loss can be devastating.

Dormitory Task (Paper-and-Pencil)

On paper, create a multiplication table from 1x1 to 5x5. Mark which cells would need absolute references if you built this in a spreadsheet. Write the formula for cell B2 2x2 using absolute references.

Resources Needed:
- Computer with spreadsheet
- expense tracker template
- chart examples
- formula reference

Differentiation

Approaching: Use a pre-made template with AutoFill already demonstrated
On Level: Create the expense tracker with all features independently
Advanced: Add pivot table analysis and multiple chart types with formatting
SEND: Paper-based spreadsheet simulation, guided formula writing

Formative Assessment

Method: Exit ticket and self-assessment
CT Skill Assessed: Algorithm Design — planning the spreadsheet structure before building it

Teacher Reflection (Post-Lesson)

What worked well:

What to improve:

Connection to next lesson:

Lesson 3.08	Duration: 40 minutes

WHERE Conditions

Learning Objectives

- Use comparison operators to filter records in SQL queries
- Apply logical operators AND, OR, and NOT to combine conditions
- Use BETWEEN, IN, and LIKE operators for advanced filtering

Timing Breakdown

0-5 min	Engage	Show a sales report with 100 rows. Ask: 'How would you find the top 10 sales? Th...
5-15 min	**Explore**	Demonstrate AutoFill: enter 1, 2, select, drag to 20. Show number series, odd nu...
15-25 min	**Explain**	Define referencing types: relative changes when copied, absolute \$ sign, stay...
25-35 min	**Elaborate**	Task: 'Create a monthly expense tracker. Use AutoFill for dates, absolute refere...
35-40 min	**Evaluate**	Exit ticket: '1. What is the difference between relative and absolute referencin...

Engage (5-7 minutes)

Teacher Activity

Show a sales report with 100 rows. Ask: 'How would you find the top 10 sales? The average? The trend over time?' Demonstrate AutoFill and charts.

Student Activity

Consider the challenge. Recognize manual methods would be too slow. Express interest in automated solutions.

Explore (10-12 minutes)

Teacher Activity

Demonstrate AutoFill: enter 1, 2, select, drag to 20. Show number series, odd numbers, days, months. Then show relative vs absolute referencing with a multiplication table.

Student Activity

Follow along. Try AutoFill with different patterns. Create a multiplication table using absolute references.

Explain (10-12 minutes)

Teacher Activity

Define referencing types: relative changes when copied, absolute $ sign, stays fixed, mixed $A1 or A$1. Explain linking: cells within same sheet, between sheets, between workbooks. Show chart types: column, bar, line, pie.

Student Activity

Take notes. Understand when to use each referencing type. See how charts visualize data.

Elaborate (8-10 minutes)

Teacher Activity

Task: 'Create a monthly expense tracker. Use AutoFill for dates, absolute reference for tax rate, link to a summary sheet, and create a pie chart of expense categories.'

Student Activity

Work independently. Apply all techniques. Create the complete tracker with charts.

Evaluate (5-8 minutes)

Teacher Activity

Exit ticket: '1. What is the difference between relative and absolute referencing? 2. When would you use a pie chart vs a bar chart? 3. How do you link cells between sheets?'

Student Activity

Answer the questions. Review their spreadsheet for completeness.

Cross-Curricular Connections

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Digital Citizen Reminder: Back up important spreadsheets regularly — data loss can be devastating.

Dormitory Task (Paper-and-Pencil)

On paper, create a multiplication table from 1x1 to 5x5. Mark which cells would need absolute references if you built this in a spreadsheet. Write the formula for cell B2 2x2 using absolute references.

Resources Needed:
- Computer with spreadsheet
- expense tracker template
- chart examples
- formula reference

Differentiation

Approaching: Use a pre-made template with AutoFill already demonstrated
On Level: Create the expense tracker with all features independently
Advanced: Add pivot table analysis and multiple chart types with formatting
SEND: Paper-based spreadsheet simulation, guided formula writing

Formative Assessment

Method: Exit ticket and self-assessment
CT Skill Assessed: Algorithm Design — planning the spreadsheet structure before building it

Teacher Reflection (Post-Lesson)

What worked well:

What to improve:

Connection to next lesson:

Lesson 3.09	Duration: 40 minutes

Aggregate Functions

Learning Objectives

- Apply COUNT, SUM, AVG, MAX, and MIN functions to calculate data summaries
- Use aliases with aggregate functions for meaningful result columns
- Combine aggregate functions with WHERE clauses for filtered calculations

Timing Breakdown

0-5 min	**Engage**	Show a sales report with 100 rows. Ask: 'How would you find the top 10 sales? Th...
5-15 min	**Explore**	Demonstrate AutoFill: enter 1, 2, select, drag to 20. Show number series, odd nu...
15-25 min	**Explain**	Define referencing types: relative changes when copied, absolute \$ sign, stay...
25-35 min	**Elaborate**	Task: 'Create a monthly expense tracker. Use AutoFill for dates, absolute refere...
35-40 min	**Evaluate**	Exit ticket: '1. What is the difference between relative and absolute referencin...

Engage (5-7 minutes)

Teacher Activity

Show a sales report with 100 rows. Ask: 'How would you find the top 10 sales? The average? The trend over time?' Demonstrate AutoFill and charts.

Student Activity

Consider the challenge. Recognize manual methods would be too slow. Express interest in automated solutions.

Explore (10-12 minutes)

Teacher Activity

Demonstrate AutoFill: enter 1, 2, select, drag to 20. Show number series, odd numbers, days, months. Then show relative vs absolute referencing with a multiplication table.

Student Activity

Follow along. Try AutoFill with different patterns. Create a multiplication table using absolute references.

Explain (10-12 minutes)

Teacher Activity

Define referencing types: relative changes when copied, absolute $ sign, stays fixed, mixed $A1 or A$1. Explain linking: cells within same sheet, between sheets, between workbooks. Show chart types: column, bar, line, pie.

Student Activity

Take notes. Understand when to use each referencing type. See how charts visualize data.

Elaborate (8-10 minutes)

Teacher Activity

Task: ‘Create a monthly expense tracker. Use AutoFill for dates, absolute reference for tax rate, link to a summary sheet, and create a pie chart of expense categories.’

Student Activity

Work independently. Apply all techniques. Create the complete tracker with charts.

Evaluate (5-8 minutes)

Teacher Activity

Exit ticket: ‘1. What is the difference between relative and absolute referencing? 2. When would you use a pie chart vs a bar chart? 3. How do you link cells between sheets?’

Student Activity

Answer the questions. Review their spreadsheet for completeness.

Cross-Curricular Connections

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Digital Citizen Reminder: Back up important spreadsheets regularly — data loss can be devastating.

Dormitory Task (Paper-and-Pencil)

On paper, create a multiplication table from 1x1 to 5x5. Mark which cells would need absolute references if you built this in a spreadsheet. Write the formula for cell B2 2x2 using absolute references.

Resources Needed:

- Computer with spreadsheet
- expense tracker template
- chart examples
- formula reference

Differentiation

Approaching: Use a pre-made template with AutoFill already demonstrated
On Level: Create the expense tracker with all features independently
Advanced: Add pivot table analysis and multiple chart types with formatting
SEND: Paper-based spreadsheet simulation, guided formula writing

Formative Assessment

Method: Exit ticket and self-assessment
CT Skill Assessed: Algorithm Design — planning the spreadsheet structure before building it

Teacher Reflection (Post-Lesson)

What worked well:

What to improve:

Connection to next lesson:

Lesson 3.10	Duration: 40 minutes

GROUP BY

Learning Objectives

- Use GROUP BY to categorize data into meaningful groups
- Apply HAVING to filter groups based on aggregate conditions
- Combine GROUP BY with aggregate functions for group-level summaries

Timing Breakdown

0-5 min	Engage	Show a sales report with 100 rows. Ask: 'How would you find the top 10 sales? Th...
5-15 min	**Explore**	Demonstrate AutoFill: enter 1, 2, select, drag to 20. Show number series, odd nu...
15-25 min	**Explain**	Define referencing types: relative changes when copied, absolute \$ sign, stay...
25-35 min	**Elaborate**	Task: 'Create a monthly expense tracker. Use AutoFill for dates, absolute refere...
35-40 min	**Evaluate**	Exit ticket: '1. What is the difference between relative and absolute referencin...

Engage (5-7 minutes)

Teacher Activity

Show a sales report with 100 rows. Ask: 'How would you find the top 10 sales? The average? The trend over time?' Demonstrate AutoFill and charts.

Student Activity

Consider the challenge. Recognize manual methods would be too slow. Express interest in automated solutions.

Explore (10-12 minutes)

Teacher Activity

Demonstrate AutoFill: enter 1, 2, select, drag to 20. Show number series, odd numbers, days, months. Then show relative vs absolute referencing with a multiplication table.

Student Activity

Follow along. Try AutoFill with different patterns. Create a multiplication table using absolute references.

Explain (10-12 minutes)

Teacher Activity

Define referencing types: relative changes when copied, absolute $ sign, stays fixed, mixed $A1 or A$1. Explain linking: cells within same sheet, between sheets, between workbooks. Show chart types: column, bar, line, pie.

Student Activity

Take notes. Understand when to use each referencing type. See how charts visualize data.

Elaborate (8-10 minutes)

Teacher Activity

Task: 'Create a monthly expense tracker. Use AutoFill for dates, absolute reference for tax rate, link to a summary sheet, and create a pie chart of expense categories.'

Student Activity

Work independently. Apply all techniques. Create the complete tracker with charts.

Evaluate (5-8 minutes)

Teacher Activity

Exit ticket: '1. What is the difference between relative and absolute referencing? 2. When would you use a pie chart vs a bar chart? 3. How do you link cells between sheets?'

Student Activity

Answer the questions. Review their spreadsheet for completeness.

Cross-Curricular Connections

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Dormitory Task (Paper-and-Pencil)

On paper, create a multiplication table from 1x1 to 5x5. Mark which cells would need absolute references if you built this in a spreadsheet. Write the formula for cell B2 2x2 using absolute references.

Resources Needed:
- Computer with spreadsheet
- expense tracker template
- chart examples
- formula reference

Differentiation

Approaching: Use a pre-made template with AutoFill already demonstrated
On Level: Create the expense tracker with all features independently
Advanced: Add pivot table analysis and multiple chart types with formatting
SEND: Paper-based spreadsheet simulation, guided formula writing

Formative Assessment

Method: Exit ticket and self-assessment
CT Skill Assessed: Algorithm Design — planning the spreadsheet structure before building it

Teacher Reflection (Post-Lesson)

What worked well:

What to improve:

Connection to next lesson:

Lesson 3.11	Duration: 40 minutes

Project - Library Database

Learning Objectives

- Design a relational database schema for a library management system
- Write SQL queries to perform all CRUD operations on the library data
- Generate reports using aggregate functions and GROUP BY

Timing Breakdown

0-5 min	Engage	Show a sales report with 100 rows. Ask: 'How would you find the top 10 sales? Th...
5-15 min	**Explore**	Demonstrate AutoFill: enter 1, 2, select, drag to 20. Show number series, odd nu...
15-25 min	**Explain**	Define referencing types: relative changes when copied, absolute \$ sign, stay...
25-35 min	**Elaborate**	Task: 'Create a monthly expense tracker. Use AutoFill for dates, absolute refere...
35-40 min	**Evaluate**	Exit ticket: '1. What is the difference between relative and absolute referencin...

Engage (5-7 minutes)

Teacher Activity

Show a sales report with 100 rows. Ask: 'How would you find the top 10 sales? The average? The trend over time?' Demonstrate AutoFill and charts.

Student Activity

Consider the challenge. Recognize manual methods would be too slow. Express interest in automated solutions.

Explore (10-12 minutes)

Teacher Activity

Demonstrate AutoFill: enter 1, 2, select, drag to 20. Show number series, odd numbers, days, months. Then show relative vs absolute referencing with a multiplication table.

Student Activity

Follow along. Try AutoFill with different patterns. Create a multiplication table using absolute references.

Explain (10-12 minutes)

Teacher Activity

Define referencing types: relative changes when copied, absolute $ sign, stays fixed, mixed $A1 or A$1. Explain linking: cells within same sheet, between sheets, between workbooks. Show chart types: column, bar, line, pie.

Student Activity

Take notes. Understand when to use each referencing type. See how charts visualize data.

Elaborate (8-10 minutes)

Teacher Activity

Task: 'Create a monthly expense tracker. Use AutoFill for dates, absolute reference for tax rate, link to a summary sheet, and create a pie chart of expense categories.'

Student Activity

Work independently. Apply all techniques. Create the complete tracker with charts.

Evaluate (5-8 minutes)

Teacher Activity

Exit ticket: '1. What is the difference between relative and absolute referencing? 2. When would you use a pie chart vs a bar chart? 3. How do you link cells between sheets?'

Student Activity

Answer the questions. Review their spreadsheet for completeness.

Cross-Curricular Connections

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Dormitory Task (Paper-and-Pencil)

On paper, create a multiplication table from 1x1 to 5x5. Mark which cells would need absolute references if you built this in a spreadsheet. Write the formula for cell B2 2x2 using absolute references.

Resources Needed:

- Computer with spreadsheet
- expense tracker template
- chart examples
- formula reference

Differentiation

Approaching: Use a pre-made template with AutoFill already demonstrated
On Level: Create the expense tracker with all features independently
Advanced: Add pivot table analysis and multiple chart types with formatting
SEND: Paper-based spreadsheet simulation, guided formula writing

Formative Assessment

Method: Exit ticket and self-assessment
CT Skill Assessed: Algorithm Design — planning the spreadsheet structure before building it

Teacher Reflection (Post-Lesson)

What worked well:

What to improve:

Connection to next lesson:

Lesson 3.12	Duration: 40 minutes

Chapter 3 Review

Learning Objectives

- Consolidate understanding of database concepts and SQL operations
- Demonstrate proficiency by solving 综合 SQL challenges
- Reflect on real-world applications of database management

Timing Breakdown

0-5 min	**Engage**	Show a sales report with 100 rows. Ask: 'How would you find the top 10 sales? Th...
5-15 min	**Explore**	Demonstrate AutoFill: enter 1, 2, select, drag to 20. Show number series, odd nu...
15-25 min	**Explain**	Define referencing types: relative changes when copied, absolute \$ sign, stay...
25-35 min	**Elaborate**	Task: 'Create a monthly expense tracker. Use AutoFill for dates, absolute refere...
35-40 min	**Evaluate**	Exit ticket: '1. What is the difference between relative and absolute referencin...

Engage (5-7 minutes)

Teacher Activity

Show a sales report with 100 rows. Ask: 'How would you find the top 10 sales? The average? The trend over time?' Demonstrate AutoFill and charts.

Student Activity

Consider the challenge. Recognize manual methods would be too slow. Express interest in automated solutions.

Explore (10-12 minutes)

Teacher Activity

Demonstrate AutoFill: enter 1, 2, select, drag to 20. Show number series, odd numbers, days, months. Then show relative vs absolute referencing with a multiplication table.

Student Activity

Follow along. Try AutoFill with different patterns. Create a multiplication table using absolute references.

Explain (10-12 minutes)

Teacher Activity

Define referencing types: relative changes when copied, absolute $ sign, stays fixed, mixed $A1 or A$1. Explain linking: cells within same sheet, between sheets, between workbooks. Show chart types: column, bar, line, pie.

Student Activity

Take notes. Understand when to use each referencing type. See how charts visualize data.

Elaborate (8-10 minutes)

Teacher Activity

Task: 'Create a monthly expense tracker. Use AutoFill for dates, absolute reference for tax rate, link to a summary sheet, and create a pie chart of expense categories.'

Student Activity

Work independently. Apply all techniques. Create the complete tracker with charts.

Evaluate (5-8 minutes)

Teacher Activity

Exit ticket: '1. What is the difference between relative and absolute referencing? 2. When would you use a pie chart vs a bar chart? 3. How do you link cells between sheets?'

Student Activity

Answer the questions. Review their spreadsheet for completeness.

Cross-Curricular Connections

Mathematics: Absolute referencing uses the \$ symbol like algebraic constants; charts use coordinate systems
Science: Scientists use spreadsheets for data analysis — graphs show experimental trends
Language: Presenting data in charts supports evidence-based arguments in reports

Digital Citizen Reminder: Back up important spreadsheets regularly — data loss can be devastating.

Dormitory Task (Paper-and-Pencil)

On paper, create a multiplication table from 1x1 to 5x5. Mark which cells would need absolute references if you built this in a spreadsheet. Write the formula for cell B2 2x2 using absolute references.

Resources Needed:
- Computer with spreadsheet
- expense tracker template
- chart examples
- formula reference

Differentiation

Approaching: Use a pre-made template with AutoFill already demonstrated
On Level: Create the expense tracker with all features independently
Advanced: Add pivot table analysis and multiple chart types with formatting
SEND: Paper-based spreadsheet simulation, guided formula writing

Formative Assessment

Method: Exit ticket and self-assessment
CT Skill Assessed: Algorithm Design — planning the spreadsheet structure before building it

Teacher Reflection (Post-Lesson)

What worked well:

What to improve:

Connection to next lesson:

Lesson 4.01	**Duration: 40 minutes**

How the Web Works

Learning Objectives

- Explain the client-server model and how web pages are delivered
- Describe the role of IP addresses, DNS, and HTTP in web communication
- Differentiate between static and dynamic web pages

Timing Breakdown

0-5 min	**Engage**	Show a Python program that prints 'Hello World'. Then show the same in Scratch. ...
5-15 min	**Explore**	Open Python IDLE. Type: print'Hello World'. Show variables: x = 10, printx. ...
15-25 min	**Explain**	Define Python basics: the print function, variables, the input function, data ty...
25-35 min	**Elaborate**	Task: 'Write a program that asks for two numbers and prints their sum, differenc...
35-40 min	**Evaluate**	Exit ticket: '1. What does print do? 2. What is the difference between = and =...

Engage (5-7 minutes)

Teacher Activity

Show a Python program that prints 'Hello World'. Then show the same in Scratch. Ask: 'What's different? Why might someone prefer text-based programming?' Discuss transition from blocks to text.

Student Activity

Compare both programs. Identify differences: syntax, typing, precision. Discuss advantages: Python runs everywhere, faster execution.

Explore (10-12 minutes)

Teacher Activity

Open Python IDLE. Type: print'Hello World'. Show variables: x = 10, printx. Show input: name = input'Name: ', print'Hello ' + name.

Student Activity

Follow along. Type each command. See the output. Try changing values and running again.

Explain (10-12 minutes)

Teacher Activity

Define Python basics: the print function, variables, the input function, data types including integer, float, string, and boolean. Show arithmetic operators. Show comparison operators. Explain indentation and colons for code blocks.

Student Activity

Take notes. Understand data types and operators. Practice writing simple expressions.

Elaborate (8-10 minutes)

Teacher Activity

Task: 'Write a program that asks for two numbers and prints their sum, difference, product, and quotient.' Show the expected output format.

Student Activity

Work independently. Write the program. Test with different inputs. Debug any errors.

Evaluate (5-8 minutes)

Teacher Activity

Exit ticket: '1. What does print do? 2. What is the difference between = and ==? 3. Why does Python need indentation?'

Student Activity

Answer the questions. Show their program output to a partner.

Cross-Curricular Connections

Mathematics: Programming uses algebraic operations, variables, and logical comparisons
Science: Algorithms model scientific processes — step-by-step experiments follow programming logic
Language: Python syntax follows English-like conventions — keywords, punctuation, structure

Digital Citizen Reminder: Code responsibly — do not create programs that harm others' computers or steal data.

Dormitory Task (Paper-and-Pencil)

On paper, write a Python program without computer that asks for your name and age, then prints: 'Hello [name], you are [age] years old.' Write the exact code with correct syntax.

Resources Needed:
- Computer with Python IDLE
- reference card
- practice worksheet
- error examples

Differentiation

Approaching: Complete a partially written program by filling in missing lines
On Level: Write the calculator program independently as described
Advanced: Add input validation, error handling, and a menu system
SEND: Paper-based code tracing, trace through given programs and predict output

Formative Assessment

Method: Exit ticket and self-assessment
CT Skill Assessed: Debugging — finding and fixing errors in Python syntax and logic

Teacher Reflection (Post-Lesson)

What worked well:

What to improve:

Connection to next lesson:

Lesson 4.02	Duration: 40 minutes

HTML Introduction

Learning Objectives

- Write a basic HTML document using proper structure and syntax
- Explain the purpose of DOCTYPE, html, head, and body tags
- Use a text editor and browser to create and preview web pages

Timing Breakdown

0-5 min	Engage	Show a Python program that prints 'Hello World'. Then show the same in Scratch. ...
5-15 min	**Explore**	Open Python IDLE. Type: print'Hello World'. Show variables: x = 10, printx. ...
15-25 min	**Explain**	Define Python basics: the print function, variables, the input function, data ty...
25-35 min	**Elaborate**	Task: 'Write a program that asks for two numbers and prints their sum, differenc...
35-40 min	**Evaluate**	Exit ticket: '1. What does print do? 2. What is the difference between = and =...

Engage (5-7 minutes)

Teacher Activity

Show a Python program that prints 'Hello World'. Then show the same in Scratch. Ask: 'What's different? Why might someone prefer text-based programming?' Discuss transition from blocks to text.

Student Activity

Compare both programs. Identify differences: syntax, typing, precision. Discuss advantages: Python runs everywhere, faster execution.

Explore (10-12 minutes)

Teacher Activity

Open Python IDLE. Type: print'Hello World'. Show variables: x = 10, printx. Show input: name = input'Name: ', print'Hello ' + name.

Student Activity

Follow along. Type each command. See the output. Try changing values and running again.

Explain (10-12 minutes)

Teacher Activity

Define Python basics: the print function, variables, the input function, data types including integer, float, string, and boolean. Show arithmetic operators. Show comparison operators. Explain indentation and colons for code blocks.

Student Activity

Take notes. Understand data types and operators. Practice writing simple expressions.

Elaborate (8-10 minutes)

Teacher Activity

Task: 'Write a program that asks for two numbers and prints their sum, difference, product, and quotient.' Show the expected output format.

Student Activity

Work independently. Write the program. Test with different inputs. Debug any errors.

Evaluate (5-8 minutes)

Teacher Activity

Exit ticket: '1. What does print do? 2. What is the difference between = and ==? 3. Why does Python need indentation?'

Student Activity

Answer the questions. Show their program output to a partner.

Cross-Curricular Connections

Mathematics: Programming uses algebraic operations, variables, and logical comparisons
Science: Algorithms model scientific processes — step-by-step experiments follow programming logic
Language: Python syntax follows English-like conventions — keywords, punctuation, structure

Digital Citizen Reminder: Code responsibly — do not create programs that harm others' computers or steal data.

Dormitory Task (Paper-and-Pencil)

On paper, write a Python program without computer that asks for your name and age, then prints: 'Hello [name], you are [age] years old.' Write the exact code with correct syntax.

Resources Needed:
- Computer with Python IDLE
- reference card
- practice worksheet
- error examples

Differentiation

Approaching: Complete a partially written program by filling in missing lines
On Level: Write the calculator program independently as described
Advanced: Add input validation, error handling, and a menu system
SEND: Paper-based code tracing, trace through given programs and predict output

Formative Assessment

Method: Exit ticket and self-assessment
CT Skill Assessed: Debugging — finding and fixing errors in Python syntax and logic

Teacher Reflection (Post-Lesson)

What worked well:

What to improve:

Connection to next lesson:

Lesson 4.03	Duration: 40 minutes

Text Elements

Learning Objectives

- Use heading, paragraph, and formatting tags to structure text content
- Apply semantic HTML tags to convey meaning and improve accessibility
- Create well-organized content using lists and block-level elements

Timing Breakdown

0-5 min	**Engage**	Show a Python program that prints 'Hello World'. Then show the same in Scratch. ...
5-15 min	**Explore**	Open Python IDLE. Type: print'Hello World'. Show variables: x = 10, printx. ...
15-25 min	**Explain**	Define Python basics: the print function, variables, the input function, data ty...
25-35 min	**Elaborate**	Task: 'Write a program that asks for two numbers and prints their sum, differenc...
35-40 min	**Evaluate**	Exit ticket: '1. What does print do? 2. What is the difference between = and =...

Engage (5-7 minutes)

Teacher Activity

Show a Python program that prints 'Hello World'. Then show the same in Scratch. Ask: 'What's different? Why might someone prefer text-based programming?' Discuss transition from blocks to text.

Student Activity

Compare both programs. Identify differences: syntax, typing, precision. Discuss advantages: Python runs everywhere, faster execution.

Explore (10-12 minutes)

Teacher Activity

Open Python IDLE. Type: print'Hello World'. Show variables: x = 10, printx. Show input: name = input'Name: ', print'Hello ' + name.

Student Activity

Follow along. Type each command. See the output. Try changing values and running again.

Explain (10-12 minutes)

Teacher Activity

Define Python basics: the print function, variables, the input function, data types including integer, float, string, and boolean. Show arithmetic operators. Show comparison operators. Explain indentation and colons for code blocks.

Student Activity

Take notes. Understand data types and operators. Practice writing simple expressions.

Elaborate (8-10 minutes)

Teacher Activity

Task: 'Write a program that asks for two numbers and prints their sum, difference, product, and quotient.' Show the expected output format.

Student Activity

Work independently. Write the program. Test with different inputs. Debug any errors.

Evaluate (5-8 minutes)

Teacher Activity

Exit ticket: '1. What does print do? 2. What is the difference between = and ==? 3. Why does Python need indentation?'

Student Activity

Answer the questions. Show their program output to a partner.

Cross-Curricular Connections

Mathematics: Programming uses algebraic operations, variables, and logical comparisons
Science: Algorithms model scientific processes — step-by-step experiments follow programming logic
Language: Python syntax follows English-like conventions — keywords, punctuation, structure

Digital Citizen Reminder: Code responsibly — do not create programs that harm others' computers or steal data.

Dormitory Task (Paper-and-Pencil)

On paper, write a Python program without computer that asks for your name and age, then prints: 'Hello [name], you are [age] years old.' Write the exact code with correct syntax.

Resources Needed:

- Computer with Python IDLE
- reference card
- practice worksheet
- error examples

Differentiation

Approaching: Complete a partially written program by filling in missing lines
On Level: Write the calculator program independently as described
Advanced: Add input validation, error handling, and a menu system
SEND: Paper-based code tracing, trace through given programs and predict output

Formative Assessment

Method: Exit ticket and self-assessment
CT Skill Assessed: Debugging — finding and fixing errors in Python syntax and logic

Teacher Reflection (Post-Lesson)

What worked well:

What to improve:

Connection to next lesson:

Lesson 4.04	Duration: 40 minutes

Links and Images

Learning Objectives

- Create hyperlinks using anchor tags with relative and absolute paths
- Embed images using img tags with proper attributes
- Use alt text and accessibility best practices for web media

Timing Breakdown

0-5 min	Engage	Show a Python program that prints 'Hello World'. Then show the same in Scratch. ...
5-15 min	Explore	Open Python IDLE. Type: print'Hello World'. Show variables: x = 10, printx. ...
15-25 min	Explain	Define Python basics: the print function, variables, the input function, data ty...
25-35 min	Elaborate	Task: 'Write a program that asks for two numbers and prints their sum, differenc...
35-40 min	Evaluate	Exit ticket: '1. What does print do? 2. What is the difference between = and =...

Engage (5-7 minutes)

Teacher Activity

Show a Python program that prints ‘Hello World’. Then show the same in Scratch. Ask: ‘What’s different? Why might someone prefer text-based programming?‘ Discuss transition from blocks to text.

Student Activity

Compare both programs. Identify differences: syntax, typing, precision. Discuss advantages: Python runs everywhere, faster execution.

Explore (10-12 minutes)

Teacher Activity

Open Python IDLE. Type: print’Hello World’. Show variables: x = 10, printx. Show input: name = input’Name: ‘, print’Hello ‘ + name.

Student Activity

Follow along. Type each command. See the output. Try changing values and running again.

Explain (10-12 minutes)

Teacher Activity

Define Python basics: the print function, variables, the input function, data types including integer, float, string, and boolean. Show arithmetic operators. Show comparison operators. Explain indentation and colons for code blocks.

Student Activity

Take notes. Understand data types and operators. Practice writing simple expressions.

Elaborate (8-10 minutes)

Teacher Activity

Task: ‘Write a program that asks for two numbers and prints their sum, difference, product, and quotient.’ Show the expected output format.

Student Activity

Work independently. Write the program. Test with different inputs. Debug any errors.

Evaluate (5-8 minutes)

Teacher Activity

Exit ticket: ‘1. What does print do? 2. What is the difference between = and ==? 3. Why does Python need indentation?’

Student Activity

Answer the questions. Show their program output to a partner.

Cross-Curricular Connections

Mathematics: Programming uses algebraic operations, variables, and logical comparisons
Science: Algorithms model scientific processes — step-by-step experiments follow programming logic
Language: Python syntax follows English-like conventions — keywords, punctuation, structure

Digital Citizen Reminder: Code responsibly — do not create programs that harm others' computers or steal data.

Dormitory Task (Paper-and-Pencil)

On paper, write a Python program without computer that asks for your name and age, then prints: ‘Hello [name], you are [age] years old.’ Write the exact code with correct syntax.

Resources Needed:
- Computer with Python IDLE
- reference card
- practice worksheet
- error examples

Differentiation

Approaching: Complete a partially written program by filling in missing lines
On Level: Write the calculator program independently as described
Advanced: Add input validation, error handling, and a menu system
SEND: Paper-based code tracing, trace through given programs and predict output

Formative Assessment

Method: Exit ticket and self-assessment
CT Skill Assessed: Debugging — finding and fixing errors in Python syntax and logic

Teacher Reflection (Post-Lesson)

What worked well:

What to improve:

Connection to next lesson:

Lesson 4.05	Duration: 40 minutes

Tables

Learning Objectives

- Create HTML tables using table, tr, th, and td elements
- Use colspan and rowspan to merge cells in complex tables
- Apply semantic table structure with thead, tbody, and tfoot

Timing Breakdown

0-5 min	Engage	Show a Python program that prints 'Hello World'. Then show the same in Scratch. ...
5-15 min	**Explore**	Open Python IDLE. Type: print'Hello World'. Show variables: x = 10, printx. ...
15-25 min	**Explain**	Define Python basics: the print function, variables, the input function, data ty...
25-35 min	**Elaborate**	Task: 'Write a program that asks for two numbers and prints their sum, differenc...
35-40 min	**Evaluate**	Exit ticket: '1. What does print do? 2. What is the difference between = and =...

Engage (5-7 minutes)

Teacher Activity

Show a Python program that prints ‘Hello World’. Then show the same in Scratch. Ask: ‘What’s different? Why might someone prefer text-based programming?‘ Discuss transition from blocks to text.

Student Activity

Compare both programs. Identify differences: syntax, typing, precision. Discuss advantages: Python runs everywhere, faster execution.

Explore (10-12 minutes)

Teacher Activity

Open Python IDLE. Type: print’Hello World’. Show variables: x = 10, printx. Show input: name = input’Name: ‘, print’Hello ‘ + name.

Student Activity

Follow along. Type each command. See the output. Try changing values and running again.

Explain (10-12 minutes)

Teacher Activity

Define Python basics: the print function, variables, the input function, data types including integer, float, string, and boolean. Show arithmetic operators. Show comparison operators. Explain indentation and colons for code blocks.

Student Activity

Take notes. Understand data types and operators. Practice writing simple expressions.

Elaborate (8-10 minutes)

Teacher Activity

Task: ‘Write a program that asks for two numbers and prints their sum, difference, product, and quotient.’ Show the expected output format.

Student Activity

Work independently. Write the program. Test with different inputs. Debug any errors.

Evaluate (5-8 minutes)

Teacher Activity

Exit ticket: ‘1. What does print do? 2. What is the difference between = and ==? 3. Why does Python need indentation?’

Student Activity

Answer the questions. Show their program output to a partner.

Cross-Curricular Connections

Mathematics: Programming uses algebraic operations, variables, and logical comparisons
Science: Algorithms model scientific processes — step-by-step experiments follow programming logic
Language: Python syntax follows English-like conventions — keywords, punctuation, structure

Digital Citizen Reminder: Code responsibly — do not create programs that harm others' computers or steal data.

Dormitory Task (Paper-and-Pencil)

On paper, write a Python program without computer that asks for your name and age, then prints: ‘Hello [name], you are [age] years old.’ Write the exact code with correct syntax.

Resources Needed:

- Computer with Python IDLE
- reference card
- practice worksheet
- error examples

Differentiation

Approaching: Complete a partially written program by filling in missing lines
On Level: Write the calculator program independently as described
Advanced: Add input validation, error handling, and a menu system
SEND: Paper-based code tracing, trace through given programs and predict output

Formative Assessment

Method: Exit ticket and self-assessment
CT Skill Assessed: Debugging — finding and fixing errors in Python syntax and logic

Teacher Reflection (Post-Lesson)

What worked well:

What to improve:

Connection to next lesson:

Lesson 4.06	Duration: 40 minutes

Forms

Learning Objectives

- Create HTML forms using the form element and action attribute
- Explain the purpose of form methods GET and POST
- Structure forms with labels, fieldsets, and legends for accessibility

Timing Breakdown

0-5 min	**Engage**	Show a Python program that prints 'Hello World'. Then show the same in Scratch. ...
5-15 min	**Explore**	Open Python IDLE. Type: print'Hello World'. Show variables: x = 10, printx. ...
15-25 min	**Explain**	Define Python basics: the print function, variables, the input function, data ty...
25-35 min	**Elaborate**	Task: 'Write a program that asks for two numbers and prints their sum, differenc...
35-40 min	**Evaluate**	Exit ticket: '1. What does print do? 2. What is the difference between = and =...

Engage (5-7 minutes)

Teacher Activity

Show a Python program that prints 'Hello World'. Then show the same in Scratch. Ask: 'What's different? Why might someone prefer text-based programming?' Discuss transition from blocks to text.

Student Activity

Compare both programs. Identify differences: syntax, typing, precision. Discuss advantages: Python runs everywhere, faster execution.

Explore (10-12 minutes)

Teacher Activity

Open Python IDLE. Type: print'Hello World'. Show variables: x = 10, printx. Show input: name = input'Name: ', print'Hello ' + name.

Student Activity

Follow along. Type each command. See the output. Try changing values and running again.

Explain (10-12 minutes)

Teacher Activity

Define Python basics: the print function, variables, the input function, data types including integer, float, string, and boolean. Show arithmetic operators. Show comparison operators. Explain indentation and colons for code blocks.

Student Activity

Take notes. Understand data types and operators. Practice writing simple expressions.

Elaborate (8-10 minutes)

Teacher Activity

Task: 'Write a program that asks for two numbers and prints their sum, difference, product, and quotient.' Show the expected output format.

Student Activity

Work independently. Write the program. Test with different inputs. Debug any errors.

Evaluate (5-8 minutes)

Teacher Activity

Exit ticket: '1. What does print do? 2. What is the difference between = and ==? 3. Why does Python need indentation?'

Student Activity

Answer the questions. Show their program output to a partner.

Cross-Curricular Connections

Mathematics: Programming uses algebraic operations, variables, and logical comparisons
Science: Algorithms model scientific processes — step-by-step experiments follow programming logic
Language: Python syntax follows English-like conventions — keywords, punctuation, structure

Digital Citizen Reminder: Code responsibly — do not create programs that harm others' computers or steal data.

Dormitory Task (Paper-and-Pencil)

On paper, write a Python program without computer that asks for your name and age, then prints: 'Hello [name], you are [age] years old.' Write the exact code with correct syntax.

Resources Needed:
- Computer with Python IDLE
- reference card
- practice worksheet
- error examples

Differentiation

Approaching: Complete a partially written program by filling in missing lines
On Level: Write the calculator program independently as described
Advanced: Add input validation, error handling, and a menu system
SEND: Paper-based code tracing, trace through given programs and predict output

Formative Assessment

Method: Exit ticket and self-assessment
CT Skill Assessed: Debugging — finding and fixing errors in Python syntax and logic

Teacher Reflection (Post-Lesson)

What worked well:

What to improve:

Connection to next lesson:

Lesson 4.07	Duration: 40 minutes

Form Elements

Learning Objectives

- Implement various input types for different data collection needs
- Use select dropdowns, radio buttons, and checkboxes appropriately
- Apply validation attributes to enforce data quality at the form level

Timing Breakdown

0-5 min	Engage	Show a Python program that prints 'Hello World'. Then show the same in Scratch. ...
5-15 min	**Explore**	Open Python IDLE. Type: print'Hello World'. Show variables: x = 10, printx. ...
15-25 min	**Explain**	Define Python basics: the print function, variables, the input function, data ty...
25-35 min	**Elaborate**	Task: 'Write a program that asks for two numbers and prints their sum, differenc...
35-40 min	**Evaluate**	Exit ticket: '1. What does print do? 2. What is the difference between = and =...

Engage (5-7 minutes)

Teacher Activity

Show a Python program that prints ‘Hello World’. Then show the same in Scratch. Ask: ‘What’s different? Why might someone prefer text-based programming?‘ Discuss transition from blocks to text.

Student Activity

Compare both programs. Identify differences: syntax, typing, precision. Discuss advantages: Python runs everywhere, faster execution.

Explore (10-12 minutes)

Teacher Activity

Open Python IDLE. Type: print’Hello World’. Show variables: x = 10, printx. Show input: name = input’Name: ‘, print’Hello ‘ + name.

Student Activity

Follow along. Type each command. See the output. Try changing values and running again.

Explain (10-12 minutes)

Teacher Activity

Define Python basics: the print function, variables, the input function, data types including integer, float, string, and boolean. Show arithmetic operators. Show comparison operators. Explain indentation and colons for code blocks.

Student Activity

Take notes. Understand data types and operators. Practice writing simple expressions.

Elaborate (8-10 minutes)

Teacher Activity

Task: 'Write a program that asks for two numbers and prints their sum, difference, product, and quotient.' Show the expected output format.

Student Activity

Work independently. Write the program. Test with different inputs. Debug any errors.

Evaluate (5-8 minutes)

Teacher Activity

Exit ticket: '1. What does print do? 2. What is the difference between = and ==? 3. Why does Python need indentation?'

Student Activity

Answer the questions. Show their program output to a partner.

Cross-Curricular Connections

Mathematics: Programming uses algebraic operations, variables, and logical comparisons
Science: Algorithms model scientific processes — step-by-step experiments follow programming logic
Language: Python syntax follows English-like conventions — keywords, punctuation, structure

Digital Citizen Reminder: Code responsibly — do not create programs that harm others' computers or steal data.

Dormitory Task (Paper-and-Pencil)

On paper, write a Python program without computer that asks for your name and age, then prints: 'Hello [name], you are [age] years old.' Write the exact code with correct syntax.

Resources Needed:
- Computer with Python IDLE
- reference card
- practice worksheet
- error examples

Differentiation

Approaching: Complete a partially written program by filling in missing lines
On Level: Write the calculator program independently as described
Advanced: Add input validation, error handling, and a menu system
SEND: Paper-based code tracing, trace through given programs and predict output

Formative Assessment

Method: Exit ticket and self-assessment
CT Skill Assessed: Debugging — finding and fixing errors in Python syntax and logic

Teacher Reflection (Post-Lesson)

What worked well:

What to improve:

Connection to next lesson:

Lesson 4.08	Duration: 40 minutes

CSS Introduction

Learning Objectives

- Explain the role of CSS in separating content from presentation
- Apply inline, internal, and external CSS to style HTML elements
- Link an external stylesheet to an HTML document

Timing Breakdown

0-5 min	Engage	Show a Python program that prints 'Hello World'. Then show the same in Scratch. ...
5-15 min	**Explore**	Open Python IDLE. Type: print'Hello World'. Show variables: x = 10, printx. ...
15-25 min	**Explain**	Define Python basics: the print function, variables, the input function, data ty...
25-35 min	**Elaborate**	Task: 'Write a program that asks for two numbers and prints their sum, differenc...
35-40 min	**Evaluate**	Exit ticket: '1. What does print do? 2. What is the difference between = and =...

Engage (5-7 minutes)

Teacher Activity

Show a Python program that prints 'Hello World'. Then show the same in Scratch. Ask: 'What's different? Why might someone prefer text-based programming?' Discuss transition from blocks to text.

Student Activity

Compare both programs. Identify differences: syntax, typing, precision. Discuss advantages: Python runs everywhere, faster execution.

Explore (10-12 minutes)

Teacher Activity

Open Python IDLE. Type: print'Hello World'. Show variables: x = 10, printx. Show input: name = input'Name: ', print'Hello ' + name.

Student Activity

Follow along. Type each command. See the output. Try changing values and running again.

Explain (10-12 minutes)

Teacher Activity

Define Python basics: the print function, variables, the input function, data types including integer, float, string, and boolean. Show arithmetic operators. Show comparison operators. Explain indentation and colons for code blocks.

Student Activity

Take notes. Understand data types and operators. Practice writing simple expressions.

Elaborate (8-10 minutes)

Teacher Activity

Task: 'Write a program that asks for two numbers and prints their sum, difference, product, and quotient.' Show the expected output format.

Student Activity

Work independently. Write the program. Test with different inputs. Debug any errors.

Evaluate (5-8 minutes)

Teacher Activity

Exit ticket: '1. What does print do? 2. What is the difference between = and ==? 3. Why does Python need indentation?'

Student Activity

Answer the questions. Show their program output to a partner.

Cross-Curricular Connections

Mathematics: Programming uses algebraic operations, variables, and logical comparisons
Science: Algorithms model scientific processes — step-by-step experiments follow programming logic
Language: Python syntax follows English-like conventions — keywords, punctuation, structure

Digital Citizen Reminder: Code responsibly — do not create programs that harm others' computers or steal data.

Dormitory Task (Paper-and-Pencil)

On paper, write a Python program without computer that asks for your name and age, then prints: 'Hello [name], you are [age] years old.' Write the exact code with correct syntax.

Resources Needed:
- Computer with Python IDLE
- reference card
- practice worksheet
- error examples

Differentiation

Approaching: Complete a partially written program by filling in missing lines
On Level: Write the calculator program independently as described
Advanced: Add input validation, error handling, and a menu system
SEND: Paper-based code tracing, trace through given programs and predict output

Formative Assessment

Method: Exit ticket and self-assessment
CT Skill Assessed: Debugging — finding and fixing errors in Python syntax and logic

Teacher Reflection (Post-Lesson)

What worked well:

What to improve:

Connection to next lesson:

Lesson 4.09	Duration: 40 minutes

CSS Selectors

Learning Objectives

- Use element, class, and ID selectors to target HTML elements
- Apply attribute and pseudo-class selectors for advanced styling
- Understand selector specificity and how CSS rules are prioritized

Timing Breakdown

0-5 min	Engage	Show a Python program that prints 'Hello World'. Then show the same in Scratch. ...
5-15 min	Explore	Open Python IDLE. Type: print'Hello World'. Show variables: x = 10, printx. ...
15-25 min	Explain	Define Python basics: the print function, variables, the input function, data ty...
25-35 min	Elaborate	Task: 'Write a program that asks for two numbers and prints their sum, differenc...
35-40 min	Evaluate	Exit ticket: '1. What does print do? 2. What is the difference between = and =...

Engage (5-7 minutes)

Teacher Activity

Show a Python program that prints ‘Hello World’. Then show the same in Scratch. Ask: ‘What’s different? Why might someone prefer text-based programming?‘ Discuss transition from blocks to text.

Student Activity

Compare both programs. Identify differences: syntax, typing, precision. Discuss advantages: Python runs everywhere, faster execution.

Explore (10-12 minutes)

Teacher Activity

Open Python IDLE. Type: print’Hello World’. Show variables: x = 10, printx. Show input: name = input’Name: ‘, print’Hello ‘ + name.

Student Activity

Follow along. Type each command. See the output. Try changing values and running again.

Explain (10-12 minutes)

Teacher Activity

Define Python basics: the print function, variables, the input function, data types including integer, float, string, and boolean. Show arithmetic operators. Show comparison operators. Explain indentation and colons for code blocks.

Student Activity

Take notes. Understand data types and operators. Practice writing simple expressions.

Elaborate (8-10 minutes)

Teacher Activity

Task: 'Write a program that asks for two numbers and prints their sum, difference, product, and quotient.' Show the expected output format.

Student Activity

Work independently. Write the program. Test with different inputs. Debug any errors.

Evaluate (5-8 minutes)

Teacher Activity

Exit ticket: '1. What does print do? 2. What is the difference between = and ==? 3. Why does Python need indentation?'

Student Activity

Answer the questions. Show their program output to a partner.

Cross-Curricular Connections

Mathematics: Programming uses algebraic operations, variables, and logical comparisons
Science: Algorithms model scientific processes — step-by-step experiments follow programming logic
Language: Python syntax follows English-like conventions — keywords, punctuation, structure

Digital Citizen Reminder: Code responsibly — do not create programs that harm others' computers or steal data.

Dormitory Task (Paper-and-Pencil)

On paper, write a Python program without computer that asks for your name and age, then prints: 'Hello [name], you are [age] years old.' Write the exact code with correct syntax.

Resources Needed:

- Computer with Python IDLE
- reference card
- practice worksheet
- error examples

Differentiation

Approaching: Complete a partially written program by filling in missing lines
On Level: Write the calculator program independently as described
Advanced: Add input validation, error handling, and a menu system
SEND: Paper-based code tracing, trace through given programs and predict output

Formative Assessment

Method: Exit ticket and self-assessment
CT Skill Assessed: Debugging — finding and fixing errors in Python syntax and logic

Teacher Reflection (Post-Lesson)

What worked well:

What to improve:

Connection to next lesson:

Lesson 4.10	Duration: 40 minutes

CSS Properties

Learning Objectives

- Apply text, color, and background properties to style web content
- Use the box model to control spacing and layout of elements
- Implement basic positioning and display properties for page layout

Timing Breakdown

0-5 min	**Engage**	Show a Python program that prints 'Hello World'. Then show the same in Scratch. ...
5-15 min	**Explore**	Open Python IDLE. Type: print'Hello World'. Show variables: x = 10, printx. ...
15-25 min	**Explain**	Define Python basics: the print function, variables, the input function, data ty...
25-35 min	**Elaborate**	Task: 'Write a program that asks for two numbers and prints their sum, differenc...
35-40 min	**Evaluate**	Exit ticket: '1. What does print do? 2. What is the difference between = and =...

Engage (5-7 minutes)

Teacher Activity

Show a Python program that prints ‘Hello World’. Then show the same in Scratch. Ask: ‘What’s different? Why might someone prefer text-based programming?‘ Discuss transition from blocks to text.

Student Activity

Compare both programs. Identify differences: syntax, typing, precision. Discuss advantages: Python runs everywhere, faster execution.

Explore (10-12 minutes)

Teacher Activity

Open Python IDLE. Type: print’Hello World’. Show variables: x = 10, printx. Show input: name = input’Name: ‘, print’Hello ‘ + name.

Student Activity

Follow along. Type each command. See the output. Try changing values and running again.

Explain (10-12 minutes)

Teacher Activity

Define Python basics: the print function, variables, the input function, data types including integer, float, string, and boolean. Show arithmetic operators. Show comparison operators. Explain indentation and colons for code blocks.

Student Activity

Take notes. Understand data types and operators. Practice writing simple expressions.

Elaborate (8-10 minutes)

Teacher Activity

Task: 'Write a program that asks for two numbers and prints their sum, difference, product, and quotient.' Show the expected output format.

Student Activity

Work independently. Write the program. Test with different inputs. Debug any errors.

Evaluate (5-8 minutes)

Teacher Activity

Exit ticket: '1. What does print do? 2. What is the difference between = and ==? 3. Why does Python need indentation?'

Student Activity

Answer the questions. Show their program output to a partner.

Cross-Curricular Connections

Mathematics: Programming uses algebraic operations, variables, and logical comparisons
Science: Algorithms model scientific processes — step-by-step experiments follow programming logic
Language: Python syntax follows English-like conventions — keywords, punctuation, structure

Digital Citizen Reminder: Code responsibly — do not create programs that harm others' computers or steal data.

Dormitory Task (Paper-and-Pencil)

On paper, write a Python program without computer that asks for your name and age, then prints: 'Hello [name], you are [age] years old.' Write the exact code with correct syntax.

Resources Needed:

- Computer with Python IDLE
- reference card
- practice worksheet
- error examples

Differentiation

Approaching: Complete a partially written program by filling in missing lines
On Level: Write the calculator program independently as described
Advanced: Add input validation, error handling, and a menu system
SEND: Paper-based code tracing, trace through given programs and predict output

Formative Assessment

Method: Exit ticket and self-assessment
CT Skill Assessed: Debugging — finding and fixing errors in Python syntax and logic

Teacher Reflection (Post-Lesson)

What worked well:

What to improve:

Connection to next lesson:

Lesson 4.11	Duration: 40 minutes

Project - Personal Webpage

Learning Objectives

- Build a complete multi-section personal webpage using HTML and CSS
- Demonstrate proper use of semantic HTML elements and CSS selectors
- Apply responsive design principles for mobile and desktop viewing

Timing Breakdown

0-5 min	Engage	Show a Python program that prints 'Hello World'. Then show the same in Scratch. ...
5-15 min	**Explore**	Open Python IDLE. Type: print'Hello World'. Show variables: x = 10, printx. ...
15-25 min	**Explain**	Define Python basics: the print function, variables, the input function, data ty...
25-35 min	**Elaborate**	Task: 'Write a program that asks for two numbers and prints their sum, differenc...
35-40 min	**Evaluate**	Exit ticket: '1. What does print do? 2. What is the difference between = and =...

Engage (5-7 minutes)

Teacher Activity

Show a Python program that prints 'Hello World'. Then show the same in Scratch. Ask: 'What's different? Why might someone prefer text-based programming?' Discuss transition from blocks to text.

Student Activity

Compare both programs. Identify differences: syntax, typing, precision. Discuss advantages: Python runs everywhere, faster execution.

Explore (10-12 minutes)

Teacher Activity

Open Python IDLE. Type: print'Hello World'. Show variables: x = 10, printx. Show input: name = input'Name: ', print'Hello ' + name.

Student Activity

Follow along. Type each command. See the output. Try changing values and running again.

Explain (10-12 minutes)

Teacher Activity

Define Python basics: the print function, variables, the input function, data types including integer, float, string, and boolean. Show arithmetic operators. Show comparison operators. Explain indentation and colons for code blocks.

Student Activity

Take notes. Understand data types and operators. Practice writing simple expressions.

Elaborate (8-10 minutes)

Teacher Activity

Task: ‘Write a program that asks for two numbers and prints their sum, difference, product, and quotient.’ Show the expected output format.

Student Activity

Work independently. Write the program. Test with different inputs. Debug any errors.

Evaluate (5-8 minutes)

Teacher Activity

Exit ticket: ‘1. What does print do? 2. What is the difference between = and ==? 3. Why does Python need indentation?’

Student Activity

Answer the questions. Show their program output to a partner.

Cross-Curricular Connections

Mathematics: Programming uses algebraic operations, variables, and logical comparisons
Science: Algorithms model scientific processes — step-by-step experiments follow programming logic
Language: Python syntax follows English-like conventions — keywords, punctuation, structure

Digital Citizen Reminder: Code responsibly — do not create programs that harm others' computers or steal data.

Dormitory Task (Paper-and-Pencil)

On paper, write a Python program without computer that asks for your name and age, then prints: ‘Hello [name], you are [age] years old.’ Write the exact code with correct syntax.

Resources Needed:

- Computer with Python IDLE
- reference card
- practice worksheet
- error examples

Differentiation

Approaching: Complete a partially written program by filling in missing lines
On Level: Write the calculator program independently as described
Advanced: Add input validation, error handling, and a menu system
SEND: Paper-based code tracing, trace through given programs and predict output

Formative Assessment

Method: Exit ticket and self-assessment
CT Skill Assessed: Debugging — finding and fixing errors in Python syntax and logic

Teacher Reflection (Post-Lesson)

What worked well:

What to improve:

Connection to next lesson:

Lesson 4.12	**Duration: 40 minutes**

Chapter 4 Review

Learning Objectives

- Consolidate knowledge of HTML structure, elements, and CSS styling
- Solve 综合 web development challenges combining HTML and CSS
- Evaluate 网页 quality based on accessibility, semantics, and design principles

Timing Breakdown

0-5 min	**Engage**	Show a Python program that prints 'Hello World'. Then show the same in Scratch. ...
5-15 min	**Explore**	Open Python IDLE. Type: print'Hello World'. Show variables: x = 10, printx. ...
15-25 min	**Explain**	Define Python basics: the print function, variables, the input function, data ty...
25-35 min	**Elaborate**	Task: 'Write a program that asks for two numbers and prints their sum, differenc...
35-40 min	**Evaluate**	Exit ticket: '1. What does print do? 2. What is the difference between = and =...

Engage (5-7 minutes)

Teacher Activity

Show a Python program that prints ‘Hello World’. Then show the same in Scratch. Ask: ‘What’s different? Why might someone prefer text-based programming?‘ Discuss transition from blocks to text.

Student Activity

Compare both programs. Identify differences: syntax, typing, precision. Discuss advantages: Python runs everywhere, faster execution.

Explore (10-12 minutes)

Teacher Activity

Open Python IDLE. Type: print’Hello World’. Show variables: x = 10, printx. Show input: name = input’Name: ‘, print’Hello ‘ + name.

Student Activity

Follow along. Type each command. See the output. Try changing values and running again.

Explain (10-12 minutes)

Teacher Activity

Define Python basics: the print function, variables, the input function, data types including integer, float, string, and boolean. Show arithmetic operators. Show comparison operators. Explain indentation and colons for code blocks.

Student Activity

Take notes. Understand data types and operators. Practice writing simple expressions.

Elaborate (8-10 minutes)

Teacher Activity

Task: ‘Write a program that asks for two numbers and prints their sum, difference, product, and quotient.’ Show the expected output format.

Student Activity

Work independently. Write the program. Test with different inputs. Debug any errors.

Evaluate (5-8 minutes)

Teacher Activity

Exit ticket: ‘1. What does print do? 2. What is the difference between = and ==? 3. Why does Python need indentation?’

Student Activity

Answer the questions. Show their program output to a partner.

Cross-Curricular Connections

Mathematics: Programming uses algebraic operations, variables, and logical comparisons
Science: Algorithms model scientific processes — step-by-step experiments follow programming logic
Language: Python syntax follows English-like conventions — keywords, punctuation, structure

Digital Citizen Reminder: Code responsibly — do not create programs that harm others' computers or steal data.

Dormitory Task (Paper-and-Pencil)

On paper, write a Python program without computer that asks for your name and age, then prints: ‘Hello [name], you are [age] years old.’ Write the exact code with correct syntax.

Resources Needed:

- Computer with Python IDLE
- reference card
- practice worksheet
- error examples

Differentiation

Approaching: Complete a partially written program by filling in missing lines
On Level: Write the calculator program independently as described
Advanced: Add input validation, error handling, and a menu system
SEND: Paper-based code tracing, trace through given programs and predict output

Formative Assessment

Method: Exit ticket and self-assessment
CT Skill Assessed: Debugging — finding and fixing errors in Python syntax and logic

Teacher Reflection (Post-Lesson)

What worked well:

What to improve:

Connection to next lesson:

Lesson 5.01 **Duration: 40 minutes**

What is an Algorithm?

Learning Objectives

- Define algorithms and explain their role in computational thinking
- Identify the key characteristics of a well-formed algorithm
- Analyze real-world processes as sequences of algorithmic steps

Timing Breakdown

0-5 min	**Engage**	Show a 3D-printed object or image. Ask: 'How do you think this was designed?' ...
5-15 min	**Explore**	Open Tinkercad. Show navigation: orbit, pan, zoom. Demonstrate: place a shape, r...
15-25 min	**Explain**	Define 3D design concepts: workplane, shapes, grouping, alignment. Explain CSG ...
25-35 min	**Elaborate**	Task: 'Design a simple house using basic shapes: box for walls, triangle for roo...
35-40 min	**Evaluate**	Exit ticket: '1. What is CSG? 2. How do you group shapes in Tinkercad? 3. Name t...

Engage (5-7 minutes)

Teacher Activity

Show a 3D-printed object or image. Ask: 'How do you think this was designed?' Introduce Tinkercad as a free 3D design tool. Show the interface.

Student Activity

Examine the object. Guess the design process. Express interest in creating 3D models.

Explore (10-12 minutes)

Teacher Activity

Open Tinkercad. Show navigation: orbit, pan, zoom. Demonstrate: place a shape, resize, move, group with another shape.

Student Activity

Follow along. Place shapes, resize them, move them around. Group two shapes to create a new object.

Explain (10-12 minutes)

Teacher Activity

Define 3D design concepts: workplane, shapes, grouping, alignment. Explain CSG Constructive Solid Geometry: combining basic shapes to create complex objects. Show the Circuit workspace for electronics.

Student Activity

Take notes. Understand how basic shapes combine. See the Circuit workspace for Arduino projects.

Elaborate (8-10 minutes)

Teacher Activity

Task: 'Design a simple house using basic shapes: box for walls, triangle for roof, cylinder for chimney. Group and align all parts.'

Student Activity

Work independently. Build the house step by step. Apply grouping and alignment. Share screenshot.

Evaluate (5-8 minutes)

Teacher Activity

Exit ticket: '1. What is CSG? 2. How do you group shapes in Tinkercad? 3. Name two workspaces in Tinkercad.'

Student Activity

Answer the questions. Show their 3D model to a partner.

Cross-Curricular Connections

Mathematics: 3D design uses geometry: shapes, dimensions, coordinates, alignment
Science: Engineering design follows the design process: plan, build, test, improve
Language: Technical documentation requires precise measurements and clear instructions

Digital Citizen Reminder: Tinkercad designs can be shared — credit original creators when remixing others' work.

Dormitory Task (Paper-and-Pencil)

On graph paper, draw a floor plan of your ideal room. Label each shape you would use in Tinkercad: box walls, cylinder table, sphere lamp. Write dimensions for each.

Resources Needed:

- Computer with Tinkercad account
- shape library reference
- graph paper
- pencil

Differentiation

Approaching: Follow a step-by-step tutorial to build a pre-designed object
On Level: Design a house independently using basic shapes
Advanced: Design a complex object with multiple grouped parts, add circuit components
SEND: Paper-based shape building, arrange pre-cut shapes to create objects

Formative Assessment

Method: Exit ticket and self-assessment
CT Skill Assessed: Decomposition — breaking a complex 3D object into basic shapes

Teacher Reflection (Post-Lesson)

What worked well:

What to improve:

Connection to next lesson:

Lesson 5.02	Duration: 40 minutes

Flowcharts

Learning Objectives

- Interpret and construct flowcharts using standard symbols
- Map algorithmic processes to flowchart representations
- Evaluate when flowcharts are more effective than pseudocode for communication

Timing Breakdown

0-5 min	Engage	Show a 3D-printed object or image. Ask: 'How do you think this was designed?' ...
5-15 min	Explore	Open Tinkercad. Show navigation: orbit, pan, zoom. Demonstrate: place a shape, r...
15-25 min	Explain	Define 3D design concepts: workplane, shapes, grouping, alignment. Explain CSG ...
25-35 min	Elaborate	Task: 'Design a simple house using basic shapes: box for walls, triangle for roo...
35-40 min	Evaluate	Exit ticket: '1. What is CSG? 2. How do you group shapes in Tinkercad? 3. Name t...

Engage (5-7 minutes)

Teacher Activity

Show a 3D-printed object or image. Ask: 'How do you think this was designed?' Introduce Tinkercad as a free 3D design tool. Show the interface.

Student Activity

Examine the object. Guess the design process. Express interest in creating 3D models.

Explore (10-12 minutes)

Teacher Activity

Open Tinkercad. Show navigation: orbit, pan, zoom. Demonstrate: place a shape, resize, move, group with another shape.

Student Activity

Follow along. Place shapes, resize them, move them around. Group two shapes to create a new object.

Explain (10-12 minutes)

Teacher Activity

Define 3D design concepts: workplane, shapes, grouping, alignment. Explain CSG Constructive Solid Geometry: combining basic shapes to create complex objects. Show the Circuit workspace for electronics.

Student Activity

Take notes. Understand how basic shapes combine. See the Circuit workspace for Arduino projects.

Elaborate (8-10 minutes)

Teacher Activity

Task: ‘Design a simple house using basic shapes: box for walls, triangle for roof, cylinder for chimney. Group and align all parts.’

Student Activity

Work independently. Build the house step by step. Apply grouping and alignment. Share screenshot.

Evaluate (5-8 minutes)

Teacher Activity

Exit ticket: ‘1. What is CSG? 2. How do you group shapes in Tinkercad? 3. Name two workspaces in Tinkercad.’

Student Activity

Answer the questions. Show their 3D model to a partner.

Cross-Curricular Connections

Mathematics: 3D design uses geometry: shapes, dimensions, coordinates, alignment
Science: Engineering design follows the design process: plan, build, test, improve
Language: Technical documentation requires precise measurements and clear instructions

Digital Citizen Reminder: Tinkercad designs can be shared — credit original creators when remixing others' work.

Dormitory Task (Paper-and-Pencil)

On graph paper, draw a floor plan of your ideal room. Label each shape you would use in Tinkercad: box walls, cylinder table, sphere lamp. Write dimensions for each.

Resources Needed:
- Computer with Tinkercad account
- shape library reference
- graph paper
- pencil

Differentiation

Approaching: Follow a step-by-step tutorial to build a pre-designed object
On Level: Design a house independently using basic shapes
Advanced: Design a complex object with multiple grouped parts, add circuit components
SEND: Paper-based shape building, arrange pre-cut shapes to create objects

Formative Assessment

Method: Exit ticket and self-assessment
CT Skill Assessed: Decomposition — breaking a complex 3D object into basic shapes

Teacher Reflection (Post-Lesson)

What worked well:

What to improve:

Connection to next lesson:

Lesson 5.03	Duration: 40 minutes

Pseudocode

Learning Objectives

- Write clear pseudocode to represent algorithms
- Translate between pseudocode, flowcharts, and natural language

Timing Breakdown

0-5 min	Engage	Show a 3D-printed object or image. Ask: 'How do you think this was designed?' ...
5-15 min	Explore	Open Tinkercad. Show navigation: orbit, pan, zoom. Demonstrate: place a shape, r...
15-25 min	Explain	Define 3D design concepts: workplane, shapes, grouping, alignment. Explain CSG ...
25-35 min	Elaborate	Task: 'Design a simple house using basic shapes: box for walls, triangle for roo...
35-40 min	Evaluate	Exit ticket: '1. What is CSG? 2. How do you group shapes in Tinkercad? 3. Name t...

Engage (5-7 minutes)

Teacher Activity

Show a 3D-printed object or image. Ask: 'How do you think this was designed?' Introduce Tinkercad as a free 3D design tool. Show the interface.

Student Activity

Examine the object. Guess the design process. Express interest in creating 3D models.

Explore (10-12 minutes)

Teacher Activity

Open Tinkercad. Show navigation: orbit, pan, zoom. Demonstrate: place a shape, resize, move, group with another shape.

Student Activity

Follow along. Place shapes, resize them, move them around. Group two shapes to create a new object.

Explain (10-12 minutes)

Teacher Activity

Define 3D design concepts: workplane, shapes, grouping, alignment. Explain CSG Constructive Solid Geometry: combining basic shapes to create complex objects. Show the Circuit workspace for electronics.

Student Activity

Take notes. Understand how basic shapes combine. See the Circuit workspace for Arduino projects.

Elaborate (8-10 minutes)

Teacher Activity

Task: 'Design a simple house using basic shapes: box for walls, triangle for roof, cylinder for chimney. Group and align all parts.'

Student Activity

Work independently. Build the house step by step. Apply grouping and alignment. Share screenshot.

Evaluate (5-8 minutes)

Teacher Activity

Exit ticket: '1. What is CSG? 2. How do you group shapes in Tinkercad? 3. Name two workspaces in Tinkercad.'

Student Activity

Answer the questions. Show their 3D model to a partner.

Cross-Curricular Connections

Mathematics: 3D design uses geometry: shapes, dimensions, coordinates, alignment
Science: Engineering design follows the design process: plan, build, test, improve
Language: Technical documentation requires precise measurements and clear instructions

Digital Citizen Reminder: Tinkercad designs can be shared — credit original creators when remixing others' work.

Dormitory Task (Paper-and-Pencil)

On graph paper, draw a floor plan of your ideal room. Label each shape you would use in Tinkercad: box walls, cylinder table, sphere lamp. Write dimensions for each.

Resources Needed:
- Computer with Tinkercad account
- shape library reference
- graph paper
- pencil

Differentiation

Approaching: Follow a step-by-step tutorial to build a pre-designed object
On Level: Design a house independently using basic shapes
Advanced: Design a complex object with multiple grouped parts, add circuit components
SEND: Paper-based shape building, arrange pre-cut shapes to create objects

Formative Assessment

Method: Exit ticket and self-assessment
CT Skill Assessed: Decomposition — breaking a complex 3D object into basic shapes

Teacher Reflection (Post-Lesson)

What worked well:

What to improve:

Connection to next lesson:

Lesson 5.04	Duration: 40 minutes

Sequential Algorithms

Learning Objectives

- Identify and construct sequential algorithms in real contexts
- Understand how step order affects algorithm outcomes
- Design sequential solutions for multi-step computational problems

Timing Breakdown

0-5 min	**Engage**	Show a 3D-printed object or image. Ask: 'How do you think this was designed?' ...
5-15 min	**Explore**	Open Tinkercad. Show navigation: orbit, pan, zoom. Demonstrate: place a shape, r...
15-25 min	**Explain**	Define 3D design concepts: workplane, shapes, grouping, alignment. Explain CSG ...
25-35 min	**Elaborate**	Task: 'Design a simple house using basic shapes: box for walls, triangle for roo...
35-40 min	**Evaluate**	Exit ticket: '1. What is CSG? 2. How do you group shapes in Tinkercad? 3. Name t...

Engage (5-7 minutes)

Teacher Activity

Show a 3D-printed object or image. Ask: 'How do you think this was designed?' Introduce Tinkercad as a free 3D design tool. Show the interface.

Student Activity

Examine the object. Guess the design process. Express interest in creating 3D models.

Explore (10-12 minutes)

Teacher Activity

Open Tinkercad. Show navigation: orbit, pan, zoom. Demonstrate: place a shape, resize, move, group with another shape.

Student Activity

Follow along. Place shapes, resize them, move them around. Group two shapes to create a new object.

Explain (10-12 minutes)

Teacher Activity

Define 3D design concepts: workplane, shapes, grouping, alignment. Explain CSG Constructive Solid Geometry: combining basic shapes to create complex objects. Show the Circuit workspace for electronics.

Student Activity

Take notes. Understand how basic shapes combine. See the Circuit workspace for Arduino projects.

Elaborate (8-10 minutes)

Teacher Activity

Task: 'Design a simple house using basic shapes: box for walls, triangle for roof, cylinder for chimney. Group and align all parts.'

Student Activity

Work independently. Build the house step by step. Apply grouping and alignment. Share screenshot.

Evaluate (5-8 minutes)

Teacher Activity

Exit ticket: '1. What is CSG? 2. How do you group shapes in Tinkercad? 3. Name two workspaces in Tinkercad.'

Student Activity

Answer the questions. Show their 3D model to a partner.

Cross-Curricular Connections

Mathematics: 3D design uses geometry: shapes, dimensions, coordinates, alignment
Science: Engineering design follows the design process: plan, build, test, improve
Language: Technical documentation requires precise measurements and clear instructions

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Dormitory Task (Paper-and-Pencil)

On graph paper, draw a floor plan of your ideal room. Label each shape you would use in Tinkercad: box walls, cylinder table, sphere lamp. Write dimensions for each.

Resources Needed:
- Computer with Tinkercad account
- shape library reference
- graph paper
- pencil

Differentiation

Approaching: Follow a step-by-step tutorial to build a pre-designed object
On Level: Design a house independently using basic shapes
Advanced: Design a complex object with multiple grouped parts, add circuit components
SEND: Paper-based shape building, arrange pre-cut shapes to create objects

Formative Assessment

Method: Exit ticket and self-assessment
CT Skill Assessed: Decomposition — breaking a complex 3D object into basic shapes

Teacher Reflection (Post-Lesson)

What worked well:

What to improve:

Connection to next lesson:

Lesson 5.05	Duration: 40 minutes

Selection Algorithms

Learning Objectives

Timing Breakdown

0-5 min	Engage	Show a 3D-printed object or image. Ask: 'How do you think this was designed?' ...
5-15 min	Explore	Open Tinkercad. Show navigation: orbit, pan, zoom. Demonstrate: place a shape, r...
15-25 min	Explain	Define 3D design concepts: workplane, shapes, grouping, alignment. Explain CSG ...
25-35 min	Elaborate	Task: 'Design a simple house using basic shapes: box for walls, triangle for roo...
35-40 min	Evaluate	Exit ticket: '1. What is CSG? 2. How do you group shapes in Tinkercad? 3. Name t...

Engage (5-7 minutes)

Teacher Activity

Show a 3D-printed object or image. Ask: 'How do you think this was designed?' Introduce Tinkercad as a free 3D design tool. Show the interface.

Student Activity

Examine the object. Guess the design process. Express interest in creating 3D models.

Explore (10-12 minutes)

Teacher Activity

Open Tinkercad. Show navigation: orbit, pan, zoom. Demonstrate: place a shape, resize, move, group with another shape.

Student Activity

Follow along. Place shapes, resize them, move them around. Group two shapes to create a new object.

Explain (10-12 minutes)

Teacher Activity

Define 3D design concepts: workplane, shapes, grouping, alignment. Explain CSG Constructive Solid Geometry: combining basic shapes to create complex objects. Show the Circuit workspace for electronics.

Student Activity

Take notes. Understand how basic shapes combine. See the Circuit workspace for Arduino projects.

Elaborate (8-10 minutes)

Teacher Activity

Task: 'Design a simple house using basic shapes: box for walls, triangle for roof, cylinder for chimney. Group and align all parts.'

Student Activity

Work independently. Build the house step by step. Apply grouping and alignment. Share screenshot.

Evaluate (5-8 minutes)

Teacher Activity

Exit ticket: '1. What is CSG? 2. How do you group shapes in Tinkercad? 3. Name two workspaces in Tinkercad.'

Student Activity

Answer the questions. Show their 3D model to a partner.

Cross-Curricular Connections

Mathematics: 3D design uses geometry: shapes, dimensions, coordinates, alignment
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Language: Technical documentation requires precise measurements and clear instructions

Digital Citizen Reminder: Tinkercad designs can be shared — credit original creators when remixing others' work.

Dormitory Task (Paper-and-Pencil)

On graph paper, draw a floor plan of your ideal room. Label each shape you would use in Tinkercad: box walls, cylinder table, sphere lamp. Write dimensions for each.

Resources Needed:
- Computer with Tinkercad account
- shape library reference
- graph paper
- pencil

Differentiation

Approaching: Follow a step-by-step tutorial to build a pre-designed object
On Level: Design a house independently using basic shapes

Advanced: Design a complex object with multiple grouped parts, add circuit components
SEND: Paper-based shape building, arrange pre-cut shapes to create objects

Formative Assessment

Method: Exit ticket and self-assessment
CT Skill Assessed: Decomposition — breaking a complex 3D object into basic shapes

Teacher Reflection (Post-Lesson)

What worked well:

What to improve:

Connection to next lesson:

Lesson 5.06	Duration: 40 minutes

Iteration Algorithms

Learning Objectives

Timing Breakdown

0-5 min	**Engage**	Show a 3D-printed object or image. Ask: 'How do you think this was designed?' ...
5-15 min	**Explore**	Open Tinkercad. Show navigation: orbit, pan, zoom. Demonstrate: place a shape, r...
15-25 min	**Explain**	Define 3D design concepts: workplane, shapes, grouping, alignment. Explain CSG ...
25-35 min	**Elaborate**	Task: 'Design a simple house using basic shapes: box for walls, triangle for roo...
35-40 min	**Evaluate**	Exit ticket: '1. What is CSG? 2. How do you group shapes in Tinkercad? 3. Name t...

Engage (5-7 minutes)

Teacher Activity

Show a 3D-printed object or image. Ask: 'How do you think this was designed?' Introduce Tinkercad as a free 3D design tool. Show the interface.

Student Activity

Examine the object. Guess the design process. Express interest in creating 3D models.

Explore (10-12 minutes)

Teacher Activity

Open Tinkercad. Show navigation: orbit, pan, zoom. Demonstrate: place a shape, resize, move, group with another shape.

Student Activity

Follow along. Place shapes, resize them, move them around. Group two shapes to create a new object.

Explain (10-12 minutes)

Teacher Activity

Define 3D design concepts: workplane, shapes, grouping, alignment. Explain CSG Constructive Solid Geometry: combining basic shapes to create complex objects. Show the Circuit workspace for electronics.

Student Activity

Take notes. Understand how basic shapes combine. See the Circuit workspace for Arduino projects.

Elaborate (8-10 minutes)

Teacher Activity

Task: 'Design a simple house using basic shapes: box for walls, triangle for roof, cylinder for chimney. Group and align all parts.'

Student Activity

Work independently. Build the house step by step. Apply grouping and alignment. Share screenshot.

Evaluate (5-8 minutes)

Teacher Activity

Exit ticket: '1. What is CSG? 2. How do you group shapes in Tinkercad? 3. Name two workspaces in Tinkercad.'

Student Activity

Answer the questions. Show their 3D model to a partner.

Cross-Curricular Connections

Mathematics: 3D design uses geometry: shapes, dimensions, coordinates, alignment
Science: Engineering design follows the design process: plan, build, test, improve
Language: Technical documentation requires precise measurements and clear instructions

Digital Citizen Reminder: Tinkercad designs can be shared — credit original creators when remixing others' work.

Dormitory Task (Paper-and-Pencil)

On graph paper, draw a floor plan of your ideal room. Label each shape you would use in Tinkercad: box walls, cylinder table, sphere lamp. Write dimensions for each.

Resources Needed:
- Computer with Tinkercad account
- shape library reference
- graph paper
- pencil

Differentiation

Approaching: Follow a step-by-step tutorial to build a pre-designed object
On Level: Design a house independently using basic shapes

Advanced: Design a complex object with multiple grouped parts, add circuit components
SEND: Paper-based shape building, arrange pre-cut shapes to create objects

Formative Assessment

Method: Exit ticket and self-assessment
CT Skill Assessed: Decomposition — breaking a complex 3D object into basic shapes

Teacher Reflection (Post-Lesson)

What worked well:

What to improve:

Connection to next lesson:

Lesson 5.07	**Duration: 40 minutes**

Sorting Algorithms

Learning Objectives

- Implement bubble sort and insertion sort algorithms
- Compare the time complexity of different sorting approaches
- Select appropriate sorting algorithms based on data characteristics

Timing Breakdown

0-5 min	**Engage**	Show a 3D-printed object or image. Ask: 'How do you think this was designed?' ...
5-15 min	**Explore**	Open Tinkercad. Show navigation: orbit, pan, zoom. Demonstrate: place a shape, r...
15-25 min	**Explain**	Define 3D design concepts: workplane, shapes, grouping, alignment. Explain CSG ...
25-35 min	**Elaborate**	Task: 'Design a simple house using basic shapes: box for walls, triangle for roo...
35-40 min	**Evaluate**	Exit ticket: '1. What is CSG? 2. How do you group shapes in Tinkercad? 3. Name t...

Engage (5-7 minutes)

Teacher Activity

Show a 3D-printed object or image. Ask: 'How do you think this was designed?' Introduce Tinkercad as a free 3D design tool. Show the interface.

Student Activity

Examine the object. Guess the design process. Express interest in creating 3D models.

Explore (10-12 minutes)

Teacher Activity

Open Tinkercad. Show navigation: orbit, pan, zoom. Demonstrate: place a shape, resize, move, group with another shape.

Student Activity

Follow along. Place shapes, resize them, move them around. Group two shapes to create a new object.

Explain (10-12 minutes)

Teacher Activity

Define 3D design concepts: workplane, shapes, grouping, alignment. Explain CSG Constructive Solid Geometry: combining basic shapes to create complex objects. Show the Circuit workspace for electronics.

Student Activity

Take notes. Understand how basic shapes combine. See the Circuit workspace for Arduino projects.

Elaborate (8-10 minutes)

Teacher Activity

Task: 'Design a simple house using basic shapes: box for walls, triangle for roof, cylinder for chimney. Group and align all parts.'

Student Activity

Work independently. Build the house step by step. Apply grouping and alignment. Share screenshot.

Evaluate (5-8 minutes)

Teacher Activity

Exit ticket: '1. What is CSG? 2. How do you group shapes in Tinkercad? 3. Name two workspaces in Tinkercad.'

Student Activity

Answer the questions. Show their 3D model to a partner.

Cross-Curricular Connections

Mathematics: 3D design uses geometry: shapes, dimensions, coordinates, alignment
Science: Engineering design follows the design process: plan, build, test, improve
Language: Technical documentation requires precise measurements and clear instructions

Digital Citizen Reminder: Tinkercad designs can be shared — credit original creators when remixing others' work.

Dormitory Task (Paper-and-Pencil)

On graph paper, draw a floor plan of your ideal room. Label each shape you would use in Tinkercad: box walls, cylinder table, sphere lamp. Write dimensions for each.

Resources Needed:

- Computer with Tinkercad account
- shape library reference
- graph paper
- pencil

Differentiation

Approaching: Follow a step-by-step tutorial to build a pre-designed object
On Level: Design a house independently using basic shapes
Advanced: Design a complex object with multiple grouped parts, add circuit components
SEND: Paper-based shape building, arrange pre-cut shapes to create objects

Formative Assessment

Method: Exit ticket and self-assessment
CT Skill Assessed: Decomposition — breaking a complex 3D object into basic shapes

Teacher Reflection (Post-Lesson)

What worked well:

What to improve:

Connection to next lesson:

Lesson 5.08	Duration: 40 minutes

Searching Algorithms

Learning Objectives

- Implement linear search and binary search algorithms
- Analyze the performance advantage of binary search on sorted data
- Determine which search algorithm to use based on data structure and requirements

Timing Breakdown

0-5 min	Engage	Show a 3D-printed object or image. Ask: 'How do you think this was designed?' ...
5-15 min	**Explore**	Open Tinkercad. Show navigation: orbit, pan, zoom. Demonstrate: place a shape, r...
15-25 min	**Explain**	Define 3D design concepts: workplane, shapes, grouping, alignment. Explain CSG ...
25-35 min	**Elaborate**	Task: 'Design a simple house using basic shapes: box for walls, triangle for roo...
35-40 min	**Evaluate**	Exit ticket: '1. What is CSG? 2. How do you group shapes in Tinkercad? 3. Name t...

Engage (5-7 minutes)

Teacher Activity

Show a 3D-printed object or image. Ask: ‘How do you think this was designed?’ Introduce Tinkercad as a free 3D design tool. Show the interface.

Student Activity

Examine the object. Guess the design process. Express interest in creating 3D models.

Explore (10-12 minutes)

Teacher Activity

Open Tinkercad. Show navigation: orbit, pan, zoom. Demonstrate: place a shape, resize, move, group with another shape.

Student Activity

Follow along. Place shapes, resize them, move them around. Group two shapes to create a new object.

Explain (10-12 minutes)

Teacher Activity

Define 3D design concepts: workplane, shapes, grouping, alignment. Explain CSG Constructive Solid Geometry: combining basic shapes to create complex objects. Show the Circuit workspace for electronics.

Student Activity

Take notes. Understand how basic shapes combine. See the Circuit workspace for Arduino projects.

Elaborate (8-10 minutes)

Teacher Activity

Task: 'Design a simple house using basic shapes: box for walls, triangle for roof, cylinder for chimney. Group and align all parts.'

Student Activity

Work independently. Build the house step by step. Apply grouping and alignment. Share screenshot.

Evaluate (5-8 minutes)

Teacher Activity

Exit ticket: '1. What is CSG? 2. How do you group shapes in Tinkercad? 3. Name two workspaces in Tinkercad.'

Student Activity

Answer the questions. Show their 3D model to a partner.

Cross-Curricular Connections

Mathematics: 3D design uses geometry: shapes, dimensions, coordinates, alignment
Science: Engineering design follows the design process: plan, build, test, improve
Language: Technical documentation requires precise measurements and clear instructions

Digital Citizen Reminder: Tinkercad designs can be shared — credit original creators when remixing others' work.

Dormitory Task (Paper-and-Pencil)

On graph paper, draw a floor plan of your ideal room. Label each shape you would use in Tinkercad: box walls, cylinder table, sphere lamp. Write dimensions for each.

Resources Needed:

- Computer with Tinkercad account
- shape library reference
- graph paper
- pencil

Differentiation

Approaching: Follow a step-by-step tutorial to build a pre-designed object
On Level: Design a house independently using basic shapes
Advanced: Design a complex object with multiple grouped parts, add circuit components
SEND: Paper-based shape building, arrange pre-cut shapes to create objects

Formative Assessment

Method: Exit ticket and self-assessment
CT Skill Assessed: Decomposition — breaking a complex 3D object into basic shapes

Teacher Reflection (Post-Lesson)

What worked well:

What to improve:

Connection to next lesson:

Lesson 5.09 **Duration: 40 minutes**

Algorithm Efficiency

Learning Objectives

- Explain time and space complexity using Big-O notation
- Compare algorithms by analyzing their growth rates
- Apply complexity analysis to choose efficient solutions for given problems

Timing Breakdown

0-5 min	**Engage**	Show a 3D-printed object or image. Ask: 'How do you think this was designed?' ...
5-15 min	**Explore**	Open Tinkercad. Show navigation: orbit, pan, zoom. Demonstrate: place a shape, r...
15-25 min	**Explain**	Define 3D design concepts: workplane, shapes, grouping, alignment. Explain CSG ...
25-35 min	**Elaborate**	Task: 'Design a simple house using basic shapes: box for walls, triangle for roo...
35-40 min	**Evaluate**	Exit ticket: '1. What is CSG? 2. How do you group shapes in Tinkercad? 3. Name t...

Engage (5-7 minutes)

Teacher Activity

Show a 3D-printed object or image. Ask: 'How do you think this was designed?' Introduce Tinkercad as a free 3D design tool. Show the interface.

Student Activity

Examine the object. Guess the design process. Express interest in creating 3D models.

Explore (10-12 minutes)

Teacher Activity

Open Tinkercad. Show navigation: orbit, pan, zoom. Demonstrate: place a shape, resize, move, group with another shape.

Student Activity

Follow along. Place shapes, resize them, move them around. Group two shapes to create a new object.

Explain (10-12 minutes)

Teacher Activity

Define 3D design concepts: workplane, shapes, grouping, alignment. Explain CSG Constructive Solid Geometry: combining basic shapes to create complex objects. Show the Circuit workspace for electronics.

Student Activity

Take notes. Understand how basic shapes combine. See the Circuit workspace for Arduino projects.

Elaborate (8-10 minutes)

Teacher Activity

Task: 'Design a simple house using basic shapes: box for walls, triangle for roof, cylinder for chimney. Group and align all parts.'

Student Activity

Work independently. Build the house step by step. Apply grouping and alignment. Share screenshot.

Evaluate (5-8 minutes)

Teacher Activity

Exit ticket: '1. What is CSG? 2. How do you group shapes in Tinkercad? 3. Name two workspaces in Tinkercad.'

Student Activity

Answer the questions. Show their 3D model to a partner.

Cross-Curricular Connections

Mathematics: 3D design uses geometry: shapes, dimensions, coordinates, alignment
Science: Engineering design follows the design process: plan, build, test, improve
Language: Technical documentation requires precise measurements and clear instructions

Digital Citizen Reminder: Tinkercad designs can be shared — credit original creators when remixing others' work.

Dormitory Task (Paper-and-Pencil)

On graph paper, draw a floor plan of your ideal room. Label each shape you would use in Tinkercad: box walls, cylinder table, sphere lamp. Write dimensions for each.

Resources Needed:

- Computer with Tinkercad account
- shape library reference
- graph paper
- pencil

Differentiation

Approaching: Follow a step-by-step tutorial to build a pre-designed object
On Level: Design a house independently using basic shapes
Advanced: Design a complex object with multiple grouped parts, add circuit components
SEND: Paper-based shape building, arrange pre-cut shapes to create objects

Formative Assessment

Method: Exit ticket and self-assessment
CT Skill Assessed: Decomposition — breaking a complex 3D object into basic shapes

Teacher Reflection (Post-Lesson)

What worked well:

What to improve:

Connection to next lesson:

Lesson 5.10	Duration: 40 minutes

Problem Decomposition

Learning Objectives

- Break complex problems into manageable sub-problems
- Identify dependencies and relationships between sub-problems
- Apply top-down and bottom-up decomposition strategies

Timing Breakdown

0-5 min	Engage	Show a 3D-printed object or image. Ask: 'How do you think this was designed?' ...
5-15 min	**Explore**	Open Tinkercad. Show navigation: orbit, pan, zoom. Demonstrate: place a shape, r...
15-25 min	**Explain**	Define 3D design concepts: workplane, shapes, grouping, alignment. Explain CSG ...
25-35 min	**Elaborate**	Task: 'Design a simple house using basic shapes: box for walls, triangle for roo...
35-40 min	**Evaluate**	Exit ticket: '1. What is CSG? 2. How do you group shapes in Tinkercad? 3. Name t...

Engage (5-7 minutes)

Teacher Activity

Show a 3D-printed object or image. Ask: ‘How do you think this was designed?’ Introduce Tinkercad as a free 3D design tool. Show the interface.

Student Activity

Examine the object. Guess the design process. Express interest in creating 3D models.

Explore (10-12 minutes)

Teacher Activity

Open Tinkercad. Show navigation: orbit, pan, zoom. Demonstrate: place a shape, resize, move, group with another shape.

Student Activity

Follow along. Place shapes, resize them, move them around. Group two shapes to create a new object.

Explain (10-12 minutes)

Teacher Activity

Define 3D design concepts: workplane, shapes, grouping, alignment. Explain CSG Constructive Solid Geometry: combining basic shapes to create complex objects. Show the Circuit workspace for electronics.

Student Activity

Take notes. Understand how basic shapes combine. See the Circuit workspace for Arduino projects.

Elaborate (8-10 minutes)

Teacher Activity

Task: 'Design a simple house using basic shapes: box for walls, triangle for roof, cylinder for chimney. Group and align all parts.'

Student Activity

Work independently. Build the house step by step. Apply grouping and alignment. Share screenshot.

Evaluate (5-8 minutes)

Teacher Activity

Exit ticket: '1. What is CSG? 2. How do you group shapes in Tinkercad? 3. Name two workspaces in Tinkercad.'

Student Activity

Answer the questions. Show their 3D model to a partner.

Cross-Curricular Connections

Mathematics: 3D design uses geometry: shapes, dimensions, coordinates, alignment
Science: Engineering design follows the design process: plan, build, test, improve
Language: Technical documentation requires precise measurements and clear instructions

Digital Citizen Reminder: Tinkercad designs can be shared — credit original creators when remixing others' work.

Dormitory Task (Paper-and-Pencil)

On graph paper, draw a floor plan of your ideal room. Label each shape you would use in Tinkercad: box walls, cylinder table, sphere lamp. Write dimensions for each.

Resources Needed:
- Computer with Tinkercad account
- shape library reference
- graph paper
- pencil

Differentiation

Approaching: Follow a step-by-step tutorial to build a pre-designed object
On Level: Design a house independently using basic shapes
Advanced: Design a complex object with multiple grouped parts, add circuit components
SEND: Paper-based shape building, arrange pre-cut shapes to create objects

Formative Assessment

Method: Exit ticket and self-assessment
CT Skill Assessed: Decomposition — breaking a complex 3D object into basic shapes

Teacher Reflection (Post-Lesson)

What worked well:

What to improve:

Connection to next lesson:

Lesson 5.11	Duration: 40 minutes

Project - Algorithm Design

Learning Objectives

- Design a complete algorithmic solution for a real-world problem

Timing Breakdown

0-5 min	Engage	Show a 3D-printed object or image. Ask: 'How do you think this was designed?' ...
5-15 min	**Explore**	Open Tinkercad. Show navigation: orbit, pan, zoom. Demonstrate: place a shape, r...
15-25 min	**Explain**	Define 3D design concepts: workplane, shapes, grouping, alignment. Explain CSG ...
25-35 min	**Elaborate**	Task: 'Design a simple house using basic shapes: box for walls, triangle for roo...
35-40 min	**Evaluate**	Exit ticket: '1. What is CSG? 2. How do you group shapes in Tinkercad? 3. Name t...

Engage (5-7 minutes)

Teacher Activity

Show a 3D-printed object or image. Ask: 'How do you think this was designed?' Introduce Tinkercad as a free 3D design tool. Show the interface.

Student Activity

Examine the object. Guess the design process. Express interest in creating 3D models.

Explore (10-12 minutes)

Teacher Activity

Open Tinkercad. Show navigation: orbit, pan, zoom. Demonstrate: place a shape, resize, move, group with another shape.

Student Activity

Follow along. Place shapes, resize them, move them around. Group two shapes to create a new object.

Explain (10-12 minutes)

Teacher Activity

Define 3D design concepts: workplane, shapes, grouping, alignment. Explain CSG Constructive Solid Geometry: combining basic shapes to create complex objects. Show the Circuit workspace for electronics.

Student Activity

Take notes. Understand how basic shapes combine. See the Circuit workspace for Arduino projects.

Elaborate (8-10 minutes)

Teacher Activity

Task: 'Design a simple house using basic shapes: box for walls, triangle for roof, cylinder for chimney. Group and align all parts.'

Student Activity

Work independently. Build the house step by step. Apply grouping and alignment. Share screenshot.

Evaluate (5-8 minutes)

Teacher Activity

Exit ticket: '1. What is CSG? 2. How do you group shapes in Tinkercad? 3. Name two workspaces in Tinkercad.'

Student Activity

Answer the questions. Show their 3D model to a partner.

Cross-Curricular Connections

Mathematics: 3D design uses geometry: shapes, dimensions, coordinates, alignment
Science: Engineering design follows the design process: plan, build, test, improve
Language: Technical documentation requires precise measurements and clear instructions

Digital Citizen Reminder: Tinkercad designs can be shared — credit original creators when remixing others' work.

Dormitory Task (Paper-and-Pencil)

On graph paper, draw a floor plan of your ideal room. Label each shape you would use in Tinkercad: box walls, cylinder table, sphere lamp. Write dimensions for each.

Resources Needed:
- Computer with Tinkercad account
- shape library reference
- graph paper
- pencil

Differentiation

Approaching: Follow a step-by-step tutorial to build a pre-designed object
On Level: Design a house independently using basic shapes
Advanced: Design a complex object with multiple grouped parts, add circuit components
SEND: Paper-based shape building, arrange pre-cut shapes to create objects

Formative Assessment

Method: Exit ticket and self-assessment
CT Skill Assessed: Decomposition — breaking a complex 3D object into basic shapes

Teacher Reflection (Post-Lesson)

What worked well:

What to improve:

Connection to next lesson:

Lesson 5.12	Duration: 40 minutes

Chapter 5 Review

Learning Objectives

- Consolidate understanding of all algorithmic concepts from Chapter 5
- Apply algorithmic thinking to novel, unfamiliar problems
- Evaluate and compare different algorithmic approaches to the same problem

Timing Breakdown

0-5 min	Engage	Show a 3D-printed object or image. Ask: 'How do you think this was designed?' ...
5-15 min	**Explore**	Open Tinkercad. Show navigation: orbit, pan, zoom. Demonstrate: place a shape, r...
15-25 min	**Explain**	Define 3D design concepts: workplane, shapes, grouping, alignment. Explain CSG ...
25-35 min	**Elaborate**	Task: 'Design a simple house using basic shapes: box for walls, triangle for roo...
35-40 min	**Evaluate**	Exit ticket: '1. What is CSG? 2. How do you group shapes in Tinkercad? 3. Name t...

Engage (5-7 minutes)

Teacher Activity

Show a 3D-printed object or image. Ask: ‘How do you think this was designed?’ Introduce Tinkercad as a free 3D design tool. Show the interface.

Student Activity

Examine the object. Guess the design process. Express interest in creating 3D models.

Explore (10-12 minutes)

Teacher Activity

Open Tinkercad. Show navigation: orbit, pan, zoom. Demonstrate: place a shape, resize, move, group with another shape.

Student Activity

Follow along. Place shapes, resize them, move them around. Group two shapes to create a new object.

Explain (10-12 minutes)

Teacher Activity

Define 3D design concepts: workplane, shapes, grouping, alignment. Explain CSG Constructive Solid Geometry: combining basic shapes to create complex objects. Show the Circuit workspace for electronics.

Student Activity

Take notes. Understand how basic shapes combine. See the Circuit workspace for Arduino projects.

Elaborate (8-10 minutes)

Teacher Activity

Task: 'Design a simple house using basic shapes: box for walls, triangle for roof, cylinder for chimney. Group and align all parts.'

Student Activity

Work independently. Build the house step by step. Apply grouping and alignment. Share screenshot.

Evaluate (5-8 minutes)

Teacher Activity

Exit ticket: '1. What is CSG? 2. How do you group shapes in Tinkercad? 3. Name two workspaces in Tinkercad.'

Student Activity

Answer the questions. Show their 3D model to a partner.

Cross-Curricular Connections

Mathematics: 3D design uses geometry: shapes, dimensions, coordinates, alignment
Science: Engineering design follows the design process: plan, build, test, improve
Language: Technical documentation requires precise measurements and clear instructions

Digital Citizen Reminder: Tinkercad designs can be shared — credit original creators when remixing others' work.

Dormitory Task (Paper-and-Pencil)

On graph paper, draw a floor plan of your ideal room. Label each shape you would use in Tinkercad: box walls, cylinder table, sphere lamp. Write dimensions for each.

Resources Needed:

- Computer with Tinkercad account
- shape library reference
- graph paper
- pencil

Differentiation

Approaching: Follow a step-by-step tutorial to build a pre-designed object
On Level: Design a house independently using basic shapes
Advanced: Design a complex object with multiple grouped parts, add circuit components
SEND: Paper-based shape building, arrange pre-cut shapes to create objects

Formative Assessment

Method: Exit ticket and self-assessment
CT Skill Assessed: Decomposition — breaking a complex 3D object into basic shapes

Teacher Reflection (Post-Lesson)

What worked well:

What to improve:

Connection to next lesson:

Lesson 6.01	Duration: 40 minutes

What is AI?

Learning Objectives

- Define artificial intelligence and distinguish between narrow and general AI
- Identify characteristics that define intelligent behavior in machines
- Evaluate current AI capabilities against human cognitive abilities

Timing Breakdown

0-5 min	Engage	Show an AI chatbot conversation or image generator output. Ask: 'Is this intel...
5-15 min	Explore	Activity: 'The Human Robot' — one student gives step-by-step commands, another f...
15-25 min	Explain	Define AI: systems that learn from data to make decisions. Key concepts: data f...
25-35 min	Elaborate	Task: 'Create a simple decision tree on paper: Should I bring an umbrella? Branc...
35-40 min	Evaluate	Exit ticket: '1. What is AI? 2. What is bias in AI? 3. Why is data important for...

Engage (5-7 minutes)

Teacher Activity

Show an AI chatbot conversation or image generator output. Ask: 'Is this intelligent? How does it work?' Discuss what makes AI different from regular software.

Student Activity

Observe the AI output. Discuss: 'It seems smart but follows rules.' Brainstorm examples of AI in daily life: voice assistants, recommendations.

Explore (10-12 minutes)

Teacher Activity

Activity: 'The Human Robot' — one student gives step-by-step commands, another follows literally. Show how AI follows instructions exactly, without common sense.

Student Activity

Participate in the game. Experience how literal instructions produce unexpected results. Discuss the gap between human and AI understanding.

Explain (10-12 minutes)

Teacher Activity

Define AI: systems that learn from data to make decisions. Key concepts: data fuel, patterns learning, algorithms rules. Show simple decision tree. Explain bias: AI learns from human data, which may contain biases.

Student Activity

Take notes. Understand data ⟶ pattern ⟶ decision pipeline. See how bias enters through training data.

Elaborate (8-10 minutes)

Teacher Activity

Task: ‘Create a simple decision tree on paper: Should I bring an umbrella? Branches: Is it raining? Yes ⟶ Bring it, No ⟶ Check forecast, etc.’ Discuss how AI makes decisions similarly.

Student Activity

Work in pairs. Create decision trees for different scenarios: ‘What should I eat for lunch?’, ‘Should I study now?’ Present to class.

Evaluate (5-8 minutes)

Teacher Activity

Exit ticket: ‘1. What is AI? 2. What is bias in AI? 3. Why is data important for AI?’

Student Activity

Answer the questions. Discuss the ethical implications of AI.

Cross-Curricular Connections

Mathematics: Decision trees use logical conditions if/then; probability determines AI confidence
Science: AI learns from data like scientists form hypotheses from observations
Language: Natural Language Processing enables AI to understand and generate human language

Digital Citizen Reminder: Be aware of AI bias — AI systems can make unfair decisions if trained on biased data.

Dormitory Task (Paper-and-Pencil)

Create a decision tree on paper for ‘Should I play outside?’ Include at least 4 decision points weather, homework, friends available, time of day. Draw branches for yes/no answers.

Resources Needed:

- Decision tree examples

- AI demo if available
- graph paper
- scenario cards

Differentiation

Approaching: Use a pre-made decision tree template and fill in the branches
On Level: Create a decision tree with 4+ decision points independently
Advanced: Research one AI application and present its benefits and risks to the class
SEND: Card-based decision activity, arrange yes/no cards in order

Formative Assessment

Method: Exit ticket and self-assessment
CT Skill Assessed: Abstraction — simplifying complex decisions into clear if/then branches

Teacher Reflection (Post-Lesson)

What worked well:

What to improve:

Connection to next lesson:

Lesson 6.02	Duration: 40 minutes

Machine Learning Basics

Learning Objectives

- Explain the fundamental concept of learning from data
- Distinguish between supervised, unsupervised, and reinforcement learning
- Analyze real-world datasets to identify learning task types

Timing Breakdown

0-5 min	Engage	Show an AI chatbot conversation or image generator output. Ask: 'Is this intel...
5-15 min	Explore	Activity: 'The Human Robot' — one student gives step-by-step commands, another f...
15-25 min	Explain	Define AI: systems that learn from data to make decisions. Key concepts: data f...
25-35 min	Elaborate	Task: 'Create a simple decision tree on paper: Should I bring an umbrella? Branc...
35-40 min	Evaluate	Exit ticket: '1. What is AI? 2. What is bias in AI? 3. Why is data important for...

Engage (5-7 minutes)

Teacher Activity

Show an AI chatbot conversation or image generator output. Ask: 'Is this intelligent? How does it work?' Discuss what makes AI different from regular software.

Student Activity

Observe the AI output. Discuss: 'It seems smart but follows rules.' Brainstorm examples of AI in daily life: voice assistants, recommendations.

Explore (10-12 minutes)

Teacher Activity

Activity: 'The Human Robot' — one student gives step-by-step commands, another follows literally. Show how AI follows instructions exactly, without common sense.

Student Activity

Participate in the game. Experience how literal instructions produce unexpected results. Discuss the gap between human and AI understanding.

Explain (10-12 minutes)

Teacher Activity

Define AI: systems that learn from data to make decisions. Key concepts: data fuel, patterns learning, algorithms rules. Show simple decision tree. Explain bias: AI learns from human data, which may contain biases.

Student Activity

Take notes. Understand data ⟶ pattern ⟶ decision pipeline. See how bias enters through training data.

Elaborate (8-10 minutes)

Teacher Activity

Task: 'Create a simple decision tree on paper: Should I bring an umbrella? Branches: Is it raining? Yes ⟶ Bring it, No ⟶ Check forecast, etc.' Discuss how AI makes decisions similarly.

Student Activity

Work in pairs. Create decision trees for different scenarios: 'What should I eat for lunch?', 'Should I study now?' Present to class.

Evaluate (5-8 minutes)

Teacher Activity

Exit ticket: '1. What is AI? 2. What is bias in AI? 3. Why is data important for AI?'

Student Activity

Answer the questions. Discuss the ethical implications of AI.

Cross-Curricular Connections

Mathematics: Decision trees use logical conditions if/then; probability determines AI confidence
Science: AI learns from data like scientists form hypotheses from observations
Language: Natural Language Processing enables AI to understand and generate human language

Digital Citizen Reminder: Be aware of AI bias — AI systems can make unfair decisions if trained on biased data.

Dormitory Task (Paper-and-Pencil)

Create a decision tree on paper for 'Should I play outside?' Include at least 4 decision points weather, homework, friends available, time of day. Draw branches for yes/no answers.

Resources Needed:
- Decision tree examples

- AI demo if available
- graph paper
- scenario cards

Differentiation

Approaching: Use a pre-made decision tree template and fill in the branches
On Level: Create a decision tree with 4+ decision points independently
Advanced: Research one AI application and present its benefits and risks to the class
SEND: Card-based decision activity, arrange yes/no cards in order

Formative Assessment

Method: Exit ticket and self-assessment
CT Skill Assessed: Abstraction — simplifying complex decisions into clear if/then branches

Teacher Reflection (Post-Lesson)

What worked well:

What to improve:

Connection to next lesson:

Lesson 6.03	Duration: 40 minutes

AI Applications

Learning Objectives

- Identify AI applications across diverse industries
- Analyze how AI transforms traditional processes in specific domains
- Evaluate the societal impact of AI deployment in everyday life

Timing Breakdown

0-5 min	Engage	Show an AI chatbot conversation or image generator output. Ask: 'Is this intel...
5-15 min	Explore	Activity: 'The Human Robot' — one student gives step-by-step commands, another f...
15-25 min	Explain	Define AI: systems that learn from data to make decisions. Key concepts: data f...
25-35 min	Elaborate	Task: 'Create a simple decision tree on paper: Should I bring an umbrella? Branc...
35-40 min	Evaluate	Exit ticket: '1. What is AI? 2. What is bias in AI? 3. Why is data important for...

Engage (5-7 minutes)

Teacher Activity

Show an AI chatbot conversation or image generator output. Ask: ‘Is this intelligent? How does it work?’ Discuss what makes AI different from regular software.

Student Activity

Observe the AI output. Discuss: ‘It seems smart but follows rules.’ Brainstorm examples of AI in daily life: voice assistants, recommendations.

Explore (10-12 minutes)

Teacher Activity

Activity: ‘The Human Robot’ — one student gives step-by-step commands, another follows literally. Show how AI follows instructions exactly, without common sense.

Student Activity

Participate in the game. Experience how literal instructions produce unexpected results. Discuss the gap between human and AI understanding.

Explain (10-12 minutes)

Teacher Activity

Define AI: systems that learn from data to make decisions. Key concepts: data fuel, patterns learning, algorithms rules. Show simple decision tree. Explain bias: AI learns from human data, which may contain biases.

Student Activity

Take notes. Understand data ⟶ pattern ⟶ decision pipeline. See how bias enters through training data.

Elaborate (8-10 minutes)

Teacher Activity

Task: 'Create a simple decision tree on paper: Should I bring an umbrella? Branches: Is it raining? Yes ⟶ Bring it, No ⟶ Check forecast, etc.' Discuss how AI makes decisions similarly.

Student Activity

Work in pairs. Create decision trees for different scenarios: 'What should I eat for lunch?', 'Should I study now?' Present to class.

Evaluate (5-8 minutes)

Teacher Activity

Exit ticket: '1. What is AI? 2. What is bias in AI? 3. Why is data important for AI?'

Student Activity

Answer the questions. Discuss the ethical implications of AI.

Cross-Curricular Connections

Mathematics: Decision trees use logical conditions if/then; probability determines AI confidence
Science: AI learns from data like scientists form hypotheses from observations
Language: Natural Language Processing enables AI to understand and generate human language

Digital Citizen Reminder: Be aware of AI bias — AI systems can make unfair decisions if trained on biased data.

Dormitory Task (Paper-and-Pencil)

Create a decision tree on paper for 'Should I play outside?' Include at least 4 decision points weather, homework, friends available, time of day. Draw branches for yes/no answers.

Resources Needed:
- Decision tree examples

- AI demo if available
- graph paper
- scenario cards

Differentiation

Approaching: Use a pre-made decision tree template and fill in the branches
On Level: Create a decision tree with 4+ decision points independently
Advanced: Research one AI application and present its benefits and risks to the class
SEND: Card-based decision activity, arrange yes/no cards in order

Formative Assessment

Method: Exit ticket and self-assessment
CT Skill Assessed: Abstraction — simplifying complex decisions into clear if/then branches

Teacher Reflection (Post-Lesson)

What worked well:

What to improve:

Connection to next lesson:

Lesson 6.04	Duration: 40 minutes

Internet of Things

Learning Objectives

- Define IoT and identify its core architectural components
- Explain how sensors, connectivity, and data processing work together in IoT systems
- Design a basic IoT system for a practical application

Timing Breakdown

0-5 min	Engage	Show an AI chatbot conversation or image generator output. Ask: 'Is this intel...
5-15 min	Explore	Activity: 'The Human Robot' — one student gives step-by-step commands, another f...
15-25 min	Explain	Define AI: systems that learn from data to make decisions. Key concepts: data f...
25-35 min	Elaborate	Task: 'Create a simple decision tree on paper: Should I bring an umbrella? Branc...
35-40 min	Evaluate	Exit ticket: '1. What is AI? 2. What is bias in AI? 3. Why is data important for...

Engage (5-7 minutes)

Teacher Activity

Show an AI chatbot conversation or image generator output. Ask: 'Is this intelligent? How does it work?' Discuss what makes AI different from regular software.

Student Activity

Observe the AI output. Discuss: 'It seems smart but follows rules.' Brainstorm examples of AI in daily life: voice assistants, recommendations.

Explore (10-12 minutes)

Teacher Activity

Activity: 'The Human Robot' — one student gives step-by-step commands, another follows literally. Show how AI follows instructions exactly, without common sense.

Student Activity

Participate in the game. Experience how literal instructions produce unexpected results. Discuss the gap between human and AI understanding.

Explain (10-12 minutes)

Teacher Activity

Define AI: systems that learn from data to make decisions. Key concepts: data fuel, patterns learning, algorithms rules. Show simple decision tree. Explain bias: AI learns from human data, which may contain biases.

Student Activity

Take notes. Understand data ⟶ pattern ⟶ decision pipeline. See how bias enters through training data.

Elaborate (8-10 minutes)

Teacher Activity

Task: ‘Create a simple decision tree on paper: Should I bring an umbrella? Branches: Is it raining? Yes ⟶ Bring it, No ⟶ Check forecast, etc.’ Discuss how AI makes decisions similarly.

Student Activity

Work in pairs. Create decision trees for different scenarios: ‘What should I eat for lunch?’, ‘Should I study now?’ Present to class.

Evaluate (5-8 minutes)

Teacher Activity

Exit ticket: ‘1. What is AI? 2. What is bias in AI? 3. Why is data important for AI?’

Student Activity

Answer the questions. Discuss the ethical implications of AI.

Cross-Curricular Connections

Mathematics: Decision trees use logical conditions if/then; probability determines AI confidence
Science: AI learns from data like scientists form hypotheses from observations
Language: Natural Language Processing enables AI to understand and generate human language

Digital Citizen Reminder: Be aware of AI bias — AI systems can make unfair decisions if trained on biased data.

Dormitory Task (Paper-and-Pencil)

Create a decision tree on paper for ‘Should I play outside?’ Include at least 4 decision points weather, homework, friends available, time of day. Draw branches for yes/no answers.

Resources Needed:
- Decision tree examples

- AI demo if available
- graph paper
- scenario cards

Differentiation

Approaching: Use a pre-made decision tree template and fill in the branches
On Level: Create a decision tree with 4+ decision points independently
Advanced: Research one AI application and present its benefits and risks to the class
SEND: Card-based decision activity, arrange yes/no cards in order

Formative Assessment

Method: Exit ticket and self-assessment
CT Skill Assessed: Abstraction — simplifying complex decisions into clear if/then branches

Teacher Reflection (Post-Lesson)

What worked well:

What to improve:

Connection to next lesson:

Lesson 6.05	Duration: 40 minutes

IoT in Agriculture

Learning Objectives

- Analyze how IoT technology addresses challenges in modern agriculture
- Evaluate the economic and environmental benefits of precision farming
- Design an IoT solution for a specific agricultural problem

Timing Breakdown

0-5 min	**Engage**	Show an AI chatbot conversation or image generator output. Ask: 'Is this intel...
5-15 min	**Explore**	Activity: 'The Human Robot' — one student gives step-by-step commands, another f...
15-25 min	**Explain**	Define AI: systems that learn from data to make decisions. Key concepts: data f...
25-35 min	**Elaborate**	Task: 'Create a simple decision tree on paper: Should I bring an umbrella? Branc...
35-40 min	**Evaluate**	Exit ticket: '1. What is AI? 2. What is bias in AI? 3. Why is data important for...

Engage (5-7 minutes)

Teacher Activity

Show an AI chatbot conversation or image generator output. Ask: 'Is this intelligent? How does it work?' Discuss what makes AI different from regular software.

Student Activity

Observe the AI output. Discuss: 'It seems smart but follows rules.' Brainstorm examples of AI in daily life: voice assistants, recommendations.

Explore (10-12 minutes)

Teacher Activity

Activity: 'The Human Robot' — one student gives step-by-step commands, another follows literally. Show how AI follows instructions exactly, without common sense.

Student Activity

Participate in the game. Experience how literal instructions produce unexpected results. Discuss the gap between human and AI understanding.

Explain (10-12 minutes)

Teacher Activity

Define AI: systems that learn from data to make decisions. Key concepts: data fuel, patterns learning, algorithms rules. Show simple decision tree. Explain bias: AI learns from human data, which may contain biases.

Student Activity

Take notes. Understand data → pattern → decision pipeline. See how bias enters through training data.

Elaborate (8-10 minutes)

Teacher Activity

Task: 'Create a simple decision tree on paper: Should I bring an umbrella? Branches: Is it raining? Yes → Bring it, No → Check forecast, etc.' Discuss how AI makes decisions similarly.

Student Activity

Work in pairs. Create decision trees for different scenarios: 'What should I eat for lunch?', 'Should I study now?' Present to class.

Evaluate (5-8 minutes)

Teacher Activity

Exit ticket: '1. What is AI? 2. What is bias in AI? 3. Why is data important for AI?'

Student Activity

Answer the questions. Discuss the ethical implications of AI.

Cross-Curricular Connections

Mathematics: Decision trees use logical conditions if/then; probability determines AI confidence
Science: AI learns from data like scientists form hypotheses from observations
Language: Natural Language Processing enables AI to understand and generate human language

Digital Citizen Reminder: Be aware of AI bias — AI systems can make unfair decisions if trained on biased data.

Dormitory Task (Paper-and-Pencil)

Create a decision tree on paper for 'Should I play outside?' Include at least 4 decision points weather, homework, friends available, time of day. Draw branches for yes/no answers.

Resources Needed:
- Decision tree examples

- AI demo if available
- graph paper
- scenario cards

Differentiation

Approaching: Use a pre-made decision tree template and fill in the branches
On Level: Create a decision tree with 4+ decision points independently
Advanced: Research one AI application and present its benefits and risks to the class
SEND: Card-based decision activity, arrange yes/no cards in order

Formative Assessment

Method: Exit ticket and self-assessment
CT Skill Assessed: Abstraction — simplifying complex decisions into clear if/then branches

Teacher Reflection (Post-Lesson)

What worked well:

What to improve:

Connection to next lesson:

Lesson 6.06	Duration: 40 minutes

Cloud Computing

Learning Objectives

Timing Breakdown

0-5 min	**Engage**	Show an AI chatbot conversation or image generator output. Ask: 'Is this intel...
5-15 min	**Explore**	Activity: 'The Human Robot' — one student gives step-by-step commands, another f...
15-25 min	**Explain**	Define AI: systems that learn from data to make decisions. Key concepts: data f...
25-35 min	**Elaborate**	Task: 'Create a simple decision tree on paper: Should I bring an umbrella? Branc...
35-40 min	**Evaluate**	Exit ticket: '1. What is AI? 2. What is bias in AI? 3. Why is data important for...

Engage (5-7 minutes)

Teacher Activity

Show an AI chatbot conversation or image generator output. Ask: 'Is this intelligent? How does it work?' Discuss what makes AI different from regular software.

Student Activity

Observe the AI output. Discuss: 'It seems smart but follows rules.' Brainstorm examples of AI in daily life: voice assistants, recommendations.

Explore (10-12 minutes)

Teacher Activity

Activity: 'The Human Robot' — one student gives step-by-step commands, another follows literally. Show how AI follows instructions exactly, without common sense.

Student Activity

Participate in the game. Experience how literal instructions produce unexpected results. Discuss the gap between human and AI understanding.

Explain (10-12 minutes)

Teacher Activity

Define AI: systems that learn from data to make decisions. Key concepts: data fuel, patterns learning, algorithms rules. Show simple decision tree. Explain bias: AI learns from human data, which may contain biases.

Student Activity

Take notes. Understand data ⟶ pattern ⟶ decision pipeline. See how bias enters through training data.

Elaborate (8-10 minutes)

Teacher Activity

Task: 'Create a simple decision tree on paper: Should I bring an umbrella? Branches: Is it raining? Yes ⟶ Bring it, No ⟶ Check forecast, etc.' Discuss how AI makes decisions similarly.

Student Activity

Work in pairs. Create decision trees for different scenarios: 'What should I eat for lunch?', 'Should I study now?' Present to class.

Evaluate (5-8 minutes)

Teacher Activity

Exit ticket: '1. What is AI? 2. What is bias in AI? 3. Why is data important for AI?'

Student Activity

Answer the questions. Discuss the ethical implications of AI.

Cross-Curricular Connections

Mathematics: Decision trees use logical conditions if/then; probability determines AI confidence
Science: AI learns from data like scientists form hypotheses from observations
Language: Natural Language Processing enables AI to understand and generate human language

Digital Citizen Reminder: Be aware of AI bias — AI systems can make unfair decisions if trained on biased data.

Dormitory Task (Paper-and-Pencil)

Create a decision tree on paper for 'Should I play outside?' Include at least 4 decision points weather, homework, friends available, time of day. Draw branches for yes/no answers.

Resources Needed:
- Decision tree examples
- AI demo if available
- graph paper
- scenario cards

Differentiation

Approaching: Use a pre-made decision tree template and fill in the branches
On Level: Create a decision tree with 4+ decision points independently
Advanced: Research one AI application and present its benefits and risks to the class
SEND: Card-based decision activity, arrange yes/no cards in order

Formative Assessment

Method: Exit ticket and self-assessment
CT Skill Assessed: Abstraction — simplifying complex decisions into clear if/then branches

Teacher Reflection (Post-Lesson)

What worked well:

What to improve:

Connection to next lesson:

Lesson 6.07	Duration: 40 minutes

Cloud Services

Learning Objectives

- Compare major cloud platforms and their service offerings
- Identify appropriate cloud services for different application needs
- Evaluate cloud service reliability, security, and cost considerations

Timing Breakdown

0-5 min	**Engage**	Show an AI chatbot conversation or image generator output. Ask: 'Is this intel...
5-15 min	**Explore**	Activity: 'The Human Robot' — one student gives step-by-step commands, another f...
15-25 min	**Explain**	Define AI: systems that learn from data to make decisions. Key concepts: data f...
25-35 min	**Elaborate**	Task: 'Create a simple decision tree on paper: Should I bring an umbrella? Branc...
35-40 min	**Evaluate**	Exit ticket: '1. What is AI? 2. What is bias in AI? 3. Why is data important for...

Engage (5-7 minutes)

Teacher Activity

Show an AI chatbot conversation or image generator output. Ask: 'Is this intelligent? How does it work?' Discuss what makes AI different from regular software.

Student Activity

Observe the AI output. Discuss: 'It seems smart but follows rules.' Brainstorm examples of AI in daily life: voice assistants, recommendations.

Explore (10-12 minutes)

Teacher Activity

Activity: 'The Human Robot' — one student gives step-by-step commands, another follows literally. Show how AI follows instructions exactly, without common sense.

Student Activity

Participate in the game. Experience how literal instructions produce unexpected results. Discuss the gap between human and AI understanding.

Explain (10-12 minutes)

Teacher Activity

Define AI: systems that learn from data to make decisions. Key concepts: data fuel, patterns learning, algorithms rules. Show simple decision tree. Explain bias: AI learns from human data, which may contain biases.

Student Activity

Take notes. Understand data → pattern → decision pipeline. See how bias enters through training data.

Elaborate (8-10 minutes)

Teacher Activity

Task: 'Create a simple decision tree on paper: Should I bring an umbrella? Branches: Is it raining? Yes → Bring it, No → Check forecast, etc.' Discuss how AI makes decisions similarly.

Student Activity

Work in pairs. Create decision trees for different scenarios: 'What should I eat for lunch?', 'Should I study now?' Present to class.

Evaluate (5-8 minutes)

Teacher Activity

Exit ticket: '1. What is AI? 2. What is bias in AI? 3. Why is data important for AI?'

Student Activity

Answer the questions. Discuss the ethical implications of AI.

Cross-Curricular Connections

Mathematics: Decision trees use logical conditions if/then; probability determines AI confidence
Science: AI learns from data like scientists form hypotheses from observations
Language: Natural Language Processing enables AI to understand and generate human language

Digital Citizen Reminder: Be aware of AI bias — AI systems can make unfair decisions if trained on biased data.

Dormitory Task (Paper-and-Pencil)

Create a decision tree on paper for 'Should I play outside?' Include at least 4 decision points weather, homework, friends available, time of day. Draw branches for yes/no answers.

Resources Needed:

- Decision tree examples

- AI demo if available
- graph paper
- scenario cards

Differentiation

Approaching: Use a pre-made decision tree template and fill in the branches
On Level: Create a decision tree with 4+ decision points independently
Advanced: Research one AI application and present its benefits and risks to the class
SEND: Card-based decision activity, arrange yes/no cards in order

Formative Assessment

Method: Exit ticket and self-assessment
CT Skill Assessed: Abstraction — simplifying complex decisions into clear if/then branches

Teacher Reflection (Post-Lesson)

What worked well:

What to improve:

Connection to next lesson:

Lesson 6.08	Duration: 40 minutes

Blockchain Basics

Learning Objectives

- Explain the fundamental principles of blockchain technology
- Describe how cryptographic hashing ensures data integrity and security
- Evaluate blockchain applications beyond cryptocurrency

Timing Breakdown

0-5 min	**Engage**	Show an AI chatbot conversation or image generator output. Ask: 'Is this intel...
5-15 min	**Explore**	Activity: 'The Human Robot' — one student gives step-by-step commands, another f...
15-25 min	**Explain**	Define AI: systems that learn from data to make decisions. Key concepts: data f...
25-35 min	**Elaborate**	Task: 'Create a simple decision tree on paper: Should I bring an umbrella? Branc...
35-40 min	**Evaluate**	Exit ticket: '1. What is AI? 2. What is bias in AI? 3. Why is data important for...

Engage (5-7 minutes)

Teacher Activity

Show an AI chatbot conversation or image generator output. Ask: 'Is this intelligent? How does it work?' Discuss what makes AI different from regular software.

Student Activity

Observe the AI output. Discuss: 'It seems smart but follows rules.' Brainstorm examples of AI in daily life: voice assistants, recommendations.

Explore (10-12 minutes)

Teacher Activity

Activity: 'The Human Robot' — one student gives step-by-step commands, another follows literally. Show how AI follows instructions exactly, without common sense.

Student Activity

Participate in the game. Experience how literal instructions produce unexpected results. Discuss the gap between human and AI understanding.

Explain (10-12 minutes)

Teacher Activity

Define AI: systems that learn from data to make decisions. Key concepts: data fuel, patterns learning, algorithms rules. Show simple decision tree. Explain bias: AI learns from human data, which may contain biases.

Student Activity

Take notes. Understand data ⟶ pattern ⟶ decision pipeline. See how bias enters through training data.

Elaborate (8-10 minutes)

Teacher Activity

Task: 'Create a simple decision tree on paper: Should I bring an umbrella? Branches: Is it raining? Yes ⟶ Bring it, No ⟶ Check forecast, etc.' Discuss how AI makes decisions similarly.

Student Activity

Work in pairs. Create decision trees for different scenarios: 'What should I eat for lunch?', 'Should I study now?' Present to class.

Evaluate (5-8 minutes)

Teacher Activity

Exit ticket: '1. What is AI? 2. What is bias in AI? 3. Why is data important for AI?'

Student Activity

Answer the questions. Discuss the ethical implications of AI.

Cross-Curricular Connections

Mathematics: Decision trees use logical conditions if/then; probability determines AI confidence
Science: AI learns from data like scientists form hypotheses from observations
Language: Natural Language Processing enables AI to understand and generate human language

Digital Citizen Reminder: Be aware of AI bias — AI systems can make unfair decisions if trained on biased data.

Dormitory Task (Paper-and-Pencil)

Create a decision tree on paper for 'Should I play outside?' Include at least 4 decision points weather, homework, friends available, time of day. Draw branches for yes/no answers.

Resources Needed:
- Decision tree examples

- AI demo if available
- graph paper
- scenario cards

Differentiation

Approaching: Use a pre-made decision tree template and fill in the branches
On Level: Create a decision tree with 4+ decision points independently
Advanced: Research one AI application and present its benefits and risks to the class
SEND: Card-based decision activity, arrange yes/no cards in order

Formative Assessment

Method: Exit ticket and self-assessment
CT Skill Assessed: Abstraction — simplifying complex decisions into clear if/then branches

Teacher Reflection (Post-Lesson)

What worked well:

What to improve:

Connection to next lesson:

Lesson 6.09	Duration: 40 minutes

5G Technology

Learning Objectives

- Explain 5G technical characteristics and how they differ from previous generations
- Analyze how 5G enables new technologies and applications
- Evaluate the societal and economic implications of 5G deployment

Timing Breakdown

0-5 min	Engage	Show an AI chatbot conversation or image generator output. Ask: 'Is this intel...
5-15 min	Explore	Activity: 'The Human Robot' — one student gives step-by-step commands, another f...
15-25 min	Explain	Define AI: systems that learn from data to make decisions. Key concepts: data f...
25-35 min	Elaborate	Task: 'Create a simple decision tree on paper: Should I bring an umbrella? Branc...
35-40 min	Evaluate	Exit ticket: '1. What is AI? 2. What is bias in AI? 3. Why is data important for...

Engage (5-7 minutes)

Teacher Activity

Show an AI chatbot conversation or image generator output. Ask: 'Is this intelligent? How does it work?' Discuss what makes AI different from regular software.

Student Activity

Observe the AI output. Discuss: 'It seems smart but follows rules.' Brainstorm examples of AI in daily life: voice assistants, recommendations.

Explore (10-12 minutes)

Teacher Activity

Activity: 'The Human Robot' — one student gives step-by-step commands, another follows literally. Show how AI follows instructions exactly, without common sense.

Student Activity

Participate in the game. Experience how literal instructions produce unexpected results. Discuss the gap between human and AI understanding.

Explain (10-12 minutes)

Teacher Activity

Define AI: systems that learn from data to make decisions. Key concepts: data fuel, patterns learning, algorithms rules. Show simple decision tree. Explain bias: AI learns from human data, which may contain biases.

Student Activity

Take notes. Understand data ⟶ pattern ⟶ decision pipeline. See how bias enters through training data.

Elaborate (8-10 minutes)

Teacher Activity

Task: 'Create a simple decision tree on paper: Should I bring an umbrella? Branches: Is it raining? Yes ⟶ Bring it, No ⟶ Check forecast, etc.' Discuss how AI makes decisions similarly.

Student Activity

Work in pairs. Create decision trees for different scenarios: 'What should I eat for lunch?', 'Should I study now?' Present to class.

Evaluate (5-8 minutes)

Teacher Activity

Exit ticket: '1. What is AI? 2. What is bias in AI? 3. Why is data important for AI?'

Student Activity

Answer the questions. Discuss the ethical implications of AI.

Cross-Curricular Connections

Mathematics: Decision trees use logical conditions if/then; probability determines AI confidence
Science: AI learns from data like scientists form hypotheses from observations
Language: Natural Language Processing enables AI to understand and generate human language

Digital Citizen Reminder: Be aware of AI bias — AI systems can make unfair decisions if trained on biased data.

Dormitory Task (Paper-and-Pencil)

Create a decision tree on paper for 'Should I play outside?' Include at least 4 decision points weather, homework, friends available, time of day. Draw branches for yes/no answers.

Resources Needed:
- Decision tree examples

- AI demo if available
- graph paper
- scenario cards

Differentiation

Approaching: Use a pre-made decision tree template and fill in the branches
On Level: Create a decision tree with 4+ decision points independently
Advanced: Research one AI application and present its benefits and risks to the class
SEND: Card-based decision activity, arrange yes/no cards in order

Formative Assessment

Method: Exit ticket and self-assessment
CT Skill Assessed: Abstraction — simplifying complex decisions into clear if/then branches

Teacher Reflection (Post-Lesson)

What worked well:

What to improve:

Connection to next lesson:

Lesson 6.10	Duration: 40 minutes

Ethics in Technology

Learning Objectives

- Identify ethical dilemmas arising from emerging technologies
- Apply ethical frameworks to evaluate technology decisions
- Propose responsible technology development practices

Timing Breakdown

0-5 min	Engage	Show an AI chatbot conversation or image generator output. Ask: 'Is this intel...
5-15 min	Explore	Activity: 'The Human Robot' — one student gives step-by-step commands, another f...
15-25 min	Explain	Define AI: systems that learn from data to make decisions. Key concepts: data f...
25-35 min	Elaborate	Task: 'Create a simple decision tree on paper: Should I bring an umbrella? Branc...
35-40 min	Evaluate	Exit ticket: '1. What is AI? 2. What is bias in AI? 3. Why is data important for...

Engage (5-7 minutes)

Teacher Activity

Show an AI chatbot conversation or image generator output. Ask: 'Is this intelligent? How does it work?' Discuss what makes AI different from regular software.

Student Activity

Observe the AI output. Discuss: 'It seems smart but follows rules.' Brainstorm examples of AI in daily life: voice assistants, recommendations.

Explore (10-12 minutes)

Teacher Activity

Activity: 'The Human Robot' — one student gives step-by-step commands, another follows literally. Show how AI follows instructions exactly, without common sense.

Student Activity

Participate in the game. Experience how literal instructions produce unexpected results. Discuss the gap between human and AI understanding.

Explain (10-12 minutes)

Teacher Activity

Define AI: systems that learn from data to make decisions. Key concepts: data fuel, patterns learning, algorithms rules. Show simple decision tree. Explain bias: AI learns from human data, which may contain biases.

Student Activity

Take notes. Understand data ⟶ pattern ⟶ decision pipeline. See how bias enters through training data.

Elaborate (8-10 minutes)

Teacher Activity

Task: 'Create a simple decision tree on paper: Should I bring an umbrella? Branches: Is it raining? Yes ⟶ Bring it, No ⟶ Check forecast, etc.' Discuss how AI makes decisions similarly.

Student Activity

Work in pairs. Create decision trees for different scenarios: 'What should I eat for lunch?', 'Should I study now?' Present to class.

Evaluate (5-8 minutes)

Teacher Activity

Exit ticket: '1. What is AI? 2. What is bias in AI? 3. Why is data important for AI?'

Student Activity

Answer the questions. Discuss the ethical implications of AI.

Cross-Curricular Connections

Mathematics: Decision trees use logical conditions if/then; probability determines AI confidence
Science: AI learns from data like scientists form hypotheses from observations
Language: Natural Language Processing enables AI to understand and generate human language

Digital Citizen Reminder: Be aware of AI bias — AI systems can make unfair decisions if trained on biased data.

Dormitory Task (Paper-and-Pencil)

Create a decision tree on paper for 'Should I play outside?' Include at least 4 decision points weather, homework, friends available, time of day. Draw branches for yes/no answers.

Resources Needed:
- Decision tree examples

- AI demo if available
- graph paper
- scenario cards

Differentiation

Approaching: Use a pre-made decision tree template and fill in the branches
On Level: Create a decision tree with 4+ decision points independently
Advanced: Research one AI application and present its benefits and risks to the class
SEND: Card-based decision activity, arrange yes/no cards in order

Formative Assessment

Method: Exit ticket and self-assessment
CT Skill Assessed: Abstraction — simplifying complex decisions into clear if/then branches

Teacher Reflection (Post-Lesson)

What worked well:

What to improve:

Connection to next lesson:

Lesson 6.11	Duration: 40 minutes

Project - Tech Research

Learning Objectives

- Conduct independent research on an emerging technology topic
- Synthesize information from multiple authoritative sources
- Present technology analysis with balanced technical and ethical perspectives

Timing Breakdown

0-5 min	**Engage**	Show an AI chatbot conversation or image generator output. Ask: 'Is this intel...
5-15 min	**Explore**	Activity: 'The Human Robot' — one student gives step-by-step commands, another f...
15-25 min	**Explain**	Define AI: systems that learn from data to make decisions. Key concepts: data f...
25-35 min	**Elaborate**	Task: 'Create a simple decision tree on paper: Should I bring an umbrella? Branc...
35-40 min	**Evaluate**	Exit ticket: '1. What is AI? 2. What is bias in AI? 3. Why is data important for...

Engage (5-7 minutes)

Teacher Activity

Show an AI chatbot conversation or image generator output. Ask: 'Is this intelligent? How does it work?' Discuss what makes AI different from regular software.

Student Activity

Observe the AI output. Discuss: 'It seems smart but follows rules.' Brainstorm examples of AI in daily life: voice assistants, recommendations.

Explore (10-12 minutes)

Teacher Activity

Activity: 'The Human Robot' — one student gives step-by-step commands, another follows literally. Show how AI follows instructions exactly, without common sense.

Student Activity

Participate in the game. Experience how literal instructions produce unexpected results. Discuss the gap between human and AI understanding.

Explain (10-12 minutes)

Teacher Activity

Define AI: systems that learn from data to make decisions. Key concepts: data fuel, patterns learning, algorithms rules. Show simple decision tree. Explain bias: AI learns from human data, which may contain biases.

Student Activity

Take notes. Understand data ⟶ pattern ⟶ decision pipeline. See how bias enters through training data.

Elaborate (8-10 minutes)

Teacher Activity

Task: 'Create a simple decision tree on paper: Should I bring an umbrella? Branches: Is it raining? Yes ⟶ Bring it, No ⟶ Check forecast, etc.' Discuss how AI makes decisions similarly.

Student Activity

Work in pairs. Create decision trees for different scenarios: 'What should I eat for lunch?', 'Should I study now?' Present to class.

Evaluate (5-8 minutes)

Teacher Activity

Exit ticket: '1. What is AI? 2. What is bias in AI? 3. Why is data important for AI?'

Student Activity

Answer the questions. Discuss the ethical implications of AI.

Cross-Curricular Connections

Mathematics: Decision trees use logical conditions if/then; probability determines AI confidence
Science: AI learns from data like scientists form hypotheses from observations
Language: Natural Language Processing enables AI to understand and generate human language

Digital Citizen Reminder: Be aware of AI bias — AI systems can make unfair decisions if trained on biased data.

Dormitory Task (Paper-and-Pencil)

Create a decision tree on paper for 'Should I play outside?' Include at least 4 decision points weather, homework, friends available, time of day. Draw branches for yes/no answers.

Resources Needed:
- Decision tree examples

- AI demo if available
- graph paper
- scenario cards

Differentiation

Approaching: Use a pre-made decision tree template and fill in the branches
On Level: Create a decision tree with 4+ decision points independently
Advanced: Research one AI application and present its benefits and risks to the class
SEND: Card-based decision activity, arrange yes/no cards in order

Formative Assessment

Method: Exit ticket and self-assessment
CT Skill Assessed: Abstraction — simplifying complex decisions into clear if/then branches

Teacher Reflection (Post-Lesson)

What worked well:

What to improve:

Connection to next lesson:

Lesson 6.12	Duration: 40 minutes

Chapter 6 Review

Learning Objectives

- Consolidate knowledge of all emerging technologies covered in Chapter 6
- Analyze the interconnected nature of modern technology ecosystems
- Evaluate emerging technologies critically with technical and ethical awareness

Timing Breakdown

0-5 min	Engage	Show an AI chatbot conversation or image generator output. Ask: 'Is this intel...
5-15 min	**Explore**	Activity: 'The Human Robot' — one student gives step-by-step commands, another f...
15-25 min	**Explain**	Define AI: systems that learn from data to make decisions. Key concepts: data f...
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35-40 min	**Evaluate**	Exit ticket: '1. What is AI? 2. What is bias in AI? 3. Why is data important for...

Engage (5-7 minutes)

Teacher Activity

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Elaborate (8-10 minutes)

Teacher Activity

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Student Activity

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Teacher Activity

Exit ticket: '1. What is AI? 2. What is bias in AI? 3. Why is data important for AI?'

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Teacher Reflection (Post-Lesson)

What worked well:

What to improve:

Connection to next lesson: